

# $p$ -ADIC HEIGHTS ON SHIMURA CURVES AND HIGHER $p$ -ADIC GREEN FUNCTIONS

PAOLO BORDIGNON, HAZEM HASSAN

ABSTRACT. Let  $X$  be a Shimura curve with Čerednik–Drinfeld uniformization at a prime  $p$  and let  $\mathcal{M}_n$  be the motive attached to modular forms of even weight  $n + 2$ . We study Nekovář’s local  $p$ -adic height of Heegner cycles on  $\mathcal{M}_n$ . The central result expresses this height as a Coleman integral of a canonical vector valued meromorphic differential on the  $p$ -adic upper half plane. We then construct this differential by means of Poincaré series and obtain an explicit formula in terms of  $p$ -adic logarithms. The final section proves that sum of higher  $p$ -adic Green functions over all the Heegner points is equal to the log of a rational number.

## CONTENTS

|                                                                |    |
|----------------------------------------------------------------|----|
| 1. Introduction                                                | 1  |
| 2. Filtered $(\varphi, N)$ -modules and Galois representations | 4  |
| 3. Motives, $p$ -adic local systems and $F$ -isocrystals       | 7  |
| 4. de Rham cohomology and Bruhat-Tits cocycles                 | 19 |
| 5. Nekovář’s $p$ -adic heights                                 | 24 |
| 6. Higher $p$ -adic Green functions                            | 42 |
| 7. A $p$ -adic Gross–Kohnen–Zagier result                      | 47 |
| References                                                     | 48 |

## 1. INTRODUCTION

The aim of this paper is to relate the local  $p$ -adic height of Heegner cycles on Shimura curves to higher  $p$ -adic Green functions. The point of view is both arithmetic and analytic. On the arithmetic side, Heegner cycles define extensions of semistable Galois representations and therefore biextensions in the category of filtered  $(\varphi, N)$ -modules. On the analytic side, the Čerednik–Drinfeld uniformization identifies the Shimura curve at  $p$  with a quotient of the  $p$ -adic upper half plane. This permits us to describe the relevant filtered modules, their monodromy, and their Poincaré pairing in terms of harmonic cocycles and Coleman primitives.

Let  $N = pN^-N^+$ , where  $N^-$  is squarefree with an odd number of prime divisors and  $N^+$  is relatively prime to  $pN^-$ . Let  $X$  be the Shimura curve attached to the indefinite quaternion algebra of discriminant  $pN^-$ . We fix an even integer  $n = 2m$  and consider the motive  $\mathcal{M}_n$  over  $X$  which realizes the local system of modular forms of weight  $n + 2$ . If  $K$  is an imaginary quadratic field satisfying the Heegner hypothesis of Section 3.9, CM points on  $X$  give cycles in  $\mathrm{CH}^{m+1}(\mathcal{M}_n)$ . Their  $p$ -adic Abel–Jacobi images determine extensions of the Galois representation

$$V = H_{\mathrm{\acute{e}t}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G.$$

Pairing two such extensions gives the biextension whose local height we compute.

The construction belongs to the framework introduced by Nekovář in [Nek93, Nek95]. Our geometric input is the description of the motive, the semistable realization, and the monodromy module developed by Iovita and Spieß in [IS03]. The relation between Nekovář’s height and integration on curves with semistable reduction is close in spirit to the work of Besser and of Besser and Zerbes [Bes22, BZ23]. The analytic construction is motivated by the treatment of higher Green

functions by Brown and Fonseca [BF25], while the geometry of Green functions on Mumford curves is developed by Wei in [Wei18].

**Relation with previous work.** The seminal work of Gross–Zagier on the BSD conjecture [GZ86] and the following work [GKZ87] have shed light on interesting connections between automorphic functions, intersection theory and modular objects in general. The higher Green functions  $G_k(z, w)$  are one of those cornerstone objects in the proof and subsequent works since they connect the hyperbolic Laplace operator to intersection theory of Heegner cycles. The value of these functions at CM points has been conjectured, in the latter paper, to be equal to the logarithm of an algebraic number. This has sparked research in several directions and has now recently been fully resolved in many instances by fairly different methods. The nature of the object has indeed attracted researchers both from a more analytic side [BEY21], [Via11] and [Li23] via the theory of Borcherds lifts and analytic Maass forms and from a geometric one [Zha97], [Mel08] and [BF25] using a more motivic approach. Our paper has been inspired by this latter point of view; it gave us the idea of bridging the  $p$ -adic heights of Heegner cycles with explicit analytic description of the higher Green functions using cohomological tools.

The local formula belongs first to the theory of Coleman and Gross [CG89]. Their height on a curve is described by intersection theory away from  $p$  and by Coleman integration of normalized differentials of the third kind above  $p$ . Nekovář recast  $p$ -adic heights in terms of mixed extensions of Galois representations and applied this formalism to Heegner cycles [Nek93, Nek95]. Besser proved that the Coleman–Gross and Nekovář constructions agree for curves [Bes04]; his later work extends this comparison to semistable reduction by means of Vologodsky integration [Bes22]. Our construction uses Nekovář’s mixed extension formalism, but the resulting local formula has the shape of a Coleman–Gross integral with coefficients in  $\mathcal{V}_n$ .

The closest higher weight antecedent is Nekovář’s formula for Heegner cycles [Nek95]. In that work  $p$  does not divide the level and the modular form is ordinary at  $p$ , so the local height is defined using the ordinary splitting. Here  $p$  divides the level exactly once, the local representation is Steinberg, and the Shimura curve is described by Čerednik–Drinfeld uniformization. The purpose of Sections 4 through 6 is to replace the ordinary calculation at  $p$  by a semistable calculation in terms of Frobenius, monodromy, Bruhat–Tits cocycles, and Coleman primitives. The monodromy module calculations of Iovita and Spieß and their extension by Masdeu provide the essential precedent for this passage [IS03, Mas11].

The formulas of Perrin–Riou, Nekovář, and Shnidman relate  $p$ -adic heights of Heegner points or generalized Heegner cycles to derivatives of  $p$ -adic  $L$ -functions [PR87, Nek95, Shn16]. Disegni places this circle of ideas on Shimura curves with potentially ordinary, including semistable, reduction and subsequently treats primes which are nonsplit in the CM field [Dis17, Dis23]. The aim here is different and complementary. We compute the local height at the Steinberg prime directly and identify it with a higher  $p$ -adic Green function. We do not construct a  $p$ -adic  $L$ -function or prove a height formula for its derivative. In the prospective global statement of Section 7, the relevant height is the cyclotomic component. The Hecke operator is expected to annihilate the middle representation and isolate a Kummer extension, leading to an algebraicity statement rather than to a  $p$ -adic Gross–Zagier formula.

On the analytic side, Bertolini, Darmon, Iovita, and Spieß relate the Čerednik–Drinfeld uniformization and boundary distributions to Coleman integration [BDIS02], while Iovita and Spieß use monodromy modules to study Heegner cycles [IS03]. Brown and Fonseca interpret higher Green functions through motivic biextensions in the complex setting [BF25], and Wei develops Green functions on Mumford curves in the  $p$ -adic setting [Wei18]. The contribution sought here is the direct bridge between these two aspects: the mixed extension attached to higher weight Heegner cycles is evaluated by an explicit vector valued Poincaré series on the  $p$ -adic upper half plane.

The new feature needed here is the simultaneous treatment of nontrivial coefficients, semistable reduction at  $p$ , and relative cohomology for complements of CM divisors. Let  $D_1$  and  $D_2$  be disjoint effective divisors and let  $Q_i$  be the residue data determined by the corresponding Heegner cycles.

The relative de Rham groups

$$H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n) \quad \text{and} \quad H_{\text{dR}}^1(X_\Gamma - D_2, D_1, \mathcal{V}_n)$$

carry compatible Frobenius, monodromy, filtration, and Poincaré pairings. The slope decomposition determines a canonical complement to the relevant Hodge filtration. This is the complement used in Nekovář's construction of the local height.

For residue data  $Q$  we denote by  $\Psi_Q$  the unique class with prescribed residue, lying in the top filtered piece and having trivial monodromy. We call it the Coleman–Gross differential associated to  $Q$ . The main result of the paper is the following formula.

**Theorem A** (Local height formula). *Let  $D_1$  and  $D_2$  be disjoint reduced divisors defined over a field  $H$  and let  $Q_1$  and  $Q_2$  be residue classes defining the corresponding Heegner extension  $e_1, e_2$  and a biextension  $E = E(Q_1, Q_2)$ . Assume that the points in  $D_1$  reduce to one irreducible component and that*

$$\sum_{z \in D_1} Q_{1,z} / \# \Gamma_z = 0.$$

*Then the local height at  $v|p$  is given by*

$$h(e_1, e_2)_{E,v} = -\text{Tr}_{H_v/\mathbb{Q}_p} \left( \int_{D_1} (\Psi_{Q_2}, Q_1) \right).$$

*Equivalently, after choosing lifts to the  $p$ -adic upper half plane and a Coleman primitive  $F_{\Psi_{Q_2}}$ , the right hand side is the sum of the values*

$$-\text{Tr}_{H_v/\mathbb{Q}_p} \left( \sum_{z \in D_1} \frac{(F_{\Psi_{Q_2}}(z), Q_{1,z})_\Gamma}{\# \Gamma_z} \right).$$

The proof identifies the semistable realization of the biextension, computes its Bloch–Kato class, and compares the Frobenius and monodromy splitting with the splitting defined by the Hodge filtration. Poincaré duality then reduces the height to the displayed Coleman integral. This is carried out in Section 5.

The second result gives an analytic realization of the differential in Theorem A. For two points  $w_1, w_2$  on the  $p$ -adic upper half plane and  $i + j = n$ , consider the Poincaré series

$$\Psi_\Gamma^{i,j}(z; w_1, w_2) = \sum_{\gamma \in \Gamma} \frac{1}{j(\gamma, z)^{n+2}} \frac{w_1 - w_2}{(\gamma z - w_1)^{i+1} (\gamma z - w_2)^{j+1}}.$$

We prove that this series converges to a rigid meromorphic modular form. The associated vector valued differential has the prescribed residues and trivial monodromy. It is therefore the Coleman–Gross differential entering the height formula. Its Coleman primitive can be written in terms of derivatives of

$$\frac{(z - w_1)^j (z - w_2)^i}{(w_1 - w_2)^n} \log_{\mathcal{L}} \left( \frac{z - w_1}{z - w_2} \right).$$

As a consequence, the local height is expressed by a convergent Poincaré sum of explicit  $p$ -adic logarithmic terms. This is the higher  $p$ -adic Green function studied in Section 6.

The global conclusion is an algebraicity statement in the spirit of Gross, Kohnen, and Zagier [GKZ87]. If a finite combination of Hecke operators annihilates the middle Galois representation  $V$ , the Heegner biextension is reduced to a Kummer extension. The sum of all the local heights above  $p$  is consequently the  $p$ -adic logarithm of the norm of an algebraic number in the compositum of the relevant Hilbert class fields. This is explained and proven in Section 7.

**Organization of the paper.** Section 2 recalls semistable representations and proves the extension formulas used later. Section 3.1 introduces the Shimura curves and their Čerednik–Drinfeld uniformization. Sections 3.2 through 3.9 describe the motive  $\mathcal{M}_n$ , its  $p$ -adic and de Rham realizations, and the Heegner cycles. Section 4 studies relative de Rham cohomology through Bruhat–Tits cocycles. Section 5 specializes Nekovář's height machinery and proves Theorem A. Section 6 constructs the Poincaré series and derives the analytic formula. Section 7 states the expected global algebraicity result and explains the strategy toward it.

## 2. FILTERED $(\varphi, N)$ -MODULES AND GALOIS REPRESENTATIONS

Before delving into the  $p$ -adic aspects of Shimura curves and their cohomology, we will start recalling some basic fact about  $p$ -adic Galois representations and the associated category of abstract linear algebra objects introduced by [Fon79]. This will help us set up the notation and already establish the categories where the realizations of certain motives will live. We will follow the exposition of [Nek93].

**2.1. Filtered  $(\varphi, N)$ -modules.** Let  $F$  be a finite extension of  $\mathbb{Q}_p$  with ring of integers  $\mathcal{O}_F$  and residue field  $k$ . Denote by  $F_0 \subseteq F$  the maximal unramified subextension that can be identified with the fraction field of the ring of Witt vectors  $W(k)$ . Let  $\sigma : F_0 \rightarrow F_0$  the absolute Frobenius automorphism. Write  $G_F$  for the absolute Galois group  $G_F = \text{Gal}(\bar{F}/F)$ .

**Definition 2.1.** Define the category  $\text{MF}_F(\varphi, N)$  of filtered  $(\varphi, N)$ -modules over  $F$  to be the category of finite dimensional  $F_0$ -vector spaces equipped with:

- a  $\sigma$ -semilinear bijective map  $\varphi : D \rightarrow D$  called Frobenius,
- a  $F_0$ -linear map  $N : D \rightarrow D$  satisfying  $N\varphi = p\varphi N$  called monodromy operator,
- a decreasing filtration  $\text{Fil}^i D_F$  on the vector space  $D_F = D \otimes_{F_0} F$  satisfying  $\bigcup \text{Fil}^i D_F = D_F$  and  $\bigcap \text{Fil}^i D_F = 0$ .

The morphisms are defined to be compatible with  $\varphi, N$  and  $\text{Fil}^\bullet$ .

The category  $\text{MF}_F(\varphi, N)$  is an additive tensor category admitting kernels and cokernels. Given a filtered  $(\varphi, N)$ -module  $D = (D, \text{Fil}_D^\bullet, \varphi_D, N_D)$  we can define its  $j$ -th Tate twist to be the object

$$(2.1) \quad D[j] = (D, \text{Fil}_D^{\bullet-j}, p^j \varphi_D, N_D).$$

Consider  $D$  a filtered  $(\varphi, N)$ -module. For a rational number  $\lambda = n/m$  with  $n, m \in \mathbb{Z}$  and  $(n, m) = 1$ , denote  $D_\lambda$  the largest subspace of  $D$  which has an  $\mathcal{O}_{F_0}$ -stable lattice  $M$  satisfying

$$(2.2) \quad \varphi^m(M) = p^n(M).$$

The subspace  $D_\lambda$  is called *isotypical component* of  $D$  of slope  $\lambda$ . By definition of the monodromy operator we have  $N(D_\lambda) \subseteq D_{\lambda-1}$  for all  $\lambda \in \mathbb{Q}$ .

A  $p$ -adic representation  $V$  of  $G_F$  is a finite dimensional vector space over  $\mathbb{Q}_p$  endowed with a continuous linear action of  $G_F$ . We denote by  $\underline{\text{Rep}}_{\mathbb{Q}_p}(G_F)$  the category of such representations with obvious morphisms. It is an abelian tensor category with twists given by tensoring with Tate representation  $\mathbb{Q}_p(1)$ .

Fontaine [Fon94] constructed topological rings endowed with continuous  $G_F$  action

$$(2.3) \quad B_{\text{cris}} \subset B_{\text{st}} \subset B_{\text{dR}}$$

and certain extra structure. The embedding  $B_{\text{st}} \hookrightarrow B_{\text{dR}}$  depends on the choice of a branch of a  $p$ -adic logarithm.

Consider the following functors

$$(2.4) \quad \begin{aligned} D_{\text{cris}} : \underline{\text{Rep}}_{\mathbb{Q}_p}(G_F) &\rightarrow \text{MF}_F(\varphi, N) \\ V &\mapsto D_{\text{cris}}(V) := (V \otimes B_{\text{cris}})^{G_F}, \end{aligned}$$

$$(2.5) \quad \begin{aligned} D_{\text{st}} : \underline{\text{Rep}}_{\mathbb{Q}_p}(G_F) &\rightarrow \text{MF}_F(\varphi, N) \\ V &\mapsto D_{\text{st}}(V) := (V \otimes B_{\text{st}})^{G_F}, \end{aligned}$$

$$(2.6) \quad \begin{aligned} \text{DR} : \underline{\text{Rep}}_{\mathbb{Q}_p}(G_F) &\rightarrow \text{FilVec}_F \\ V &\mapsto D_{\text{dR}}(V) := (V \otimes B_{\text{dR}})^{G_F} \end{aligned}$$

where  $\text{FilVec}_F$  denotes the category of filtered  $F$ -vector spaces.

**Definition 2.2.** A  $p$ -adic representation  $V$  is called

- crystalline if  $\dim_{F_0} D_{\text{cris}}(V) = \dim_{\mathbb{Q}_p}(V)$ ,

- semistable if  $\dim_{F_0} D_{\text{st}}(V) = \dim_{\mathbb{Q}_p}(V)$ ,
- de Rham if  $\dim_F \text{DR}(V) = \dim_{\mathbb{Q}_p}(V)$ ,

The category of crystalline (resp. semistable, resp. de Rham) representations, denoted  $\underline{\text{Rep}}_{\text{cris}}(G_F)$  (resp.  $\underline{\text{Rep}}_{\text{st}}(G_F)$ , resp.  $\underline{\text{Rep}}_{\text{dR}}(G_F)$ ) is the full subcategory of  $\underline{\text{Rep}}_{\mathbb{Q}_p}(G_F)$  of crystalline (resp. semistable, resp. de Rham) objects.

The Tate twist  $D(n)$  is given by

$$(2.7) \quad D(n) := D_{\text{cris}}(\mathbb{Q}_p(n)) = D_{\text{st}}(\mathbb{Q}_p(n)) = (F_0, \text{Fil}^{-n}, p^{-n}\sigma, 0)$$

with the filtration concentrated in degree  $[-n]$ . The de Rham realization consists in the filtered vector space  $\text{DR}(\mathbb{Q}_p(n)) = (F, \text{Fil}^{\bullet-n})$ . The effect of tensoring by the Tate twist in  $\underline{\text{Rep}}_{\mathbb{Q}_p}(G_F)$  corresponds to taking the tensor product by (2.7) in the category of filtered  $(\varphi, N)$ -modules. In particular, according to our notation, for every  $p$ -adic representation  $V$  we have the following relation:

$$(2.8) \quad D_{\text{st}}(V(n)) = D_{\text{st}}(V)[-n].$$

**Definition 2.3.** A filtered  $(\varphi, N)$ -module  $D$  is called *admissible* if is isomorphic to  $D_{\text{st}}(V)$  for some semistable representation  $V$  of  $G_F$ . The full subcategory of admissible filtered  $(\varphi, N)$ -modules is denoted  $\text{MF}_F^{\text{ad}}(\varphi, N)$ .

**Theorem 2.1** ([CF00]). The functor  $D_{\text{st}}$  gives an equivalence of categories between  $\underline{\text{Rep}}_{\text{st}}(G_F)$  and  $\text{MF}_F^{\text{ad}}(\varphi, N)$ , which is compatible with exact sequences, tensor product and duality.

This theorem will be fundamental in the construction of the  $p$ -adic height of pairing à la Nekovář since we can establish the following group isomorphism.

**Corollary 2.1.** The equivalence of categories induces a canonical group isomorphism

$$(2.9) \quad \text{Ext}_{\underline{\text{Rep}}_{\text{st}}(G_F)}^1(V, W) \cong \text{Ext}_{\text{MF}_F^{\text{ad}}(\varphi, N)}^1(D_{\text{st}}(V), D_{\text{st}}(W))$$

for every  $V, W$  semistable  $p$ -adic representations of  $G_F$ .

**2.2. Extension of  $p$ -adic Galois representations.** One of the advantage of working with the linear algebraic category of filtered  $(\varphi, N)$ -modules is given by the characterization and explicit description that we can give of the extension group  $\text{Ext}_{\text{MF}_F^{\text{ad}}(\varphi, N)}^1$ .

**Lemma 2.1.** Let  $D$  be an admissible filtered  $(\varphi, N)$ -module and assume  $N$  induces a bijection  $N(D_0) = D_{-1}$ . Then there is a canonical isomorphism

$$(2.10) \quad \text{Ext}_{\text{MF}_F^{\text{ad}}(\varphi, N)}^1(F_0, D) \cong D_F / \text{Fil}^0 D_F$$

where  $F_0$  denotes the filtered  $(\varphi, N)$ -module  $D_{\text{st}}(\mathbb{Q}_p) = (F_0, \text{Fil}^0, \text{Id}, 0)$ . The map is often referred to as the Bloch-Kato isomorphism.

*Proof.* See [IS03, Lemma 2.1], [Mas11, Lemma 4.6]. Note the difference by a Tate shift  $[n+1]$ .  $\square$

It will be fundamental for our later computations to have a precise and explicit description of the isomorphism. We here recall the construction contained in the proof. Consider

$$(2.11) \quad e : 0 \rightarrow D \xrightarrow{l} E \xrightarrow{p} F_0 \rightarrow 0$$

an extension of filtered  $(\varphi, N)$ -modules whose class belong to  $\text{Ext}_{\text{MF}_F^{\text{ad}}(\varphi, N)}^1(F_0, D)$  and satisfies the hypothesis of the Lemma. We have a decomposition of  $E_F \cong D_F \oplus \bar{F}$  compatible with filtration

$$(2.12) \quad \text{Fil}^j E_F \cong \begin{cases} \text{Fil}^j D_F \oplus F & \text{for } i \leq 0, \\ \text{Fil}^i D_F & \text{for } i > 0 \end{cases}$$

and there exists a splitting of  $p$  compatible with Frobenius and monodromy

$$(2.13) \quad s : F_0 \rightarrow E_0$$

where  $E_0$  is the slope-0 part. We then define  $\pi_{D_F}(s(1)) + \text{Fil}^0 D_F \in D_F / \text{Fil}^0 D_F$  the element associated to the class  $[e]$  of (2.11), where  $\pi_{D_F}$  denotes the projection to  $D_F$  under the decomposition  $E_F \cong D_F \oplus F$ .

**Remark 2.1.** Lemma 2.1 fundamentally states that the class of a simple extension  $[e]$  is characterized by the comparison between the Frobenius-monodromy data and the filtration of the vector spaces.

We will also need to characterize the extension groups for other filtered  $(\varphi, N)$ -modules not necessarily satisfying the hypothesis of Lemma 2.1. Consider the following exact sequence of admissible  $(\varphi, N)$ -modules

$$(2.14) \quad 0 \rightarrow F_0[-1] \rightarrow E \rightarrow V \rightarrow 0$$

where  $V$  satisfies the hypothesis above and  $F_0[-1]$  denotes the filtered  $(\varphi, N)$ -module  $D_{\text{st}}(\mathbb{Q}_p(1))$ . Assume furthermore that  $V$  satisfies the following property

$$(2.15) \quad V^{\varphi=1, N=0} = (V^\vee[-1])^{\varphi=1, N=0} = 0.$$

**Lemma 2.2.** Let  $E$  be a filtered  $(\varphi, N)$ -module defined by (2.14). Then there is a canonical isomorphism

$$(2.16) \quad \text{Ext}_{\text{MF}_F^{\text{ad}}(\varphi, N)}^1(F_0, E) \cong E_F / \text{Fil}^0 E_F \oplus \mathbb{Q}_p$$

where  $F$  denotes the filtered  $(\varphi, N)$ -module  $D_{\text{st}}(\mathbb{Q}_p)$  as before. The map is often referred to as the Bloch-Kato isomorphism.

*Proof.* We consider the complex  $C_{\text{st}}^\bullet(E)$  defined by

$$(2.17) \quad C_{\text{st}}^\bullet(E) = [E \xrightarrow{(-i, \varphi - \text{Id}, N)} E_F / \text{Fil}^0 E_F \oplus E \oplus E[1] \xrightarrow{(0, N, \text{Id} - \varphi)} E[1]].$$

In [Nek93, Sec. 1.19], we have an isomorphism

$$(2.18) \quad \text{Ext}_{\text{MF}_F^{\text{ad}}(\varphi, N)}^1(F_0, E) \cong H^1(C_{\text{st}}^\bullet(E)).$$

Explicitly, let

$$(2.19) \quad 0 \rightarrow E \rightarrow D \rightarrow F_0 \rightarrow 0$$

be an extension of admissible  $(\varphi, N)$ -modules whose class belongs to  $\text{Ext}_{\text{MF}_F^{\text{ad}}(\varphi, N)}^1(F_0, E)$ . Then we have a commutative diagram of the form

$$(2.20) \quad \begin{array}{ccccccc} 0 & \longrightarrow & E_F & \longrightarrow & D_F & \longrightarrow & F \longrightarrow 0 \\ & & \downarrow & & \downarrow \pi & & \downarrow \\ 0 & \longrightarrow & E_F / \text{Fil}^0 E_F & \xrightarrow{\sim} & D_F / \text{Fil}^0 D_F & \longrightarrow & 0 \longrightarrow 0. \end{array}$$

Pick a  $F_0$ -section  $s : F_0 \rightarrow D$  and define a triple

$$(2.21) \quad [-\pi \circ i(s(1), (\varphi - \text{Id})s(1), Ns(1))] \in E_F / \text{Fil}^0 E_F \oplus E \oplus E[1]$$

whose cohomology class  $\rho(D) \in H^1(C_{\text{st}}^\bullet(E))$  is independent of the choice of the section  $s$ . We now need to conclude that  $H^1(C_{\text{st}}^\bullet(E)) \cong F_0 \oplus E_F / \text{Fil}^0 E_F$ . Consider the dual shifted sequence of (2.14)

$$(2.22) \quad 0 \rightarrow V^\vee[-1] \rightarrow E^\vee[-1] \rightarrow F_0 \rightarrow 0.$$

Under the assumption (2.15), there exists a unique  $(\varphi, N)$ -equivariant decomposition of  $E^\vee[-1]$  as in the proof of Lemma 2.1 and then a decomposition

$$(2.23) \quad E = F_0[-1] \oplus V.$$

We then have that

$$(2.24) \quad \text{Im}(E \xrightarrow{((\varphi - \text{Id}), N)} E \oplus E[1]) = F_0[-1] \oplus \text{Im}(V \xrightarrow{((\varphi - \text{Id}), N)} V \oplus V[1])$$

since  $\varphi$  and  $N$  act respectively by  $\sigma$  and trivially on  $F_0$ . On the other hand,

$$(2.25) \quad \ker(E \oplus E[1] \xrightarrow{(N, (\text{Id} - \varphi))} E \oplus E[1]) = F_0[-1] \oplus \mathbb{Q}_p \oplus \ker(V \oplus V[1] \xrightarrow{(N, (\text{Id} - \varphi))} V \oplus V[1]).$$

We can then express the cohomology group  $H^1(C_{\text{st}}^\bullet)$  as

$$(2.26) \quad H^1(C_{\text{st}}^\bullet(E)) = E_F / \text{Fil}^0 E_F \oplus \mathbb{Q}_p \oplus \frac{\ker(V \oplus V[1] \xrightarrow{(N, (\text{Id}-\varphi))} V \oplus V[1])}{\text{Im}(V \xrightarrow{((\varphi-\text{Id}), N)} V \oplus V[1])}.$$

Recall that under the assumption (2.15), every simple extension of  $V$  factors through a filtered  $(\varphi, N)$ -equivariant map as a direct sum  $V \oplus F_0$ . In particular, we have

$$(2.27) \quad \text{Ext}_{\text{MF}_F^{\text{ad}}(\varphi, N)}^1(F_0, V) \cong H^1(C_{\text{st}}^\bullet(V)) \cong V_F / \text{Fil}^0 V_F.$$

From this we conclude  $H^1(C_{\text{st}}^\bullet(E)) = E_F / \text{Fil}^0 E_F \oplus \mathbb{Q}_p$ .  $\square$

The explicit description of the *Bloch-Kato isomorphism* (2.16) is given as follows. First of all fix the unique  $(\varphi, N)$ -equivariant decomposition  $E = F_0[-1] \oplus V$ . Then for every extension

$$(2.28) \quad 0 \rightarrow E \rightarrow D \rightarrow F_0 \rightarrow 0$$

consider the  $\varphi$ -equivariant splitting  $s : F_0 \rightarrow D$  inducing a  $\varphi$ -equivariant decomposition

$$(2.29) \quad D \cong F_0[-1] \oplus V \oplus F_0.$$

Fix a decomposition  $D_F \cong E_F \oplus F$  as in the previous case (2.12) then we define

$$(2.30) \quad (\pi_{E_F}(s(1)), \pi_{1, F_0}(N(s(1)))) \in E_F / \text{Fil}^0 E_F \oplus \mathbb{Q}_p$$

where  $\pi_{E_F}$  denotes the projection under the decomposition  $D_F \cong E_F \oplus F$ , while  $\pi_{1, F_0}$  denotes the projection under the decomposition (2.29).

### 3. MOTIVES, $p$ -ADIC LOCAL SYSTEMS AND $F$ -ISOCRYSTALS

**3.1. Shimura curves and  $p$ -adic uniformization.** Let  $N$  be a positive integer that can be factored as  $N = pN^-N^+$  with

- $p$  prime;
- $N^-$  positive squarefree integer with odd number of prime divisors;
- $N^+$  positive integer relatively prime to  $pN^-$ .

Let  $\mathcal{B}$  be the indefinite rational quaternion algebra of discriminant  $pN^-$ . Fix  $\mathcal{R}^{\max} \subset \mathcal{B}$  a maximal order and  $\mathcal{R} \subset \mathcal{R}^{\max}$  an Eichler order of level  $N^+$ .

**Definition 3.1.** Let  $S$  be a  $\mathbb{Q}$ -scheme. An abelian surface with quaternionic multiplication (QM) by  $\mathcal{R}^{\max}$ , and level  $N^+$  structure over  $S$  is a triple  $(\mathcal{A}, i, C)$  with

- (i)  $\mathcal{A}$  is a principally polarized abelian scheme over  $S$  of relative dimension 2;
- (ii)  $i : \mathcal{R}^{\max} \hookrightarrow \text{End}_S(\mathcal{A})$  is an injective ring homomorphism;
- (iii)  $C$  is a subgroup scheme of  $\mathcal{A}$  which is étale locally isomorphic to  $(\mathbb{Z}/N^+\mathbb{Z})^2$  stable and locally cyclic under the action of  $\mathcal{R}$ .

**Definition 3.2.** The Shimura curve  $X := X_{N^+, pN^-} / \mathbb{Q}$  is the coarse moduli scheme for the moduli problem over  $\mathbb{Q}$

$$\underline{\text{Sch}} / \mathbb{Q} \rightarrow \underline{\text{Sets}}$$

$$S \mapsto \{\text{Abelian surfaces with QM by } \mathcal{R}^{\max} \text{ and level } N^+ \text{ structure over } S\} / \sim.$$

$X$  is a smooth, projective, and geometrically connected curve over  $\mathbb{Q}$ .

Let  $B$  be the definite rational quaternion algebra of discriminant  $N^-$ , let  $R$  be an Eichler  $\mathbb{Z}[1/p]$ -order of level  $N^+$  in  $B$ . Define the group  $\Gamma$  to be

$$\Gamma := \{x \in R^\times : \text{nrd}(x) = 1\}.$$

Since the quaternion algebra  $B$  splits at  $p$ , fix an isomorphism

$$\iota_p : B_p = B \otimes_{\mathbb{Q}} \mathbb{Q}_p \xrightarrow{\sim} M_2(\mathbb{Q}_p).$$

The isomorphism  $\iota_p$  identifies  $\Gamma$  with a discrete cocompact subgroup of  $SL_2(\mathbb{Q}_p)$ .

**Theorem 3.1** (Čerednik–Drinfeld). *There is an isomorphism of rigid analytic varieties over  $\mathbb{Q}_{p^2}$ .*

$$(X_{\mathbb{Q}_{p^2}})^{\text{an}} \cong X_{\Gamma} := \Gamma \setminus \mathcal{H}_p.$$

*Proof.* See [BC91, Section 5.3]. □

In order to proceed with our construction we need to work with a finely representable moduli problem and with the universal abelian scheme associated. To do this, we rigidify the problem adding full level structure.

**Definition 3.3.** *Let  $M \geq 3$  be an integer relatively prime to  $N$ . An abelian surface with quaternionic multiplication (QM) by  $\mathcal{R}^{\max}$ , level  $N^+$  structure and full level  $M$  over  $S$  is a quadruple  $(\mathcal{A}, i, G, \nu)$  such that  $(\mathcal{A}, i, G)$  satisfies the properties of definition 3.1 and  $\nu$  is a  $\mathcal{R}^{\max}$ -equivariant isomorphism*

$$\nu : (\mathcal{R}^{\max} / M\mathcal{R}^{\max})_S \xrightarrow{\sim} A[M]$$

*from the constant group scheme  $(\mathcal{R}^{\max} / M\mathcal{R}^{\max})_S$  to the group scheme of  $M$ -division points.*

The corresponding moduli problem admits a fine moduli scheme over  $\mathbb{Q}$  which we denote by  $X_M$ . This is a smooth projective curve over  $\mathbb{Q}$  but it is not geometrically connected. The map induced by forgetting the full level structure

$$q : X_M \rightarrow X$$

is a Galois covering with Galois group  $G/\{\pm 1\}$  with

$$(3.1) \quad G := G_M := (\mathcal{R}^{\max} / M\mathcal{R}^{\max})^{\times} \cong (\mathcal{R} / M\mathcal{R})^{\times} \cong GL_2(\mathbb{Z} / M\mathbb{Z}) \cong (R / MR)^{\times}.$$

Let  $\Gamma_M$  be the group

$$\Gamma_M := \{\gamma \in \Gamma : \gamma \equiv 1 \pmod{MR}\}$$

**Theorem 3.2** (Čerednik–Drinfeld). *There is an isomorphism of rigid analytic varieties over  $\mathbb{Q}_{p^f}$  for a suitable finite unramified extension*

$$(X_{M, \mathbb{Q}_{p^f}})^{\text{an}} \cong \Gamma \setminus (\mathcal{H}_p \times (R / MR)^{\times}) \cong \coprod_{(\mathbb{Z} / M\mathbb{Z})^{\times}} \Gamma_M \setminus \mathcal{H}_p.$$

*In particular,  $(X_{M, \mathbb{Q}_{p^f}})^{\text{an}}$  is the disjoint union of Mumford curves  $X_{\Gamma_M} := \Gamma_M \setminus \mathcal{H}_p$ .*

*Proof.* See [BC91, Section 5.3]. □

**3.2. Chow motives and abelian schemes.** In this section we recall some of the main constructions and properties of motives and Galois representations attached to  $p$ -adic sheaves on Shimura curves. In Appendix 10.1 [IS03], Iovita-Spieß generalized the construction of [Sch90] to the case of Shimura curves. We briefly recall some of the main results.

Consider  $\pi : \mathcal{A} \rightarrow X$  the universal abelian scheme associated with the moduli problem of Definition 3.3. The category  $\mathcal{M}^0(X_M)$  of *relative Chow motives* is an additive pseudo-abelian  $\mathbb{Q}$ -linear rigid tensor category. The objects of  $\mathcal{M}^0(X_M)$  are triples  $(Y, p, e)$  where

- $Y$  is a projective  $X_M$ -scheme,
- $p$  is an idempotent element in the ring of relative correspondences  $\text{Corr}_{X_M}(Y \times Y) := \bigoplus_i CH^1(Y_i \times_{X_M} Y)$  where  $Y_i$  are the connected components of  $Y$ ,
- $e$  is an integer.

For a general motive  $\mathcal{M}$  we can define its associated *Chow group*  $CH^i(\mathcal{M})$  to be

$$(3.2) \quad CH^i(\mathcal{M}) := \text{Hom}_{\mathcal{M}^0(X_M)}(\mathbb{1}, \mathcal{M}(i))$$

where  $\mathbb{1}$  is the motive associated to the point. For a smooth projective  $X_M$ -scheme  $Y$  we denote by  $h(Y) = (Y, \Delta_Y, 0)$  its motive. Thanks to an important Theorem of Deninger-Murre [DM91] and Künnemann [Kün94], the motive associated to the abelian scheme  $\mathcal{A}$  decomposes into its graded pieces

$$(3.3) \quad h(\mathcal{A}) = h^0(\mathcal{A}) \oplus h^1(\mathcal{A}) \oplus h^2(\mathcal{A}) \oplus h^3(\mathcal{A}) \oplus h^4(\mathcal{A})$$

such that

$$(3.4) \quad h^i(\mathcal{A}) \cong \wedge^i h^1(\mathcal{A}), \quad h^i(\mathcal{A})^\vee \cong h^{4-i}(\mathcal{A})(2).$$

The  $\mathcal{R}^{\max}$ -action induces an embedding  $\mathcal{B} \hookrightarrow \text{End}(h^1(\mathcal{A}))$ . Let  $e_2$  be the unique nonzero idempotent in  $\text{End}(h^2(\mathcal{A})) = \text{End}(\wedge^2 h^1(\mathcal{A}))$  such that

$$(3.5) \quad x^2 \cdot e_2 = N(x)e_2 \quad \text{for all } x \in \mathcal{B}$$

where  $N$  is the reduced norm. The projector  $\epsilon_2 \in \text{Corr}_{X_M}(\mathcal{A}, \mathcal{A})$  is defined by

$$(3.6) \quad (\mathcal{A}, \epsilon_2) = h^2(\mathcal{A})_- := \ker(e_2).$$

This motive should be thought as the *non-quaternionic* piece inside  $h^2(\mathcal{A})$ . Let

$$(3.7) \quad (-, -) : \text{Sym}^2(h^2(\mathcal{A})_-) \rightarrow \mathbb{Q}(-2)$$

be the restriction of the symmetric pairing

$$(3.8) \quad h^2(\mathcal{A}) \otimes h^2(\mathcal{A}) \cong \wedge^2 h^1(\mathcal{A}) \otimes \wedge^2 h^1(\mathcal{A}) \rightarrow \wedge^4 h^1(\mathcal{A}) \cong \mathbb{Q}(-2).$$

This pairing induce the *Laplace operator*

$$(3.9) \quad \Delta_m : \text{Sym}^m(h^2(\mathcal{A})_-) \rightarrow (\text{Sym}^{m-2}(h^2(\mathcal{A})_-))(-2)$$

given by

$$(3.10) \quad \Delta_m(x_1 x_2 \cdots x_m) = \sum_{1 \leq i < j \leq m} (x_i, x_j) x_1 \cdots \hat{x}_i \cdots \hat{x}_j \cdots x_m.$$

Iovita-Spieß proved that  $\ker(\Delta_m)$  exists as a motive and they denote it as

$$(3.11) \quad (\mathcal{A}^m, \epsilon_{2m}) := \ker(\Delta_m).$$

**Remark 3.1.** In Scholl's original construction [Sch90] for elliptic modular curves  $X$ , the motive of weight  $k$  cusp forms sits directly inside  $\text{Sym}^{k-2}(h^1(\mathcal{E}))$  with  $\mathcal{E} \rightarrow X$  the universal elliptic curve. In this case, the  $h^1(\mathcal{E})$  is already the standard representation of rank 2. On our Shimura curve, parametrising QM abelian surfaces,  $h^1(\mathcal{A})$  is of rank 4 and decomposes as  $V \otimes V'$  where  $V$  is the piece we are interested in. Taking the wedge product gives

$$\wedge^2 h^1(\mathcal{A}) = h^2(\mathcal{A}) = (\text{Sym}^2(V) \otimes \wedge^2 V') \oplus (\wedge^2 V \otimes \text{Sym}^2 V')$$

and consequently the idempotent  $e_2$  isolates the piece  $h^2(\mathcal{A})_- = \text{Sym}^2 V$ . For this reason, the definition of  $(\mathcal{A}^m, \epsilon_n) = \text{Sym}^n V$  is extracted from  $\text{Sym}^m(h^2(\mathcal{A})_-) = \text{Sym}^m(\text{Sym}^2 V)$  through the Laplace operator  $\Delta_m$ .

The Galois group  $G$  defined by (3.1) acts canonically as  $X$ -automorphisms on  $X_M$  and  $\mathcal{A}^m$ . In order to pass from the Shimura curve  $X_M$  to  $X$ , we consider the projector

$$(3.12) \quad p_G := \frac{1}{|G|} \sum_{g \in G} g \in \mathbb{Q}[G]$$

as an element in  $\text{Corr}_X(\mathcal{A}^m, \mathcal{A}^m)$ . The Chow motive  $\mathcal{M}_n$  on  $\mathcal{M}^0(X)$  is then defined to be

$$(3.13) \quad \mathcal{M}_n := (\mathcal{A}^m, p_G \circ \epsilon_n)$$

for  $m = n/2$ .

**3.3.  $p$ -adic local systems.** We follow Sec. 5 [IS03]. In order to determine the  $p$ -adic realization of the motive  $\mathcal{M}_n$ , we consider the  $p$ -adic local system  $R^2 \pi_* \mathbb{Q}_p$  and mimic the construction in the light of Remark 3.1. The action of  $\mathcal{R}^{\max}$  on  $\mathcal{A}$  induces an action of  $\mathcal{B}^*$  on  $R^2 \pi_* \mathbb{Q}_p$  allowing us to define

$$(3.14) \quad \mathbb{L}_2 := \bigcap_{b \in \mathcal{B}} \ker(b - N(b) : R^2 \pi_* \mathbb{Q}_p \rightarrow R^2 \pi_* \mathbb{Q}_p)$$

where  $N(b)$  denotes the reduced norm of  $b$ . Consider the Laplace operator

$$(3.15) \quad \Delta_m : \text{Sym}^m \mathbb{L}_2 \rightarrow (\text{Sym}^{m-2} \mathbb{L}_2)(-2)$$

induced by the non-degenerated pairing

$$(3.16) \quad (-, -) : \mathbb{L}_2 \otimes \mathbb{L}_2 \hookrightarrow R^2 \pi_* \mathbb{Q}_p \otimes R^2 \pi_* \mathbb{Q}_p \rightarrow R^4 \pi_* \mathbb{Q}_p \rightarrow \mathbb{Q}_p(-2).$$

For  $n > 2$ , we define  $\mathbb{L}_n$  to be the kernel of  $\Delta_m$  where  $m = n/2$ .

**Lemma 3.1** (Lemma 10.1 [IS03]). *Let  $R_p : \mathcal{M}^0(X_M) \rightarrow D^b(X_M, \mathbb{Q}_p)$  be the  $p$ -adic realization functor from  $\mathcal{M}^0(X_M)$  to the bounded derived category of  $\mathbb{Q}_p$ -sheaves. Let  $m = n/2$ , then we have*

$$R_p(\mathcal{A}^m, \epsilon_n) = \mathbb{L}_n[-n].$$

If we consider  $\mathcal{M}_n$  as a Chow motive over  $\mathbb{Q}$  under the canonical functor  $\mathcal{M}^0(X) \rightarrow \mathcal{M}^0(\mathbb{Q})$  then its  $p$ -adic realisation is given by

$$(3.17) \quad R\Gamma(X, R_p(\mathcal{M}_n)) = H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G[-n-1].$$

The étale cohomology of the  $p$ -adic sheaf  $\mathbb{L}_n$  will be correspond under the  $D_{\text{st}}$  functor to the de Rham cohomology of the  $F$ -isocrystal obtained from the relative de Rham cohomology. In particular, we want to closely study the  $p$ -adic Galois representation

$$(3.18) \quad V := H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G.$$

**3.4. Cycle map and  $p$ -adic Abel-Jacobi.** In order to compute the  $p$ -adic height pairing of Heegner cycles, we will need to construct the associated extensions of  $p$ -adic Galois representations. To do this, recall that to every element in the Chow group of a Chow motive we can associate its cycle class

$$(3.19) \quad cl : \text{CH}^i(\mathcal{M}) \rightarrow R^{2i}\Gamma(X, R_p(\mathcal{M})(i))^{G_K}$$

given by the  $p$ -adic realization of  $\text{Hom}_{\mathcal{M}^0(X_M)}(\mathbb{1}, \mathcal{M}(i))$ . For the motive  $\mathcal{M}_n = (\mathcal{A}^m, p_G \circ \epsilon_n)$  we have

$$(3.20) \quad cl : \text{CH}^i(\mathcal{M}) \rightarrow (p_G \circ \epsilon_n)_* H^{2i}(\mathcal{A}^m \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)).$$

We denote the kernel of the cycle class map by  $\text{CH}^i(\mathcal{M})_0$ . By Lemma 3.1, the codomain of the cycle map for the motive  $\mathcal{M}_n$  vanishes for every  $i$  and we then have  $\text{CH}^i(\mathcal{M}_n) = \text{CH}^i(\mathcal{M}_n)_0$ .

The  $p$ -adic Abel-Jacobi map

$$(3.21) \quad cl_0 : \text{CH}^i(\mathcal{M})_0 \rightarrow H_{\text{cont}}^1(K, R^{2i-1}\Gamma(X, R_p(\mathcal{M})(i))) = \text{Ext}_{\text{Rep}(G_K)}^1(\mathbb{Q}_p, R^{2i-1}\Gamma(X, R_p(\mathcal{M})(i)))$$

associates to every class  $z \in \text{CH}^i(\mathcal{M})_0$  an extension

$$(3.22) \quad 0 \rightarrow R^{2i-1}\Gamma(X, R_p(\mathcal{M})(i)) \rightarrow E_z \rightarrow \mathbb{Q}_p \rightarrow 0.$$

The  $p$ -adic representation  $E_z$  is defined as follows. Consider  $\mathcal{M} = (Y, p)$  a relative motive over  $X$ , let  $z \in \text{CH}^i(\mathcal{M})_0 = p_* \text{CH}^i(Y)_0$  and  $Z$  be a cycle in the Chow group  $\text{CH}^i(Y)_0$  such that  $z = p_*[Z]$ . Then we can first consider the Gysin sequence

$$(3.23) \quad 0 \rightarrow H_{\text{ét}}^{2i-1}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) \rightarrow H_{\text{ét}}^{2i-1}((Y - |Z|) \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) \rightarrow H_{\text{ét}, |Z|}^{2i}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))_0 \rightarrow 0$$

where

$$(3.24) \quad H_{\text{ét}, |Z|}^{2i}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))_0 := \ker(H_{\text{ét}, |Z|}^{2i}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) \rightarrow H_{\text{ét}}^{2i}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)))$$

and then apply the projector  $p$  and take the pullback of the exact sequence by the map  $\mathbb{Q}_p \rightarrow p_* H_{\text{ét}, |Z|}^{2i}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))$  sending 1 to  $cl(z)$

$$(3.25) \quad \begin{array}{ccccccc} 0 & \longrightarrow & R^{2i-1}\Gamma(X, R_p(\mathcal{M})(i)) & \longrightarrow & E_z & \longrightarrow & \mathbb{Q}_p \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow cl \\ 0 & \longrightarrow & p_* H_{\text{ét}}^{2i-1}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) & \longrightarrow & p_* H_{\text{ét}}^{2i-1}((Y - |Z|) \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) & \longrightarrow & p_* H_{\text{ét}, |Z|}^{2i}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))_0 \longrightarrow 0. \end{array}$$

In the case of  $\mathcal{M}_n = (\mathcal{A}^m, p_G \circ \epsilon_n)$ , using once again the results of Lemma 3.1, we can associate to every  $z \in \text{CH}^{m+1}(\mathcal{M})$  an extension

$$(3.26) \quad 0 \rightarrow H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow E_z \rightarrow \mathbb{Q}_p \rightarrow 0.$$

As discussed in Sec. 7 of [IS03], the extensions obtained for the motive  $\mathcal{M}_n$  are *semistable*, implying that the  $p$ -adic Abel-Jacobi map factors through

$$(3.27) \quad cl_0 : \mathrm{CH}^{m+1}(\mathcal{M}_n)_0 \rightarrow H_{\mathrm{st}}^1(K_v, H_{\mathrm{\acute{e}t}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G)$$

where  $v$  is a place above  $p$  and  $H_{\mathrm{st}}^1$  denotes the Selmer group introduced by Nekovář [Nek93] given by

$$(3.28) \quad H_{\mathrm{st}}^1(K_v, V) := \ker(H^1(K_v, V) \rightarrow H^1(K_v, V \otimes_{\mathbb{Q}_p} B_{\mathrm{st}}))$$

for any  $p$ -adic representation  $V$  of  $G_{K_v}$ . If  $V$  is semistable, then  $H_{\mathrm{st}}^1(K, V)$  can be identified with the group of extension classes of  $\mathbb{Q}_p$  by  $V$ . In our case we then have that the  $p$ -adic Abel-Jacobi map factors through

$$(3.29) \quad cl_0 : \mathrm{CH}^{m+1}(\mathcal{M}_n) \rightarrow \mathrm{Ext}_{\mathrm{Rep}_{\mathrm{st}}(G_K)}^1(\mathbb{Q}_p, H_{\mathrm{\acute{e}t}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G).$$

**3.5. Biextensions.** Let  $D$  be a reduced effective divisor on  $X_M$  defined over  $K$  such that all closed points reduce to distinct points on  $X$  under the map  $q : X_M \rightarrow X$ . There is then an exact sequence of  $G_K$ -modules given by the Gysin sequence

$$(3.30) \quad 0 \rightarrow H_{\mathrm{\acute{e}t}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow H_{\mathrm{\acute{e}t}}^1((X_M - |D|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(m) \rightarrow 0$$

where  $G_x$  is the stabilizer of the point  $x$ . It's important to remark that the direct sum of the stalks of the  $p$ -adic sheaf  $\mathbb{L}_n(m)$  is the  $p$ -adic realisation of the motive

$$(3.31) \quad R\Gamma^{2m}(X_{M,K}, R_p((\mathcal{A}_D^m, \epsilon_n)_K)(m)) = \bigoplus_{x \in D} (\mathbb{L}_n)_x(m)$$

where  $\mathcal{A}_D^m = \coprod_{x \in D} \mathcal{A}_x^m$  is the disjoint union of products of abelian surfaces associated with  $x \in D$ .

Its dual exact sequence of  $p$ -adic representations of  $G_K$  is given by the relative cohomology sequence

$$(3.32) \quad 0 \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x^{\Gamma_x}(m) \rightarrow H_{\mathrm{\acute{e}t}, |D|}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G \rightarrow H_{\mathrm{\acute{e}t}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G \rightarrow 0.$$

The motives  $\mathcal{M}_n$  come equipped with the symmetric pairing (3.7) and at the level of  $p$ -adic realization  $\mathbb{L}$  descends to (3.16) induced by Poincaré duality. In particular, we have

$$(-, -) : \mathbb{L}_n(m) \otimes \mathbb{L}_n(m) \rightarrow \mathbb{Q}_p.$$

The pairing between the exact sequences is then given by

$$(3.33) \quad \begin{array}{ccc} H_{\mathrm{\acute{e}t}}^1((X_M - |D|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \times H_{\mathrm{\acute{e}t}, |D|}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G & \longrightarrow & H_{\mathrm{\acute{e}t}}^2(X_M \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(1))^G \xrightarrow{\sim} \mathbb{Q}_p \\ \uparrow & & \parallel \\ H_{\mathrm{\acute{e}t}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \times H_{\mathrm{\acute{e}t}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G & \longrightarrow & H_{\mathrm{\acute{e}t}}^2(X_M \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(1))^G \xrightarrow{\sim} \mathbb{Q}_p. \end{array}$$

Consider now two effective reduced divisors  $D_1, D_2$  on  $X_M$  defined over  $K_1, K_2$  with disjoint support such that all their closed points reduce to distinct points on  $X$  under the map  $q : X_M \rightarrow X$ . Consider the two short exact Gysin sequences

$$(3.34) \quad e_{D_1} : 0 \rightarrow H_{\mathrm{\acute{e}t}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow H_{\mathrm{\acute{e}t}}^1(U_1 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow \bigoplus_{x \in D_1} (\mathbb{L}_n)_x^{G_x}(m) \rightarrow 0$$

$$(3.35) \quad e_{D_2} : 0 \rightarrow H_{\mathrm{\acute{e}t}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow H_{\mathrm{\acute{e}t}}^1(U_2 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow \bigoplus_{x \in D_2} (\mathbb{L}_n)_x^{G_x}(m) \rightarrow 0$$

where  $U_i$  is the open subscheme of  $X_M$  defined by removing a choice of lift of  $D_i$  to  $X_M$ . The cohomology group

$$(3.36) \quad H_{\mathrm{\acute{e}t}, |D_2|}^1(U_1 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G$$

represents a biextension of the exact sequence  $e_{D_1}$  and of the twisted dual  $e_{D_2}^\vee(1)$ . In particular, we obtain a diagram with exact rows and exact columns of  $p$ -adic representations of the absolute Galois group of the composite field  $G_{K_1 K_2}$

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & \bigoplus_{x \in D_2} (\mathbb{L}_n)_x^{G_x}(m+1) & \longrightarrow & H_{\text{ét}, |D_2|}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G & \longrightarrow & H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \longrightarrow 0 \\
 & & \parallel & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \bigoplus_{x \in D_2} (\mathbb{L}_n)_x^{G_x}(m+1) & \longrightarrow & H_{\text{ét}, |D_2|}^1(U_1 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G & \longrightarrow & H_{\text{ét}}^1(U_1 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \longrightarrow 0 \\
 & & & & \downarrow & & \downarrow \\
 & & & & \bigoplus_{x \in D_1} (\mathbb{L}_n)_x^{G_x}(m) & \xlongequal{\quad} & \bigoplus_{x \in D_1} (\mathbb{L}_n)_x^{G_x}(m) \\
 & & & & \downarrow & & \downarrow \\
 & & & & 0 & & 0
 \end{array}$$

We will construct maps of  $p$ -adic Galois representations  $\mathbb{Q}_p \rightarrow \bigoplus_{x \in D_1} (\mathbb{L}_n)_x^{G_x}(m)$  and  $\bigoplus_{x \in D_1} (\mathbb{L}_n)_x^{G_x}(m+1) \rightarrow \mathbb{Q}_p(1)$  and we will take the pull-back of the first one and the push-out by the second one to obtain the diagram

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E_2^\vee[-1] & \longrightarrow & V \longrightarrow 0 \\
 & & \parallel & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E & \longrightarrow & E_1 \longrightarrow 0 \\
 & & & & \downarrow & & \downarrow \\
 & & & & \mathbb{Q}_p & \xlongequal{\quad} & \mathbb{Q}_p \\
 & & & & \downarrow & & \downarrow \\
 & & & & 0 & & 0
 \end{array}$$

**Remark 3.2.** In light of equation (3.31), the problem of constructing trivial subobjects and quotients of  $\bigoplus_{x \in D_1} (\mathbb{L}_n)_x(m)$  will be addressed from the motivic point of view in Section 3.9 where we will use the construction in [IS03] of suitable classes in  $\text{CH}^m((\mathcal{A}_D^m, \epsilon_n))$  and associated trivial  $p$ -adic representations under the cycle class map (3.19).

This biextension diagram is going to be the starting point of our computations on the  $p$ -adic height pairing of Section 5.

**3.6.  $F$ -Isocrystals on the  $\mathcal{H}_p$ .** For every representation  $(V, \rho)$  in  $\text{Rep}_{\mathbb{Q}_p}(GL_2 \times GL_2)$  we can construct a convergent filtered  $F$ -isocrystal on the formal model of the  $p$ -adic upper half plane  $\widehat{\mathcal{H}}$ . We denote by  $(V, \rho_1)$  and  $(V, \rho_2)$  the elements of  $\text{Rep}_{\mathbb{Q}_p}(GL_2)$  obtained restricting  $\rho$  respectively to the first and second component. Conversely, for any  $(V_1, \rho_1), (V_2, \rho_2)$  in  $\text{Rep}_{\mathbb{Q}_p}(GL_2)$  we can form a  $GL_2 \times GL_2$  representation

$$(V_1, \rho_1) \odot (V_2, \rho_2) = (V_1 \otimes V_2, \rho_1 \otimes 1_{V_2}, 1_{V_1} \otimes \rho_2).$$

For  $V$  in  $\text{Rep}_{\mathbb{Q}_p}(GL_2)$  and  $m \in \mathbb{Z}$  we define

$$V\{m\} := V \odot \det^{\otimes m}, \quad \{m\}V := \det^{\otimes m} \odot V.$$

For  $n \geq 0$ , let  $\mathcal{P}_n$  denote the space of all polynomials in  $X, Y$  which are homogeneous of degree  $n$  over  $\mathbb{Q}_p$  with  $GL_2$ -action

$$(3.39) \quad P(X, Y) \cdot \gamma = P(aX + bY, cX + dY), \quad \text{for } \gamma^{-1} = (\det \gamma)^{-1} \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Denote the dual representation by  $V_n := \mathcal{P}_n^\vee = \text{Hom}_{\mathbb{Q}_p}(\mathcal{P}_n, \mathbb{Q}_p)$ .

Let  $\hat{\mathcal{H}}_p$  denote the canonical formal model of  $\mathcal{H}_p$  over  $\mathbb{Z}_p$ . We briefly recall the convergent filtered  $F$ -isocrystals associated to a  $GL_2 \times GL_2$  representation  $(V, \rho_1, \rho_2)$  see [IS03, Section 4] for more details.

- As an *isocrystal*,  $\mathcal{E}(V) := V \otimes_{\mathbb{Q}_p} \mathcal{O}_{\hat{\mathcal{H}}}$  is given to be the isotrivial vector bundle over the  $p$ -adic upper half plane  $\mathcal{O}_{\mathcal{H}_p}$  with the induced connection given by usual derivation  $d$ ;
- Consider the endomorphism

$$(3.40) \quad \varphi_V := \rho_2 \begin{pmatrix} 0 & p \\ 1 & 0 \end{pmatrix} : V \rightarrow V.$$

The *Frobenius* is then given by

$$(3.41) \quad \Phi = \varphi_V \otimes \Phi_{\mathcal{O}_{\mathcal{H}_p}}$$

in particular, it only depends on the  $GL_2$  representation  $\rho_2$ .

- The *filtration* is only defined in terms of  $\rho_1$ . We describe it only for the representations  $V_n$

$$(3.42) \quad F^k V_n = \sum_{j=0}^{n-k} (zx + y)^k x^j y^{n-j-k} \mathcal{O}_{\mathcal{H}_p}$$

where  $x, y \in V_n$  are the dual elements corresponding to the variables  $X, Y$  in  $\mathcal{P}_n$ . In particular, we have  $F^0 V_n = V_n \otimes_{\mathbb{Q}_p} \mathcal{O}_{\hat{\mathcal{H}}}$ ,  $F^n V_n = (zx + y)^n \mathcal{O}_{\hat{\mathcal{H}}}$  and  $F^{n+1} V_n = 0$ .

In the case where  $\Gamma \subseteq GL_2$  is a discrete, cocompact, and torsion-free subgroup, the filtered isocrystals  $\mathcal{E}(V)$  descend to filtered isocrystals on the Mumford curve  $X_\Gamma$ . In particular, the action of  $GL_2$  on  $\mathcal{E}(V)$  through  $\rho_1$  is compatible with the action on  $\mathcal{H}_p$ . We will denote these filtered isocrystals again by  $\mathcal{E}(V)$  and denote their associated locally free  $\mathcal{O}_{X_\Gamma}$ -module with connection and filtration by  $E(V)$ .

Consider now the  $GL_2 \times GL_2$ -representation  $(M_2, \rho_1, \rho_2)$  given by

$$(3.43) \quad \rho_1(A)(B) := A \cdot B, \quad \rho_2(A)(B) := B \cdot \text{adj}(A)$$

for  $A \in GL_2$  and  $B \in M_2$  where  $\text{adj}(A)$  is the adjugate matrix.

The following result explains why the  $F$ -isocrystals  $\mathcal{E}(V_n\{n/2\})$  will be the correct object of study.

**Theorem 3.3.** *Consider the  $F$ -isocrystal obtained from  $\pi : \mathcal{A} \rightarrow X$  by*

$$(3.44) \quad R_{F\text{-isoc}}(h^1(\mathcal{A})) = R^1 \pi_* \Omega_{\mathcal{A}/X_M}^\bullet.$$

*There exists an (up to scaling) canonical isomorphism of filtered isocrystals on  $X_M$*

$$(3.45) \quad R_{F\text{-isoc}}(h^1(\mathcal{A})) \otimes \mathbb{Q}_{p^{\text{unr}}} \cong \mathcal{E}(M_2, \rho_1, \rho_2) \otimes \mathbb{Q}_{p^{\text{unr}}}.$$

*This implies that the de Rham cohomology  $\mathcal{H}_{\text{dR}}^1(\mathcal{A}/X_M)$  is isomorphic to the locally free  $\mathcal{O}_{X_M}$ -module  $E(M_2)$  of rank 4.*

*Proof.* See [IS03, Lemma 5.10]. □

**Remark 3.3.** *It is important to note that the previous Theorem establishes an isomorphism of isocrystal after passing to the maximal unramified extension of  $\mathbb{Q}_p$ . In [IS03] this is needed because they explicitly use the moduli description of the formal model of the upper-half plane  $\hat{\mathcal{H}}_p \otimes \mathbb{Z}_{p^{\text{unr}}}$  of special formal groups.*

Now we can observe that  $(M_2, \rho_1, \rho_2)$  is canonically isomorphic to  $V_1 \odot V_1$ , giving

$$(3.46) \quad (\wedge^2 M_2, \wedge^2 \rho_1, \wedge^2 \rho_2) = \wedge^2 V_1 \odot \text{Sym}^2 V_1 \oplus \text{Sym}^2 V_1 \odot \wedge^2 V_1 = \{1\} V_2 \oplus V_2 \{1\}.$$

In particular, we obtain

$$(3.47) \quad \mathcal{E}(V_2\{1\}) \cong \bigcap_{b \in \mathcal{B}} \ker(b - N(b) : \mathcal{E}(\wedge^2 M_2) \rightarrow \mathcal{E}(\wedge^2 M_2))$$

where the action of  $\mathcal{B}$  is given by the isomorphism  $\mathcal{B} \otimes \mathbb{Q}_{p^2} \cong M_2(\mathbb{Q}_{p^2})$ . Observe that  $\mathcal{V}_2$  inherits a non-degenerate symmetric bilinear pairing by restriction to

$$(3.48) \quad (-, -) : V_2\{1\} \otimes V_2\{1\} \hookrightarrow \wedge^2 M_2 \otimes \wedge^2 M_2 \rightarrow \wedge^4 M_2 \cong \det^{\otimes 2} \odot \det^{\otimes 2} = \{2\}\{2\}.$$

Compare once again this to the pairings (3.7) and (3.16). We will need an explicit and computable description of this pairing that will be later use to identify Heegner cycles. We recall from [BDIS02, Sec. 1.2] the construction using the following isomorphic description.

**Lemma 3.2.** *Let  $\mathcal{U} = \{U \in M_2 : \text{tr}(U) = 0\}$  be endowed with the right  $\text{GL}_2$  action defined by*

$$(3.49) \quad U \cdot A := \bar{A}UA$$

*for  $U \in \mathcal{U}$  and  $A \in \text{GL}_2$ . There is an isomorphism of right- $\text{GL}_2$ -modules defined by*

$$(3.50) \quad \mathcal{U} \rightarrow \mathcal{P}_2$$

$$U \mapsto P_U(X, Y) := \text{tr} \left( U \begin{pmatrix} X \\ Y \end{pmatrix} \begin{pmatrix} Y & -X \end{pmatrix} \right).$$

The pairing on  $\mathcal{P}_2$  arises from the Cartan-Killing form on  $\mathcal{U}$

$$(3.51) \quad (P_{U_1}, P_{U_2}) := -\text{tr}(U_1 \bar{U}_2)$$

for every  $U_1, U_2 \in \mathcal{U}$ . This induces an isomorphism

$$(3.52) \quad \mathcal{P}_2 \rightarrow V_2$$

$$P(X, Y) \mapsto (P(X, Y), -)$$

and induces a non-degenerate symmetric pairing on  $V_2$  coinciding with (3.48). Consider the Laplace operator

$$(3.53) \quad \Delta_m : \text{Sym}^m V_2 \rightarrow \text{Sym}^{m-2} V_2 \otimes \det^{\otimes 2}$$

defined analogously to (3.9) and (3.15). The  $\text{GL}_2$ -subrepresentation  $V_{2m}$  of  $\text{Sym}^m V_2$  appears without surprise to be the kernel  $\Delta_m$ .

**3.7.  $F$ -isocrystals on  $X_\Gamma$ .** In [Fal89], [Fal92] and [Fal02] Faltings gives a proof of Fontaine's conjectures based on the work of Fontaine–Messing [FM87] generalizing the result of Tsuji [Tsu99] for cohomologies with non-trivial coefficients. In our context we can summarize their results in the following theorem.

**Theorem 3.4.** *Let  $\mathbb{L}_n$  the  $p$ -adic sheaf on  $X_M$  defined in Section 3.3. There exists an  $F$ -isocrystal on  $X_M$  such that*

$$(3.54) \quad D_{\text{st}}(H_{\text{ét}}^1(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n)^G) = H_{\text{dR}}^1(X_M \otimes \mathbb{Q}_p, \mathcal{V}_n)^G.$$

In particular, the functor  $\mathbb{D}$  sends  $R_p(h^1(\mathcal{A}))$  to the  $F$ -isocrystal given by the relative de Rham cohomology  $R^1 \pi_* \Omega_{\mathcal{A}/X_M}^\bullet$ . This then gives us an explicit description of the  $F$ -isocrystal  $\mathcal{V}_n$  when we pass to the maximal unramified extension.

**Corollary 3.1.** *Under the analytic isomorphism of Theorem 3.2, the pullback of  $\mathcal{V}_n \otimes \mathbb{Q}_{p^{\text{unr}}}$  on  $\mathcal{H}_p$  is given by the  $F$ -isocrystal  $\mathcal{E}(V_n\{n/2\})$ .*

*Proof.* Recall that Cerednik-Drinfeld Theorem gives an isomorphism

$$(3.55) \quad X_{M, \mathbb{Q}_{p^{\text{unr}}}} \cong \Gamma \setminus (\mathcal{H}_p \times (R/MR)^\times) \cong \coprod_{(\mathbb{Z}/M\mathbb{Z})^\times} \Gamma_M \setminus \mathcal{H}_p.$$

Let  $\pi_{\Gamma_M} : \mathcal{H}_p \rightarrow \Gamma_M \setminus \mathcal{H}_p$  the projection map. By the previous Theorem, we have  $\mathbb{D}(R_p(h^1(\mathcal{A}))) = R^1\pi_*\Omega_{\mathcal{A}/X_M}^\bullet$  and by Theorem 3.3 we have that the pullback to the upper-half plane is given by

$$(3.56) \quad \pi_{\Gamma_M}^*(R^1\pi_*\Omega_{\mathcal{A}/X_M}^\bullet) \cong \mathcal{E}(M_2).$$

Under the non-degenerate bilinear symmetric pairing on the  $F$ -isocrystal (3.48) descending from the motivic pairing (3.7) we have that the  $F$ -isocrystal  $\mathcal{E}(V_n\{n/2\})$  is isomorphic to the one cut from  $R^1\pi_*\Omega_{\mathcal{A}/X_M}^\bullet$  and isomorphic to the pullback  $\pi_{\Gamma_M}^*(\mathcal{V}_n)$ .  $\square$

**Remark 3.4.** *It is important to note that the  $F$ -isocrystal on the upper half plane  $\mathcal{H}_p$  associated to  $\mathcal{V}_n$  can be constructed passing to a finite extension of  $\mathbb{Q}_p$ . This isocrystal becomes isomorphic to  $\mathcal{E}(V_n\{n/2\})$  only after a base change to the maximal unramified extension. From the fact that  $\mathcal{H}_p$  is a Stein space, we obtain that the isocrystal  $\mathcal{V}_n$  is given by a free vector bundle. The unramified extension is then need to obtain an explicit horizontal basis for the Frobenius action.*

We will from now on work on a finite unramified extension of  $\mathbb{Q}_p$  where the  $p$ -adic uniformization of Cerednik–Drinfel holds. When no confusion will arise, we will then denote the isocrystal on  $X_M$  and on the upper-half plane  $\mathcal{H}_p$  in the same way.

Given such an object, we can define the de Rham cohomology of general quotients of the upper-half plane.

**Definition 3.4.** *Let  $\Gamma \subseteq \mathrm{GL}_2$  be a discrete and cocompact subgroup. If  $\Gamma$  is torsion-free we define*

$$(3.57) \quad H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n) := H^1(X_\Gamma, \mathcal{V}_n \otimes \Omega_1^\bullet)$$

where the complex  $(\mathcal{V}_n \otimes \Omega_1^\bullet)$  is induced by the integrable connection. If  $\Gamma$  is not torsion-free, we consider a torsion-free subgroup  $\Gamma' \subseteq \Gamma$  and we define

$$(3.58) \quad H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n) := H_{\mathrm{dR}}^1(X_{\Gamma'}, \mathcal{V}_n)^{\Gamma/\Gamma'}$$

In Section 4, we will study the properties of these cohomology groups. In particular, the  $F$ -isocrystal structure of  $\mathcal{V}_n$  will induce the structure of a filtered  $(\varphi, N)$ -module on  $H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n)$ . We then have a comparison isomorphism between the  $p$ -adic representations attached to  $\mathbb{L}_n$  and the hypercohomology of the vector bundle with connection  $\mathcal{V}_n$ .

**Theorem 3.5.** *Let  $\Gamma$  the discrete cocompact subgroup of  $\mathrm{GL}_2$  associated to  $X$  as in Section 3.1. The  $p$ -adic representation  $H_{\mathrm{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G$  is semistable and we have a canonical (up to scaling) isomorphisms of filtered  $(\varphi, N)$ -modules*

$$(3.59) \quad D_{\mathrm{st}, \mathbb{Q}_{p^f}}(H_{\mathrm{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G) = H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n).$$

*Proof.* It follows from Theorem 3.4 and [IS03, Theorem 5.9]  $\square$

One of the important properties of this theorem and key element inside its proof is the compatibility between the symmetric products defined over the  $p$ -adic local system  $\mathbb{L}_n$  (3.16) and the  $F$ -isocrystal  $\mathcal{V}_n$  coming from the relative de Rham cohomology. This has indeed a motivic nature and comes from the pairing (3.7). One instance of this compatibility is the fact that the pairing

$$(3.60) \quad H_{\mathrm{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n) \otimes H_{\mathrm{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n) \rightarrow \mathbb{Q}_{p^2}[n+1]$$

is mapped under Fontaine’s functor to the pairing

$$(3.61) \quad H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n) \otimes H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n) \rightarrow H_{\mathrm{dR}}^2(X_\Gamma, \mathcal{V}_n \otimes \mathcal{V}_n) \rightarrow H_{\mathrm{dR}}^2(X_\Gamma, \mathcal{O}_{X_\Gamma})[n] \cong \mathbb{Q}_{p^2}[n+1].$$

This property will be used later to pass from dual exact sequences in the category of  $p$ -adic representations to the dual of exact sequences in the de Rham realizations.

Consider now  $D$  to be an effective reduced divisor of  $X$  defined over  $K$ . Assume that  $K/\mathbb{Q}$  is nontrivial unramified at  $p$  and all the closed points  $x_1, x_2, \dots, x_r \in |D|$  are defined over  $K$  and such that all  $x_i$  reduce to distinct points on  $X$  under the map  $q : X_M \rightarrow X$ . We then have a Gysin exact sequence of étale cohomology groups

$$(3.62) \quad e_{\mathrm{ét}} : 0 \rightarrow H_{\mathrm{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G \rightarrow H_{\mathrm{ét}}^1(U \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(-1) \rightarrow 0.$$

Let  $z_1, z_2, \dots, z_r \in \mathcal{H}_p(K_p)$  points on the  $p$ -adic upper half plane such that  $\Gamma z_i = q(x_i)$  under the  $p$ -adic uniformization of  $X$  Theorem 3.1.

**Theorem 3.6.** *Assume that  $x_1, x_2, \dots, x_r$  specialize to smooth distinct points of the special fiber  $X_{\mathbb{F}_{p^2}}$ . The exact sequence obtained from (3.62) by applying the functor  $D_{\text{st}, \mathbb{Q}_{p^f}}$  is isomorphic to the sequence of filtered  $(\varphi, N)$ -modules*

$$(3.63) \quad e_{\text{dR}} : 0 \rightarrow H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) \rightarrow H_{\text{dR}}^1(X_\Gamma - D_\Gamma, \mathcal{V}_n) \rightarrow \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)_{\Gamma_z}^z[1] \rightarrow 0$$

where  $\Gamma_z$  is the stabilizer of the point  $z$ .

*Proof.* See [CI10, Theorem 7.6] and [IS03, Theorem 5.13].  $\square$

**3.8.  $\mathbb{Q}_{p^2}$ -points on  $\mathcal{H}_p$ .** It will be useful to identify the set of  $\mathbb{Q}_{p^2}$ -points on  $\mathcal{H}_p$  with the set of  $\mathbb{Q}_p$ -algebra homomorphisms  $\text{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$ . This will turn out useful when we will describe Heegner points geometrically on  $X$  and  $p$ -adic analytically on  $X_\Gamma$ .

**Lemma 3.3.** *There is a bijection between  $\text{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$  and  $\mathcal{H}_p(\mathbb{Q}_{p^2})$ .*

*Proof.* Every  $\Psi \in \text{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$  defines an action of  $\mathbb{Q}_{p^2}^*$  on  $\mathcal{H}_p(\mathbb{Q}_{p^2})$ . The corresponding point  $z_\Psi \in \mathcal{H}_p(\mathbb{Q}_{p^2})$  is defined to be the unique point such that

$$(3.64) \quad \Psi(a) \cdot \begin{pmatrix} z \\ 1 \end{pmatrix} = a \begin{pmatrix} z \\ 1 \end{pmatrix}$$

for every  $a \in \mathbb{Q}_{p^2}$ . Conversely, for every point  $z \in \mathcal{H}(\mathbb{Q}_{p^2})$  we can define the associated homomorphisms  $\Psi_z \in \text{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$  to be the linear transformation associated to multiplication by  $\mathbb{Q}_{p^2}^*$  on the projective coordinates of  $z$ . These two maps can be easily checked to be inverse of each other.  $\square$

Consider now a point  $z \in \mathcal{H}_p(\mathbb{Q}_{p^2})$  and let  $\Psi_z \in \text{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$  we want to compute the stalk of the isocrystal  $\mathcal{E}(V_n\{n/2\})_{\mathbb{Q}_{p^2}}$  at the point  $z$ . This in particular will have the structure of a *filtered Frobenius module*. As a vector space this coincide with  $V_{n, \mathbb{Q}_{p^2}} = V_n \otimes \mathbb{Q}_{p^2}$  and it is generated by  $\text{ev}_z \in V_{n, \mathbb{Q}_{p^2}} = \text{Hom}(\mathcal{P}_n, \mathbb{Q}_{p^2})$  given by

$$(3.65) \quad \text{ev}_z(P(X, Y)) := P(z, 1)$$

that has the property

$$(3.66) \quad (\Psi_z(a) \cdot \text{ev}_z)(P(X, Y)) = a^n P(z, 1).$$

To define the filtration, consider first the subspaces

$$(3.67) \quad (V_{n, \mathbb{Q}_{p^2}})_i := \{v \in V_{n, \mathbb{Q}_{p^2}} : \rho_1(\Psi(a))(v) = a^i \sigma(a)^{n-i} v \quad \forall a \in \mathbb{Q}_{p^2}\}$$

and then define

$$(3.68) \quad F^i(V_{n, \mathbb{Q}_{p^2}})_z := \bigoplus_{j \geq i} (V_{n, \mathbb{Q}_{p^2}})_j.$$

The *Frobenius endomorphism* is given by  $\phi_V \otimes \sigma$ .

**3.9. Heegner points and special cycles.** In this section we will discuss special cycles in  $\text{CH}^{m+1}(\mathcal{M}_n)$  arising from abelian surfaces with complex multiplication. Under the Čerednik-Drinfelĭ  $p$ -adic uniformization, we will also study their characterization on the  $p$ -adic upper half plane. We will follow the construction of [IS03] Sec.8.

Recall that we want to construct interesting classes in  $\text{CM}^{m+1}(\mathcal{M}_n)$  or, as explained at the end of Section 3.2, construct trivial  $p$ -adic representations in the  $p$ -adic realization of motives  $(\mathcal{A}_D^m, \varepsilon_n)$  for certain reduced effective divisors  $D$  on  $X_M$ . We will define these classes using special CM abelian surfaces.

**Definition 3.5.** Consider  $A$  an abelian surface with CM by  $\mathcal{R}^{\max}$ . The abelian surface is said to have complex multiplication (CM) if  $\text{End}_{\mathcal{R}^{\max}}(A) \neq \mathbb{Z}$ . In this case it is an order  $\mathcal{O}$  in an imaginary quadratic number field. A closed point on  $X$  is called CM if it can be represented by a triple  $(A, \iota, C)$  where  $A$  has complex multiplication. It is furthermore called Heegner point if  $(A, \iota, C)$  is CM by an order  $\mathcal{O}$  and  $C$  is  $\mathcal{O}$ -stable.

We will now fix a number field  $K$  satisfying certain prescribed properties that will be key in the construction of the Chow classes.

**Hypothesis 3.1** (Heegner). Let  $K$  be an imaginary quadratic field of discriminant  $D_K$  satisfying the following properties:

- $(D_K, N) = 1$ ,
- all prime divisors of  $N^-$  are inert in  $K$ ,
- all prime divisors of  $N^+$  are split in  $K$ .

Consider now an abelian surface  $A$  with CM by the maximal order  $\mathcal{O}_K$ . Thanks to the splitting behaviour of the primes in  $K$  we obtain

$$\text{End}(A) \cong \mathcal{O}_K \otimes \mathcal{R}^{\max} \cong M_2(\mathcal{O}_K).$$

In particular, fixing an isomorphism  $\text{End}(A) \cong M_2(\mathcal{O}_K)$ , we get an isomorphism

$$(3.69) \quad A \cong E \times E$$

where  $E$  is an elliptic curve with CM over  $H$  Hilbert class field of  $K$  and  $\text{End}(E) \cong \mathcal{O}_K$ . As usual, we fix an isomorphism between the endomorphism ring and the maximal order. The special rigid structure of  $A$  is reflected in its associated motive. From the identifications of  $\text{End}(A)_{\mathbb{Q}} = K \otimes \mathcal{R}^{\max} = M_2(K) = K \otimes M_2(\mathbb{Q})$  and  $A = E \times E$  we can explicitly describe the action of the endomorphisms on the motives. By the *Kunneth decomposition*

$$(3.70) \quad h^1(A) = h^1(E \times E) = h^1(E) \otimes_{\mathbb{Q}} \mathbb{Q}^2,$$

where  $K$  acts on  $h^1(E)$  and  $M_2(\mathbb{Q})$  acts on  $\mathbb{Q}^2$ . Considering the second piece of the motive of  $A$  and applying the decomposition once again we obtain

$$(3.71) \quad h^2(A) = \wedge^2 h^1(A) = \wedge^2(h^1(E) \otimes_{\mathbb{Q}} \mathbb{Q}^2) = (h^2(E) \otimes \text{Sym}^2(\mathbb{Q}^2)) \oplus (\text{Sym}^2 h^1(E) \otimes \wedge^2 \mathbb{Q}^2).$$

Where now the matrix group  $M_2$  acts by the determinant on the last summand showing how the action of  $\mathcal{B}^*$  cuts, in the notation of (3.6), the part

$$(3.72) \quad h^2(A)_{-} = \text{Sym}^2(h^1(E)).$$

Furthermore, since  $E$  has CM by  $\mathcal{O}_K$ , then  $h^1(E)$  acquires an action of  $K$  and in particular the base change to  $K$  will decompose into two summands

$$(3.73) \quad h^1(E) \otimes_{\mathbb{Q}} K = M(\psi) \oplus M(\bar{\psi})$$

where  $M(\psi), M(\bar{\psi})$  are the motives associated to the Hecke character  $\psi$  of  $H$  attached to  $E$  and its conjugate as in Ch. 1 [Sch88]. From the fact that  $\psi\bar{\psi} = N_{K/\mathbb{Q}}$ , the symmetric square is then given by

$$(3.74) \quad \text{Sym}^2(h^1(E)) \otimes_{\mathbb{Q}} K = M(\psi^2) \oplus \mathbb{1}(-1) \oplus M(\bar{\psi}^2).$$

Consider the cycle defined by

$$(3.75) \quad y := [\Gamma_{\sqrt{D_K}}] + D_K[\{O\} \otimes E] - [E \times \{O\}] \in CH^1(E \times E)$$

where  $\Gamma_{\sqrt{D_K}}$  is the graph of the endomorphism  $\sqrt{D_K}$  in  $\text{End}(E)$ . Sec. 8 [IS03] proves that  $y \in CH^1(A, \epsilon_2)$  and will land in the  $\mathbb{1}$ -part of (3.74) under the cycle class map.

Let  $t$  be the number of prime factors of  $N$  and  $h$  the class number of  $K$ . Consider the following subgroups of  $\text{Aut}(X_H)$ :

- $\mathcal{W} \cong (\mathbb{Z}/2\mathbb{Z})^t$  subgroup generated by Atkin-Lehner involutions  $\omega_l$  for every  $l$  prime divisor of  $N$ ,
- $\Delta = \text{Pic}(\mathcal{O}_K)$  subgroup identified with  $\text{Gal}(H/K)$  via the Artin map.

By Lemma 2.5 [BD96] there are precisely  $2^t h$  Heegner points with CM by  $\mathcal{O}_K$  which are defined over the Hilber class field  $H$  of  $K$  and the group action of  $\mathcal{W} \times \Delta$  on this set is free and transitive.

In order to construct the desired cycle classes in  $\mathrm{CH}^{m+1}(\mathcal{M}_n)$  we need once again to pass through the fine moduli scheme  $X_M$ . Consider  $P$  closed Heegner point on  $X_H$  and let  $\tilde{P}$  a closed point in the fiber  $q^{-1}(P)$  in  $(X_M)_H$ . The fiber  $\mathcal{A}_{\tilde{P}}$  above  $\tilde{P}$  is an abelian surface with CM by  $\mathcal{O}_K$ . We can then consider  $y_{\tilde{P}} \in \mathrm{CH}^1(\mathcal{A}_{\tilde{P}}, \epsilon_2)$  as defined above. For an abelian surface with CM, the cycle class map factors through algebraic equivalence, in particular we could modify the cycle  $y_{\tilde{P}}$  by any element in  $\mathrm{Pic}^0(\mathcal{A}_{\tilde{P}})$ . We choose the representatives

$$(3.76) \quad z_{\tilde{P}} \in [y_{\tilde{P}}] \in \mathrm{NS}(\mathcal{A}_{\tilde{P}}) = \mathrm{Pic}(\mathcal{A}_{\tilde{P}}) / \mathrm{Pic}^0(\mathcal{A}_{\tilde{P}})$$

for every  $\tilde{P}$  in such a way that they are compatible with the  $G$ -action, i.e.

$$(3.77) \quad g_*(z_{\tilde{P}}) = z_{g(\tilde{P})}.$$

**Definition 3.6.** Define the cycle class  $y_P = y_P^{(2)} \in \mathrm{CH}^2((\mathcal{M}_n)_H)$  to be the image of  $z_{\tilde{P}}$  under

$$(3.78) \quad \mathrm{CH}^1(\mathcal{A}_{\tilde{P}}, \epsilon_2) \xrightarrow{(p_{G \circ \iota_{\tilde{P}}})^*} \mathrm{CH}^2((\mathcal{M}_2)_H)$$

where  $\iota_{\tilde{P}} : \mathcal{A}_{\tilde{P}} \hookrightarrow \mathcal{A}_H$ . For arbitrary even  $n \geq 2$ , first consider the  $m$ -th product of cycles under the projector  $\epsilon_n$  as defined in (3.11)

$$(3.79) \quad z_{\tilde{P}}^{(n)} = (\epsilon_n)_*(z_{\tilde{P}} \times \cdots \times z_{\tilde{P}})$$

and then we define the cycle class  $y_P^{(n)} \in \mathrm{CH}^{m+1}((\mathcal{M}_n)_H)$  to be the image of  $z_{\tilde{P}}^{(n)}$  under

$$(3.80) \quad \mathrm{CH}^m(\mathcal{A}_{\tilde{P}}^m, \epsilon_n) \xrightarrow{(p_{G \circ \iota_{\tilde{P}}})^*} \mathrm{CH}^{m+1}((\mathcal{M}_n)_H).$$

Thanks to the compatibility under the  $G$ -action (3.77), the definition of  $y_P$  and  $y_P^{(n)}$  is independent of the point  $\tilde{P}$ , but still depends on the set of compatible classes  $z_{\tilde{P}}$ .

**3.10. Heegner points on  $\mathcal{H}_p$ .** We briefly recall the description of  $p$ -adic analytic Heegner points on  $\mathcal{H}_p$  following [BD98] and [IS03].

**Definition 3.7.** An embedding  $\Psi : K \rightarrow B$  is called optimal if  $\Psi(K) \cap R = \Psi(\mathcal{O}_K[1/p])$ . The group  $\Gamma$  acts by conjugation of the set of optimal embeddings and we denote  $\mathrm{emb}(\mathcal{O}_K, R)$  the set of conjugacy classes  $[\Psi]$ .

The set  $\mathrm{emb}(\mathcal{O}_K, R)$  is endowed with an action of  $\Delta$  and [BD98, Theorem 5.3] gives a bijection between the set of Heegner points on  $X_H$  and  $\mathrm{emb}(\mathcal{O}_K, R)$  compatible with the  $\Delta$ -action. Fix an embedding  $H \hookrightarrow \mathbb{C}_p$  and let  $v|p$  the corresponding place of  $H$ . Since  $p$  is inert in  $K$  and totally split in  $H$ , we then identify  $K_p$  and  $H_v$  with  $\mathbb{Q}_{p^2}$ . By Lemma 3.3, we can identify the classes in  $\mathrm{emb}(\mathcal{O}_K, R)$  with  $K_p$ -points of  $X$  via

$$(3.81) \quad \mathrm{emb}(\mathcal{O}_K, R) \hookrightarrow \Gamma \backslash \mathrm{Hom}(K_p, M_2(\mathbb{Q}_p)) = X_\Gamma(K_p).$$

An Heegner point  $P \in X_\Gamma(K_p)$  lies in the image of  $\mathrm{emb}(\mathcal{O}_K, R)$  since the corresponding embedding  $\Psi_P$  can be defined by looking at the endomorphism ring of the associate CM abelian surface and at its reduction modulo  $p$

$$(3.82) \quad \Psi_P : \mathcal{O}_K = \mathrm{End}(\mathcal{A}_P) \hookrightarrow \mathrm{End}(\mathcal{A}_P \otimes \mathbb{F}_{p^2}).$$

By the work of Čerednik–Drinfeld [BC91], we have  $\mathrm{End}(\mathcal{A}_P \otimes \mathbb{F}_{p^2})[1/p] = R[1/p]$ . This map is an optimal embedding by [GZ86, Prop. 7.3]. See [BD98] for the complete details.

#### 4. DE RHAM COHOMOLOGY AND BRUHAT-TITS COCYCLES

In order to compute the local  $p$ -adic component of Nekovar height, we need to understand the filtered  $(\phi, N)$ -structure of the following cohomology groups

$$H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n), \quad H_{\text{dR}}^1(U, \mathcal{V}_n), \quad H_{\text{dR}}^1(X_\Gamma, Z, \mathcal{V}_n), \quad H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$$

where  $U \subseteq X_\Gamma$  is an open set and  $Z \hookrightarrow X_\Gamma$  is a closed subscheme belonging to  $U$ . When  $\Gamma$  contains torsion, these groups are defined as follows: Pick a normal subgroup  $\Gamma' \subset \Gamma$  which is torsion-free. Let  $\pi : X_{\Gamma'} \rightarrow X_\Gamma$  be the projection,  $U' = \pi^{-1}(U)$  and  $Z' = \pi^{-1}(Z)$ , then define

$$\begin{aligned} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) &:= H_{\text{dR}}^1(X_{\Gamma'}, \mathcal{V}_n)^{\Gamma/\Gamma'} \\ H_{\text{dR}}^1(U, \mathcal{V}_n) &:= H_{\text{dR}}^1(U', \mathcal{V}_n)^{\Gamma/\Gamma'} \\ H_{\text{dR}}^1(X_\Gamma, Z, \mathcal{V}_n) &:= H_{\text{dR}}^1(X_{\Gamma'}, Z', \mathcal{V}_n)^{\Gamma/\Gamma'} \\ H_{\text{dR}}^1(U, Z, \mathcal{V}_n) &:= H_{\text{dR}}^1(U', Z', \mathcal{V}_n)^{\Gamma/\Gamma'}. \end{aligned}$$

These groups do not depend on the choice of  $\Gamma'$ . For the remainder of the paper, fix a choice of branch of  $p$ -adic logarithm  $\log : \mathbb{C}_p^\times \rightarrow \mathbb{C}_p$ .

**4.1. The  $(\phi, N)$ -module structure of  $H_{\text{dR}}^1$ .** In this section we recall the structure of  $H_{\text{dR}}^1(X, \mathcal{V}_n)$  as a  $(\phi, N)$ -module following [IS03] and [Mas11]. Assume for now that  $\Gamma$  is torsion-free. Then the Mumford curve  $X_\Gamma$  over  $\mathbb{Q}_p^{\text{ur}}$  is a smooth projective curve with a semistable model  $\mathfrak{X}_\Gamma$ . Let  $C_1, \dots, C_r$  be its irreducible components of the special fiber,  $C$ , of  $\mathfrak{X}_\Gamma$ . Let  $\mathfrak{G} = (\mathfrak{V}(\mathfrak{G}), \mathfrak{E}(\mathfrak{G}))$  the oriented intersection graph of  $C$ . Consider the reduction map

$$\text{red} : X_\Gamma^{\text{an}} \rightarrow C(\overline{\mathbb{F}}_p)$$

and define the admissible cover  $\{U_v\}_{v \in \mathfrak{V}}$  where  $U_v = \text{red}^{-1}(C_v)$ . For every edge  $e \in \mathfrak{E}$  denote  $s(e)$  the start vertex and  $t(e)$  the target vertex. We can associate to each edge  $e$  a wide open annulus  $A_e$  given by

$$A_e = U_{s(e)} \cap U_{t(e)} = \text{red}^{-1}(x)$$

where  $x = C_{s(e)} \cap C_{t(e)}$ .

**4.2. The general case.** Let  $D = (\tilde{z}_1) + \dots + (\tilde{z}_r)$  and  $Z = (\tilde{w}_1) + \dots + (\tilde{w}_s)$  be disjoint divisors in  $X_\Gamma$  such that each of the points  $\tilde{z}_i$  and  $\tilde{w}_j$  reduces to a smooth point in  $\mathfrak{X}_\Gamma$ . We will denote by  $U$  the open subscheme  $X_\Gamma - D$ , and by

$$i : Z \hookrightarrow X_\Gamma$$

the closed subscheme associated with the same divisor. Our aim is to describe the  $(\phi, N)$ -module structure on the relative de Rham cohomology group  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$ .

Recall that  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$  is defined as follows: Let  $\Omega_{(U, Z)}^\bullet(\mathcal{V}_n)$  denote the cone associated to the morphism of complexes

$$i^* : \Omega_U^\bullet \otimes \mathcal{V}_n \rightarrow i_*(\Omega_Z^\bullet \otimes i^* \mathcal{V}_n).$$

Then  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$  is the hypercohomology of the complex of sheaves  $\Omega_{(U, Z)}^\bullet(\mathcal{V}_n)$ .

For each  $v \in \mathfrak{V}$ , let  $U_{v, D}$  denote the wide-open affinoid  $U_v \setminus |D|$ . Then the collection  $(U_{v, D})_{v \in \mathfrak{V}}$  is an admissible covering of  $U$ . Computing  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$  by Čech hypercohomology with this choice of covering, each cohomology class is represented by a cocycle

$$((\omega_v)_{v \in \mathfrak{V}}, \alpha, (f_e)_{e \in \mathfrak{E}}) \quad \text{for } \omega_v \in H^0(U_{v, D}, \Omega_{U_{v, D}}^1 \otimes \mathcal{V}_n), \alpha \in H^0(Z, i^* \mathcal{V}_n), f_e \in H^0(A_e, \mathcal{V}_n),$$

which satisfy

$$\omega_{v_1}|_{A_e} - \omega_{v_2}|_{A_e} = df_e \quad \text{for every } e = (v_1, v_2) \in \mathfrak{E}.$$

A representative cocycle for a cohomology class is well-defined up to

$$(4.1) \quad \left( (dg_v)_{v \in \mathfrak{V}}, (g_v)|_Z, (g_{s(e)} - g_{t(e)})_{e \in \mathfrak{E}} \right) \quad \text{for } g_v \in H^0(U_{v, D}, \mathcal{V}_n).$$

This implies that the map

$$\gamma_{U, Z} : H_{\text{dR}}^1(U, Z, \mathcal{V}_n) \rightarrow \ker(H_{\mathfrak{V}, D}^1 \rightarrow H_{\mathfrak{E}}^1)$$

given by

$$((\omega_v)_{v \in \mathfrak{V}}, \alpha, (f_e)_{e \in \mathfrak{E}}) \mapsto (\omega_v)_{v \in \mathfrak{V}}$$

is well-defined and surjective, where

$$H_{\mathfrak{V},D}^i = \bigoplus_{v \in \mathfrak{V}} H_{dR}^i(U_{v,D}, \mathcal{V}_n), \quad H_{\mathfrak{E}}^i = \bigoplus_{e \in \mathfrak{E}} H_{dR}^i(A_e, \mathcal{V}_n).$$

The group  $\ker(\gamma_{U,Z})$  is generated by elements of the form

$$((0)_{v \in \mathfrak{V}}, \alpha, (f_e)_{e \in \mathfrak{E}}) \quad \text{such that } df_e = 0,$$

modulo elements of the form in (4.1), under the condition that  $dg_v = 0$  for all  $v \in \mathfrak{V}$ . Hence the kernel of  $\gamma_{U,Z}$  is naturally identified with the quotient

$$\left( (H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n) \right) / H_{\mathfrak{V},D}^0.$$

Here  $(H_{\mathfrak{E}}^0)^-$  denotes the edge cochains which change sign when the orientation of an edge is reversed. We will denote this identification by  $\iota_{U,Z}$  and note that we have an exact sequence

$$(4.2) \quad 0 \rightarrow \left( (H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n) \right) / H_{\mathfrak{V},D}^0 \xrightarrow{\iota_{U,Z}} H_{dR}^1(U, Z, \mathcal{V}_n) \xrightarrow{\gamma_{U,Z}} \ker(H_{\mathfrak{V},D}^1 \rightarrow H_{\mathfrak{E}}^1) \rightarrow 0.$$

Consider the retraction map

$$P_{U,Z} : H_{dR}^1(U, Z, \mathcal{V}_n) \longrightarrow \left( (H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n) \right) / H_{\mathfrak{V},D}^0$$

defined on representatives by

$$((\omega_v)_{v \in \mathfrak{V}}, \alpha, (f_e)_{e \in \mathfrak{E}}) \mapsto \left( (f_e - (F_{s(e)} - F_{t(e)}))_{e \in \mathfrak{E}}, \alpha - (F_v)|_Z \right),$$

where  $F_v$  is a choice of Coleman primitive of  $\omega_v$  on  $U_{v,D}$  with respect to the fixed choice of  $p$ -adic logarithm, which is well-defined up to constant sections of  $\mathcal{V}_n|_{U_{v,D}}$ . This defines a section

$$T_{U,Z} : \ker(H_{\mathfrak{V},D}^1 \rightarrow H_{\mathfrak{E}}^1) \longrightarrow H_{dR}^1(U, Z, \mathcal{V}_n)$$

by

$$\alpha \mapsto \beta - \iota_{U,Z}(P_{U,Z}(\beta)),$$

where  $\beta \in H_{dR}^1(U, Z, \mathcal{V}_n)$  is any element satisfying  $\gamma_{U,Z}(\beta) = \alpha$ .

Denote by

$$\varphi_1 : \ker(H_{\mathfrak{V},D}^1 \rightarrow H_{\mathfrak{E}}^1) \longrightarrow \ker(H_{\mathfrak{V},D}^1 \rightarrow H_{\mathfrak{E}}^1)$$

and

$$\varphi_2 : \left( (H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n) \right) / H_{\mathfrak{V},D}^0 \longrightarrow \left( (H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n) \right) / H_{\mathfrak{V},D}^0$$

the respective natural Frobenius action. The Frobenius action on  $H_{dR}^1(U, Z, \mathcal{V}_n)$  is defined by

$$\varphi(\omega) = \iota_{U,Z}(\varphi_2(P_{U,Z}(\omega))) + T_{U,Z}(\varphi_1(\gamma_{U,Z}(\omega))) \quad \text{for } \omega \in H_{dR}^1(U, Z, \mathcal{V}_n).$$

Given  $e \in \mathfrak{E}$ , there is a natural residue map

$$\text{Res}_e : H_{dR}^1(A_e, \mathcal{V}_n^{an}) \longrightarrow H_{dR}^0(A_e, \mathcal{V}_n^{an}).$$

The monodromy operator is defined as

$$(4.3) \quad N(\omega) := \iota_{U,Z}(((\text{Res}_e(\omega|_{A_e}))_{e \in \mathfrak{E}}, 0)) \quad \text{for } \omega \in H_{dR}^1(U, Z, \mathcal{V}_n).$$

**Proposition 4.1.** *The admissible filtered  $(\varphi, N)$ -module  $H_{dR}^1(U, Z, \mathcal{V}_n)$  admits a slope decomposition*

$$(4.4) \quad H_{dR}^1(U, Z, \mathcal{V}_n) = H_{dR}^1(U, Z, \mathcal{V}_n)_m \oplus H_{dR}^1(U, Z, \mathcal{V}_n)_{m+1}$$

and  $\text{im}(\iota_{U,Z}) = H_{dR}^1(U, Z, \mathcal{V}_n)_m$ ,  $\text{im}(T_{U,Z}) = H_{dR}^1(U, Z, \mathcal{V}_n)_{m+1}$ . Furthermore, there is a decomposition

$$(4.5) \quad H_{dR}^1(U, Z, \mathcal{V}_n) = F_{U,Z}^{m+1} \oplus H_{dR}^1(U, Z, \mathcal{V}_n)_m.$$

Here  $F_{U,Z}^{m+1} = H^0(X_\Gamma, \Omega_{X_\Gamma}^1(\log(D)) \otimes \mathcal{V}_n)$ .

**4.3. Pulling back to  $\mathcal{H}_p$ .** By pulling back to  $\mathcal{H}_p$ , the decomposition of  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$  obtains a more explicit combinatorial form, which allows us to define a  $(\varphi, N)$ -module for  $\Gamma$  containing torsion. For now, assume as before that  $\Gamma$  is torsion free.

Let  $\mathcal{H}_p := \mathbb{P}^1(\mathbb{C}_p) \setminus \mathbb{P}^1(\mathbb{Q}_p)$  denote the  $p$ -adic upper half-plane. It is a rigid analytic space with an action of  $\text{GL}_2(\mathbb{Q}_p)$  by Mobius transformation.

Denote by  $\mathcal{T}_p = (\mathcal{T}_p^0, \mathcal{T}_p^1)$  the Bruhat-Tits tree with vertex and oriented edge set  $\mathcal{T}_p^0$  and  $\mathcal{T}_p^1$ , respectively. It is equipped with the usual action of  $\text{GL}_2(\mathbb{Q}_p)$ , and there is a  $\text{GL}_2(\mathbb{Q}_p)$ -equivariant reduction map

$$\text{red}_p : \mathcal{H}_p \longrightarrow \mathcal{T}_p.$$

Let  $C^0(V_n)$  (resp.  $C^1(V_n)$ ) denote the  $\text{GL}_2(\mathbb{Q}_p)$  module of maps  $f : \mathcal{T}_p^0 \rightarrow V_n$  (resp.  $f : \mathcal{T}_p^1 \rightarrow V_n$  such that  $f(e) = -f(\bar{e})$  for every oriented edge  $e \in \mathcal{T}_p^1$ ). Define the map

$$\Delta : C^0(V_n) \longrightarrow C^1(V_n)$$

by

$$\Delta(f)(e) = f(s(e)) - f(t(e)) \quad \text{for } e \in \mathcal{T}_p^1,$$

where  $s(e)$  and  $t(e)$ , respectively, are the start and end vertices of  $e$ . This map sits in a short exact sequence

$$(4.6) \quad 0 \longrightarrow V_n \longrightarrow C^0(V_n) \longrightarrow C^1(V_n) \longrightarrow 0.$$

Let  $\nabla : C^1(V_n) \longrightarrow C^0(V_n)$  be the  $\text{GL}_2(\mathbb{Q}_p)$ -equivariant map obtained by

$$\nabla(f)(v) = \sum_{s(e)=v} f(e) \quad \text{for } f \in C^1(V_n), v \in \mathcal{T}_p^0,$$

where the sum runs over all edges starting at the vertex  $v$ . Let  $C_{\text{har}}^1$  denote the kernel of  $\nabla$ . It sits in the exact sequence

$$0 \longrightarrow C_{\text{har}}^1(V_n) \longrightarrow C^1(V_n) \longrightarrow C^0(V_n) \longrightarrow 0.$$

The oriented intersection graph,  $\mathfrak{G}$ , is canonically identified with  $\Gamma \backslash \mathcal{T}_p$ . For each vertex  $v \in \mathcal{T}_p^0$ , let  $N(v)$  be the neighborhood of  $v$  in  $\mathcal{T}_p$ . The reduction maps and the quotient by the action of  $\Gamma$  fit into the commutative diagram

$$\begin{array}{ccc} \mathcal{H}_p & \xrightarrow{\text{red}_p} & \mathcal{T}_p \\ \downarrow & & \downarrow \\ X_\Gamma^{\text{an}} & \xrightarrow{\text{red}} & \mathfrak{G}. \end{array}$$

It follows that for each vertex  $v \in \mathfrak{V}(\mathfrak{G})$ , the wide open affinoid domain  $U_v$  is isomorphic to  $\mathbb{P}^1$  minus a set of disjoint disks, and for each edge  $e \in \mathfrak{E}(\mathfrak{G})$ , the preimage  $A_e$  is indeed a wide open annulus. Therefore, there are compatible isomorphisms

$$C^0(V_n)^\Gamma \cong H_{\mathfrak{V}}^0$$

and

$$C^1(V_n)^\Gamma \cong (H_{\mathfrak{E}}^0)^-.$$

Let  $w_i \in \mathcal{H}_p$ , be any choice of lift of  $\tilde{w}_i \in X_\Gamma(\mathbb{C}_p) \cong \mathcal{H}_p/\Gamma$ . It follows that there is an isomorphism

$$\text{Func} \left( \bigcup_{i=1}^s \Gamma \cdot w_i, V_n \right)^\Gamma \cong H^0(Z, i^* \mathcal{V}_n).$$

Note then that  $((H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n)) / H_{\mathfrak{V}, D}^0$  is identified with the first cohomology of the total complex of the bi-complex

$$(4.7) \quad \begin{array}{ccc} C^1(V_n)^\Gamma & \longrightarrow & 0 \\ \uparrow & & \uparrow \\ C^0(V_n)^\Gamma & \longrightarrow & \text{Func}(\bigcup_i \Gamma \cdot w_i, V_n)^\Gamma. \end{array}$$

Since the complex

$$V_n \longrightarrow C^0(V_n) \longrightarrow C^1(V_n) \longrightarrow 0$$

is a free resolution of  $V_n$ , the cohomology of (4.7) computes the  $\Gamma$ -hypercohomology of the complex

$$\mathcal{K}_Z(V_n) = \left( V_n \longrightarrow \text{Func} \left( \bigcup_i \Gamma \cdot w_i, V_n \right) \right),$$

where the arrow indicates the map

$$\alpha \mapsto (w \mapsto \alpha).$$

There is an isomorphism

$$H^1(\Gamma, \mathcal{K}_Z(V_n)) \cong \left( (H_\epsilon^0)^- \oplus H^0(Z, i^* \mathcal{V}_n) \right) / H_{\mathfrak{Z}, D}^0.$$

**Definition 4.1.** Let  $\{z_1, \dots, z_r\} \subset \mathcal{H}_p$  be a fixed choice of lifts of the points  $\tilde{z}_i$  to  $\mathcal{H}_p$ , and write

$$\mathcal{H}_U := \mathcal{H}_p - \bigcup_{i=1}^r \Gamma \cdot z_i.$$

Let  $\mathcal{T}_U$  be the tree whose set of vertices is  $\mathcal{T}_p^0 \cup \bigcup_{i=1}^r \Gamma \cdot z_i$ , and whose set of edges is the set of edges of  $\mathcal{T}_p$  along with an edge  $e_w$  for each  $w \in \bigcup_{i=1}^r \Gamma \cdot z_i$ , where the edge  $e_w$  connects the vertex associated with  $w$  and the vertex  $\text{red}_p(w)$ .

Let  $\mathcal{T}_U^1$  denote the set of oriented edges of  $\mathcal{T}_U$ , and  $C_U^1(V_n)$  be the set of functions

$$c : \mathcal{T}_U^1 \longrightarrow V_n$$

satisfying

$$c(\bar{e}) = -c(e) \quad \text{for all } e \in \mathcal{T}_U^1,$$

and

$$(4.8) \quad \sum_{s(e)=v} c(e) = 0 \quad \text{for all } v \in \mathcal{T}_p^0,$$

where the sum in (4.8) runs over all edges in  $\mathcal{T}_U^1$  whose starting vertex is  $v$ .

**Proposition 4.2.** There is a natural identification

$$C_U^1(V_n)^\Gamma \cong \ker(H_{\mathfrak{Z}, D}^1 \rightarrow H_{\epsilon, D}^1).$$

*Proof.* The group  $\ker(H_{\mathfrak{Z}, D}^1 \rightarrow H_{\epsilon, D}^1)$  is isomorphic to

$$(4.9) \quad \ker \left( \left( \bigoplus_{v \in \mathcal{T}_p^0} H_{\text{dR}}^1((\text{red}_p|_{\mathcal{H}_U})^{-1}(N(v)), \mathcal{V}_n^{an}) \right)^\Gamma \rightarrow \left( \bigoplus_{e \in \mathcal{T}_p^1} H_{\text{dR}}^1(\text{red}_p^{-1}(e), \mathcal{V}_n^{an}) \right)^\Gamma \right).$$

For every  $v \in \mathcal{T}_p^0$ , the pre-image

$$(\text{red}_p|_{\mathcal{H}_U})^{-1}(N(v))$$

is  $\mathbb{P}^1(\mathbb{C}_p)$  minus a set of  $(p+1)$  disjoint disks and minus a finite set of points. Each disk corresponds to an edge in  $N(v)$  and each removed point corresponds to some  $w \in \bigcup_{i=1}^r \Gamma \cdot z_i$  with  $\text{red}_p(w) = v$ . It follows that an element of the de Rham cohomology group

$$H_{\text{dR}}^1 \left( (\text{red}_p|_{\mathcal{H}_U})^{-1}(N(v)), \mathcal{V}_n^{an} \right)$$

is determined by the residues on the removed disks and on the removed points, with the restriction that the sum of residues is 0.  $\square$

It follows that we have an exact sequence

$$(4.10) \quad 0 \rightarrow H^1(\Gamma, \mathcal{K}_Z(V_n)) \xrightarrow{\iota'_{U,Z}} H^1_{\text{dR}}(U, Z, \mathcal{V}_n) \xrightarrow{\gamma'_{U,Z}} C^1_U(V_n)^\Gamma \rightarrow 0$$

The natural restriction map

$$C^1_U(V_n)^\Gamma \longrightarrow C^1(V_n)^\Gamma$$

defines a map

$$\text{res} : C^1_U(V_n) \longrightarrow H^1(\Gamma, \mathcal{K}_Z(V_n))$$

and the monodromy operator can be described as the composition

$$N : H^1_{\text{dR}}(U, Z, \mathcal{V}_n) \xrightarrow{\gamma'_{U,Z}} C^1_U(V_n)^\Gamma \xrightarrow{\text{res}} H^1(\Gamma, \mathcal{K}_Z(V_n)) \xrightarrow{\iota'_{U,Z}} H^1_{\text{dR}}(U, Z, \mathcal{V}_n).$$

**4.4. Removing the restriction on  $\Gamma$ .** If  $\Gamma$  is not necessarily torsion free, choose some normal subgroup  $\Gamma' \subset \Gamma$ , which is torsion free and let  $G = \Gamma/\Gamma'$ . Let

$$r : X_{\Gamma'} \longrightarrow X_\Gamma$$

be the natural reduction map. Let  $D' = r^*D$ ,  $Z' = r^*Z$ , and  $U' = r^{-1}(U)$ . We define

$$H^1_{\text{dR}}(U, Z, \mathcal{V}_n) := H^1_{\text{dR}}(U', Z', \mathcal{V}_n)^G.$$

The group  $G$  also acts on  $H^1(\Gamma, \mathcal{K}_{Z'}(V_n))$  and  $C^1_{U'}(V_n)$ . It is clear that the action of  $G$  commutes with the maps Frobenius action as well as the maps  $\iota'_{U',Z'}$ ,  $\gamma'_{U',Z'}$ ,  $P'_{U',Z'}$ ,  $T'_{U',Z'}$  and therefore also the monodromy operator  $N$  on  $H^1_{\text{dR}}(U', Z', \mathcal{V}_n)$ .

**Proposition 4.3.** *There is an exact sequence*

$$0 \rightarrow H^1(\Gamma, \mathcal{K}_Z(V_n)) \xrightarrow{\iota'_{U,Z}} H^1_{\text{dR}}(U, Z, \mathcal{V}_n) \xrightarrow{\gamma'_{U,Z}} C^1_U(V_n)^\Gamma \rightarrow 0.$$

*Proof.* We begin with the analogous exact sequence for  $\Gamma'$  and take  $G$ -invariance to obtain

$$0 \rightarrow H^1(\Gamma, \mathcal{K}_{Z'}(V_n))^G \xrightarrow{\iota'_{U',Z'}} H^1_{\text{dR}}(U', Z', \mathcal{V}_n)^G \xrightarrow{\gamma'_{U',Z'}} (C^1_{U'}(V_n)^\Gamma)^G \rightarrow H^1(G, H^1(\Gamma, \mathcal{K}_Z(V_n))).$$

Since  $G$  is a finite group, and  $H^1(G, H^1(\Gamma, \mathcal{K}_Z(V_n)))$  is a vector space over a field of characteristic zero, and hence is divisible, it follows that

$$H^1(G, H^1(\Gamma, \mathcal{K}_Z(V_n))) = 0.$$

We remark that there  $\mathcal{K}_{Z'} = \mathcal{K}_Z$  and  $C^1_U(V_n) = C^1_{U'}(V_n)$ . It follows by definition that

$$H^1_{\text{dR}}(U', Z', \mathcal{V}_n)^G = H^1_{\text{dR}}(U, Z, \mathcal{V}_n)$$

and it is immediate that

$$(C^1_{U'}(V_n)^\Gamma)^G = C^1_U(V_n)^\Gamma.$$

By the inflation-restriction sequence for group hypercohomology we obtain an exact sequence

$$0 \rightarrow H^1(G, H^0(\Gamma', \mathcal{K}_{Z'})) \rightarrow H^1(\Gamma, \mathcal{K}_Z(V_n)) \rightarrow H^1(\Gamma', \mathcal{K}'_Z(V_n))^G \rightarrow H^2(G, H^0(\Gamma', \mathcal{K}_{Z'}(V_n))).$$

Again, we note that  $G$  is a finite group and  $H^0(\Gamma', \mathcal{K}_{Z'}(V_n))$  is a divisible group, and so it follows that

$$H^1(\Gamma, \mathcal{K}_Z(V_n)) \cong H^1(\Gamma', \mathcal{K}'_Z(V_n))^G,$$

and the result holds.  $\square$

**Proposition 4.4.** *For  $\Gamma$  not necessarily torsion-free,  $H^1_{\text{dR}}(U, Z, \mathcal{V}_n)$  admits a  $(\varphi, N)$ -module structure that is independent of the choice of  $\Gamma'$ , and that there is a slope decomposition*

$$H^1_{\text{dR}}(U, Z, \mathcal{V}_n) = H^1_{\text{dR}}(U, Z, \mathcal{V}_n)_m \oplus H^1_{\text{dR}}(U, Z, \mathcal{V}_n)_{m+1},$$

where

$$H^1_{\text{dR}}(U, Z, \mathcal{V}_n)_m \cong H^1(\Gamma, \mathcal{K}_Z(V_n)),$$

and

$$H^1_{\text{dR}}(U, Z, \mathcal{V}_n)_{m+1} \cong C^1_U(V_n)^\Gamma.$$

There is also a decomposition

$$H^1_{\text{dR}}(U, Z, \mathcal{V}_n) = \text{Fil}_{U,Z}^{m+1} \oplus H^1_{\text{dR}}(U, Z, \mathcal{V}_n)_m,$$

where  $\text{Fil}_{U,Z}^{m+1}$  is the space of  $\Gamma$ -invariant meromorphic differentials on  $\mathcal{H}_p$  with poles supported on the set  $\bigcup_{i=1}^r \Gamma \cdot z_i$  modulo exact differentials.

*Proof.* See [IS03, Appendix 10].  $\square$

**4.5. Poincaré Duality.** As before, let  $D_1 = (z_1) + \cdots + (z_r)$  and  $D_2 = (\tilde{w}_1) + \cdots + (\tilde{w}_s)$  be two divisors on  $X_\Gamma$  defined over a field  $K$ . Let  $U_j$  denote the open subscheme  $X_\Gamma - |D_j|$  for  $j \in \{1, 2\}$  and

$$i_j : Z_j \longrightarrow X_\Gamma$$

denote the closed subscheme associated to the divisor  $D_j$  for  $j \in \{1, 2\}$ . Let  $j, l \in \{1, 2\}$  be two distinct elements. We remark that the complex  $\Omega_{(U_j, Z_l)}^\bullet(\mathcal{V}_n)$ , whose hypercohomology computes  $H_{\text{dR}}^1(U_j, Z_l, \mathcal{V}_n)$ , is easily seen to be quasi-isomorphic to its subcomplex

$$[\mathcal{O}_{U_j}(-D_l) \otimes \mathcal{V}_n \rightarrow \Omega_{U_j}^1 \otimes \mathcal{V}_n].$$

It follows that  $H_{\text{dR}}^1(U_j, Z_l, \mathcal{V}_n)$  is computed as the hypercohomology of the complex

$$[\mathcal{O}_{X_\Gamma}(-D_l) \otimes \mathcal{V}_n \rightarrow \Omega_{X_\Gamma}^1(\log(D_j)) \otimes \mathcal{V}_n].$$

Using this complex, we have the usual cup product in de Rham cohomology

$$(4.11) \quad \langle \cdot, \cdot \rangle_{D_1, D_2} : H_{\text{dR}}^1(U_1, Z_2, \mathcal{V}_n) \times H_{\text{dR}}^1(U_2, Z_1, \mathcal{V}_n) \xrightarrow{\cup} H_{\text{dR}}^2(X_\Gamma, \mathcal{V}_n \otimes \mathcal{V}_n) \xrightarrow{\text{tr}} K[2m+1].$$

compatible with the  $(\varphi, N)$ -structure. In particular, from the slope decomposition of Proposition 4.4 we have that (4.11) induces a perfect pairing

$$(4.12) \quad \langle \cdot, \cdot \rangle_{D_1, D_2} : H_{\text{dR}}^1(U_1, Z_2, \mathcal{V}_n)_m \times H_{\text{dR}}^1(U_2, Z_1, \mathcal{V}_n)_{m+1} \rightarrow K[2m+1].$$

By the identifications of Proposition 4.4, this is a perfect pairing  $C_{U_j}^1(V_n)^\Gamma \times H^1(\Gamma, \mathcal{K}_{Z_j}) \rightarrow K$ . This is precisely the pairing described in [IS03, Section 10.2]. We recall briefly its explicit description. Fix  $\Gamma' \subseteq \Gamma$  a free subgroup with generators  $\gamma_1, \dots, \gamma_t$ . Let  $\mathcal{F}$  be a good fundamental domain for  $\mathcal{T}_p/\Gamma$  and  $b_1, \dots, b_t, c_1, \dots, c_t$  its free edges in the sense of de Shalit [Sha89]. The pairing of  $\mathfrak{f} = (f, f_1, \dots, f_r) \in C_{U_1}^1(V_n)^\Gamma$  and  $\mathfrak{g} = [(g, g_1, \dots, g_r)] \in H^1(\Gamma, \mathcal{K}_{D_1}^\bullet(V_n))$  is given by

$$(4.13) \quad (\mathfrak{f}, \mathfrak{g}) = \frac{1}{[\Gamma : \Gamma']} \sum_{i=1}^t (f(\gamma_i), g(c_i))_\Gamma - \sum_{i=1}^t \frac{(f_i, g_i)_\Gamma}{\#\Gamma_{z_i}}$$

where  $\Gamma_{z_i}$  denotes the stabilizer of  $z_i$ .

**Lemma 4.1.** *For all  $\omega \in H_{\text{dR}}^1(U_1, Z_2, \mathcal{V}_n)$  and  $\eta \in H_{\text{dR}}^1(U_2, Z_1, \mathcal{V}_n)$  we have*

$$(4.14) \quad (\omega, \eta) = (P_{D_2}(\omega), I_{U_2}(\eta))_\Gamma - (I_{U_1}(\omega), P_{D_1}(\eta))_\Gamma.$$

*Proof.* This follows from the orthogonality of the slope decomposition. See [IS03, Theorem 10.6].  $\square$

**Proposition 4.5.** *The pairing  $\langle \cdot, \cdot \rangle_{D_1, D_2}$  is non-degenerate. Furthermore,  $\text{Fil}_{U_j, Z_l}^{m+1}$  and  $\text{Fil}_{U_l, Z_j}^{m+1}$  are orthogonal complements. This induces a perfect pairing*

$$(4.15) \quad \text{Fil}_{U_j, Z_l}^{m+1} \times H_{\text{dR}}^1(U_l, Z_j, \mathcal{V}_n) / \text{Fil}^{m+1} \rightarrow K.$$

*Proof.* The key input is Serre duality which implies that the cup products

$$H^1(X_\Gamma, \mathcal{O}_{X_\Gamma}(-D_j) \otimes \mathcal{V}_n) \times H^0(X_\Gamma, \Omega_{X_\Gamma}^1(\log(D_j)) \otimes \mathcal{V}_n) \xrightarrow{\cup} H^1(X_\Gamma, \Omega_{X_\Gamma}^1) \xrightarrow{\text{tr}} K$$

are non-degenerate for each  $j \in \{1, 2\}$ . The proof follows *mutatis mutandis* the proof of Poincaré duality. See [The26]  $\square$

## 5. NEKOVÁŘ'S $p$ -ADIC HEIGHTS

In this section we recall the construction of Nekovář's  $p$ -adic height pairing in the semistable case [Nek93].

**5.1. Global  $p$ -adic height pairing.** Let  $K$  be a number field and let  $G_K = \text{Gal}(\bar{K}/K)$  be its absolute Galois group. Let  $L$  be a finite extension of  $\mathbb{Q}_p$  and consider  $V$  a  $p$ -adic Galois representation of  $G_K$  over  $L$ . We will denote by  $V_v$  the local  $p$ -adic representation of  $G_{K_v} = \text{Gal}(\bar{K}_v/K_v)$  obtained by restriction from  $V$ . Suppose that

- $V$  is unramified for all but a finite set of places of  $K$ ,
- $H^i(K_v, V) = H^i(K_v, V^\vee(1)) = 0$  for  $i = 0, 1, 2$  and for all places  $v \nmid p$ ,
- $V_v$  is a semistable representation of  $\text{Gal}(\bar{K}_v/K_v)$  for all  $v \mid p$ ,
- $D_{\text{st}}(V_v)^{\varphi=1, N=0} = D_{\text{st}}(V_v^\vee(1))^{\varphi=1, N=0} = 0$ .

The definition of the *local  $p$ -adic height* will depend on the choice of the following two objects:

- a continuous idele class character  $l_K : \mathbb{A}_K^\times / K^\times \rightarrow \mathbb{Q}_p$  such that for each  $v \mid p$  we have a factorization

$$(5.1) \quad \begin{array}{ccc} \mathcal{O}_{K_v}^\times & \xrightarrow{l_v} & \mathbb{Q}_p \\ & \searrow \log_v \quad \nearrow \text{tr} & \\ & K_v & \end{array}$$

- for each  $v \mid p$  a choice of a subspace  $W_v \subset DR(V_v)$  complementary to the filtered piece  $\text{Fil}^0$ , i.e. a decomposition  $DR(V_v) = W_v \oplus \text{Fil}^0(DR(V_v))$ .

For the remaining of the paper we fix the continuous *cyclotomic character*  $l_K^{\text{cyc}}$  defined as follows:

$$(5.2) \quad l_K^{\text{cyc}} : \mathbb{A}_K^\times / K^\times \xrightarrow{\text{rec}_K} \text{Gal}(\bar{K}/K) \xrightarrow{\chi_p^{\text{cyc}}} \mathbb{Z}_p^\times \xrightarrow{\log_p} \mathbb{Q}_p$$

where  $\text{rec}_K$  is the usual reciprocity map and  $\log_p$  is a branch of the  $p$ -adic logarithm normalized such that  $\log_p(p) = 0$ . This character satisfies the property  $l_K^{\text{cyc}} = l_{\mathbb{Q}}^{\text{cyc}} \circ N_{K/\mathbb{Q}}$  and in particular we have the following description place by place

$$(5.3) \quad l_{K,v}^{\text{cyc}}(x) = \begin{cases} \log_p(N_{K_v/\mathbb{Q}_p}) & v \nmid p, \\ -\text{ord}_v(x) \log_p(\#k_v) & v \mid p. \end{cases}$$

We will refer to  $l_K^{\text{cyc}}$  simply as  $l_K$ .

Nekovář [Nek93, Theorem 2.2] constructed a family of continuous bilinear pairings

$$(5.4) \quad h : H_f^1(K, V) \times H_f^1(K, V^\vee(1)) \rightarrow \mathbb{Q}_p.$$

Let  $e_1 \in H_f^1(K, V)$  and  $e_2 \in H_f^1(K, V^\vee(1))$  be two Selmer classes, these correspond to semistable extensions of Galois representations

$$(5.5) \quad \begin{array}{l} e_1 : \quad 0 \longrightarrow V \longrightarrow E_1 \longrightarrow \mathbb{Q}_p \longrightarrow 0 \\ e_2 : \quad 0 \longrightarrow V^\vee(1) \longrightarrow E_2 \longrightarrow \mathbb{Q}_p \longrightarrow 0. \end{array}$$

There exist a set of Galois representations called *biextensions* which fit in to a diagram with exact row and columns

$$\begin{array}{ccccccc}
& & 0 & & 0 & & \\
& & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E_2^\vee(1) & \longrightarrow & V \longrightarrow 0 \\
& & \parallel & & \downarrow & & \downarrow \\
(5.6) \quad 0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E & \longrightarrow & E_1 \longrightarrow 0 \\
& & & & \downarrow & & \downarrow \\
& & & & \mathbb{Q}_p & = & \mathbb{Q}_p \\
& & & & \downarrow & & \downarrow \\
& & & & 0 & & 0
\end{array}$$

Every choice of such biextension  $E$  determines a splitting of the height pairing in its *local components*

$$(5.7) \quad h(e_1, e_2) = \sum_v h(e_1, e_v)_{E,v}.$$

5.2.  **$p$ -adic heights at  $v \nmid p$ .** First of all observe that by continuity, every idele character restricted to  $v$

$$(5.8) \quad l_v : K_v^\times \rightarrow \mathbb{Q}_p$$

is unramified, i.e. it has to factors through  $\text{ord}_v : K_v \rightarrow \mathbb{Z}$ . Under the Kummer isomorphism, the map  $l_v$  induces an isomorphism

$$(5.9) \quad (\text{ord}_v)_* : H^1(K_v, \mathbb{Q}_p(1)) \xrightarrow{\sim} K_v^\times \hat{\otimes} \mathbb{Q}_p \xrightarrow{\sim} \mathbb{Q}_p$$

Furthermore, the assumption  $H^1(K_v, V) = H^1(K_v, V^\vee(1)) = 0$  simplifies the biextension diagram because it implies that the extensions  $e_1, e_2$  are split and consequently the biextension  $E_v$  can be written as  $E_v = V_v \oplus U_v$  where  $U_v$  is a simple extension of  $G_{K_v}$   $p$ -adic representation

$$(5.10) \quad [0 \rightarrow \mathbb{Q}_p(1) \rightarrow U_v \rightarrow \mathbb{Q}_p \rightarrow 0] \in H^1(K_v, \mathbb{Q}_p(1)).$$

The *local height pairing* is then defined to be

$$(5.11) \quad h(e_1, e_2)_{E,v} = \log_p(\#k_v) \text{ord}_v([U_v]).$$

5.3.  **$p$ -adic heights at  $v|p$ .** The case for local heights at places dividing  $p$  is more involved and uses the structure of  $(\varphi, N)$ -modules. The top short exact sequence of the local version of the biextension diagram (5.6) induces a commutative diagram (Sec. 3.5 [Nek93]) with exact rows and columns

$$\begin{array}{ccccccc}
& & 0 & & 0 & & \\
& & \downarrow & & \downarrow & & \\
& & D_{\text{st}}(E_{2,v}^\vee(1))^{\varphi=1} & \xrightarrow{\sim} & D_{\text{st}}(V_v)^{\varphi=1} & & \\
& & \downarrow (0, N, -i) & & \downarrow (N, -i) & & \\
0 & \longrightarrow & \mathbb{Q}_p \oplus K_v & \longrightarrow & \mathbb{Q}_p \oplus D_{\text{st}}(V_v(-1))^{\varphi=1} \oplus DR(E_{2,v}^\vee(1))/\text{Fil}^0 & \longrightarrow & D_{\text{st}}(V_v(-1))^{\varphi=1} \oplus DR(V_v)/\text{Fil}^0 \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & H_{\text{st}}^1(K_v, \mathbb{Q}_p(1)) & \longrightarrow & H_{\text{st}}^1(K_v, E_{2,v}^\vee(1)) & \longrightarrow & H_{\text{st}}^1(K_v, V_v) \longrightarrow 0 \\
& & & & \downarrow & & \downarrow \\
& & & & 0 & & 0
\end{array}$$

where we have denoted by  $i$  the inclusion map  $D_{\text{st}}(V_v) \hookrightarrow DR(V_v)$  and the same for  $E_{2,v}^\vee(1)$ . Recall that this map depends on the choice of a branch of the  $p$ -adic logarithm.

Under the assumption

$$(5.13) \quad D_{\text{st}}(V_v)^{\varphi=1, N=0} = D_{\text{st}}(V_v^\vee(1))^{\varphi=1, N=0} = 0$$

the embeddings of the  $DR$  components give isomorphisms

$$(5.14) \quad DR(E_{2,v}^\vee(1))/\text{Fil}^0 \cong \text{coker}(D_{\text{st}}(E_{2,v}^\vee(1))^{\varphi=1} \rightarrow D_{\text{st}}(V_v(-1))^{\varphi=1} \oplus DR(E_{2,v}^\vee(1))/\text{Fil}^0)$$

$$(5.15) \quad DR(V_v)/\text{Fil}^0 \cong \text{coker}(D_{\text{st}}(V_v)^{\varphi=1} \rightarrow D_{\text{st}}(V_v(-1))^{\varphi=1} \oplus DR(V_v)/\text{Fil}^0).$$

This implies that from the exact diagram (5.12), it can be extracted the following commutative diagram with exact rows

$$(5.16) \quad \begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Q}_p \oplus K_v & \longrightarrow & \mathbb{Q}_p \oplus DR(E_{2,v}^\vee(1))/\text{Fil}^0 & \longrightarrow & DR(V_v)/\text{Fil}^0 \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & H_{\text{st}}^1(K_v, \mathbb{Q}_p(1)) & \longrightarrow & H_{\text{st}}^1(K_v, E_{2,v}^\vee(1)) & \longrightarrow & H_{\text{st}}^1(K_v, V_v) \longrightarrow 0 \end{array}$$

where the vertical isomorphisms are often referred to as *Bloch-Kato exponential maps*. This isomorphisms are explicitly given by Lemma 2.1 and Lemma 2.2. It is important to recall that the  $\mathbb{Q}_p$  summands are mapped isomorphically under the first horizontal map leaving with a commutative diagram of the form

$$(5.17) \quad \begin{array}{ccccccc} 0 & \longrightarrow & K_v & \longrightarrow & DR(E_{2,v}^\vee(1))/\text{Fil}^0 & \longrightarrow & DR(V_v)/\text{Fil}^0 \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & H_{\text{f}}^1(K_v, \mathbb{Q}_p(1)) & \longrightarrow & H_{\text{f}}^1(K_v, E_{2,v}^\vee(1)) & \longrightarrow & H_{\text{f}}^1(K_v, V_v) \longrightarrow 0. \end{array}$$

Recall that the definition of the  $p$ -adic height depends on the choice of the following two objects the continuous homomorphism  $l_v : K_v^\times \rightarrow \mathbb{Q}_p$  and a choice of a subspace  $W_v \subset DR(V_v)$  complementary to the filtered piece  $\text{Fil}^0$ . In particular, the choice of the subspace  $W_v$  induces a splitting of the top exact sequence as follows

$$(5.18) \quad w' : DR(V_v)/\text{Fil}^0(V_v) \xrightarrow{W_v} DR(V_v) \xrightarrow{\alpha} DR(E_{2,v}^\vee(1)) \rightarrow DR(E_{2,v}^\vee(1))/\text{Fil}^0(E_{2,v}^\vee(1))$$

where the leftmost map is induced by the decomposition  $DR(V_v) = W_v \oplus \text{Fil}^0$  and  $\alpha$  denotes the unique  $(\varphi, N)$ -equivariant splitting

$$(5.19) \quad \begin{array}{ccccccc} 0 & \longrightarrow & K_v & \longrightarrow & DR(E_{2,v}^\vee(1))/\text{Fil}^0 & \longrightarrow & DR(V_v)/\text{Fil}^0 \longrightarrow 0. \\ & & & & \swarrow \text{---} w' \text{---} & & \end{array}$$

The splitting of the exact sequence induces a retraction in the bottom exact sequence

$$(5.20) \quad w : H_{\text{st}}^1(K_v, E_2) \rightarrow H_{\text{st}}^1(K_v, \mathbb{Q}_p(1)).$$

Recall that the *Kummer map* determines a canonical isomorphism

$$(5.21) \quad H_{\text{st}}^1(K_v, \mathbb{Q}_p(1)) = K_v^\times \hat{\otimes} \mathbb{Q}_p.$$

Under this identification, the inverse of the left most vertical map of the diagram (5.19) is given by

$$(5.22) \quad \begin{array}{c} H_{\text{st}}^1(K_v, \mathbb{Q}_p(1)) \rightarrow \mathbb{Q}_p \oplus K_v \\ x \mapsto (v(x), \log(x)) \end{array}$$

for the given fixed choice of the branch of the  $p$ -adic logarithm. We are now ready to give the definition of the  $p$ -adic height.

**Definition 5.1.** The local  $p$ -adic height of the biextension  $[E_v] \in H_{\text{st}}^1(K, E_2)$  as in (5.6) is defined to be

$$(5.23) \quad h(e_1, e_2)_{E,v} = -l_v(w([E_v])).$$

In the next section we will specialize this to the case of biextensions arising from Heegner cycles and we will provide an explicit formula for the local  $p$ -adic height.

**5.4.  $p$ -adic heights on the Shimura curve.** We now want to study the extensions of filtered  $(\varphi, N)$ -modules obtained from the extensions of  $p$ -adic Galois representations considered in Section 3.2. Recall that by Theorem 3.4 we have the following isomorphism

$$(5.24) \quad D_{\text{st}}(H_{\text{ét}}^1(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n)^G) = H_{\text{dR}}^1(X_M \otimes \mathbb{Q}_p, \mathcal{V}_n)^G.$$

This isomorphism is defined over the field of definition  $\mathbb{Q}$  of  $X_M$ . From the discussion and results of Section 3.7, we have that we can explicitly characterize the  $F$ -isocrystal  $\mathcal{V}_n = \mathbb{D}(\mathbb{L}_n)$  after base changing to  $\mathbb{Q}_{p^{\text{unr}}}$ . In order to construct the  $p$ -adic height of a biextension we need, as explained in the previous Section, to describe the unique splitting of (5.17). We will characterize this splitting after the base change of  $\mathbb{Q}_{p^{\text{unr}}}$  so we will not need to worry and carry on the field of definition of all the objects. Recall that under this assumption we have that we can translate the de Rham cohomology computation to the curve  $X_\Gamma$ .

Let  $K$  be a number field unramified at  $p$ . Let  $D$  be a reduced effective divisor on  $X_M$  defined over  $K$  such that all closed points reduce to distinct points on  $X$  under the map  $q : X_M \rightarrow X$ , we have an extension

$$(5.25) \quad e_{\text{ét}}^1 : 0 \rightarrow H_{\text{ét}}^1(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow H_{\text{ét}}^1((X_M - |D|) \otimes \bar{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(m) \rightarrow 0$$

applying the functor  $D_{\text{st}}$ , thanks to Theorem 3.6, we get an extension in  $\text{MF}_{K_p}(\varphi, N)$

$$(5.26) \quad e_{\text{dR}} : 0 \rightarrow H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \rightarrow H_{\text{dR}}^1(U, \mathcal{V}_n)[-m-1] \rightarrow \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)_z^{\Gamma_z}[-m] \rightarrow 0.$$

Recall from (2.8), the relation between the effect of the Tate twist in the two categories. By Theorem 2.1, the equivalence of categories between semistable  $p$ -adic representations and admissible filtered  $(\varphi, N)$ -modules given by Fontaine's functor  $D_{\text{st}}$  is compatible with exact sequences, tensor product and duality. See in particular the discussion in Subsection 3.7 for the compatibility between the different pairings. Recall that from Subsection 3.5, the dual exact sequence  $e_{\text{ét}}^\vee$  is given by

$$(5.27) \quad e_{\text{ét}}^\vee : 0 \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(m) \rightarrow H_{\text{ét}, |D|}^1(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n(m))^G \rightarrow H_{\text{ét}}^1(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n(m))^G \rightarrow 0$$

while the corresponding dual sequence  $e_{\text{dR}}^\vee$  is given by

$$(5.28) \quad e_{\text{dR}}^\vee : 0 \rightarrow \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)_z^{\Gamma_z}[-m] \rightarrow H_{\text{dR}}^1(X_\Gamma, D_\Gamma, \mathcal{V}_n)[-m] \rightarrow H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m] \rightarrow 0.$$

Considering two different reduced disjoint divisors  $D_1, D_2$  satisfying previous assumptions we can consider the biextension diagram constructed in Subsection 3.5 in the category  $\text{MF}_{K_p}(\varphi, N)$

$$(5.29) \quad \begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \bigoplus_{z \in D_2} (\mathcal{V}_n)_z^{\Gamma_z}[-m-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)[-m-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \bigoplus_{z \in D_2} (\mathcal{V}_n)_z^{\Gamma_z}[-m-1] & \longrightarrow & H_{\text{dR}}^1(U_{1,\Gamma}, D_2, \mathcal{V}_n)[-m-1] & \longrightarrow & H_{\text{dR}}^1(U_{1,\Gamma}, \mathcal{V}_n)[-m-1] \longrightarrow 0 \\ & & & & \downarrow & & \downarrow \\ & & & & \bigoplus_{z \in D_1} (\mathcal{V}_n)_z^{\Gamma_z}[-m] & \xlongequal{\quad} & \bigoplus_{z \in D_1} (\mathcal{V}_n)_z^{\Gamma_z}[-m] \\ & & & & \downarrow & & \downarrow \\ & & & & 0 & & 0 \end{array}.$$

We now want to consider only simple extensions of  $p$ -adic Galois representations by  $\mathbb{Q}_p$  and  $\mathbb{Q}_p(1)$  and the corresponding simple extensions of filtered  $(\varphi, N)$ -modules. In Section 3.9 we have

recalled the construction of special Heegner classes  $\mathrm{CH}^m(\mathcal{A}_{\bar{P}}^m, \epsilon_n)$  attached to a Heegner point  $P$ . Under the  $p$ -adic Abel-Jacobi map, these classes can be used to produce extension classes

$$(5.30) \quad \mathrm{cl}_0 : \mathrm{CH}^{m+1}(\mathcal{M}_n) \rightarrow \mathrm{Ext}_{\mathrm{Rep}_{\mathrm{st}}(G_K)}^1(\mathbb{Q}_p, H_{\mathrm{et}}^1(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n(m+1))^G)$$

Applying the functor  $D_{\mathrm{st}, \mathbb{Q}_{p^2}}$  this produces a simple extension class in

$$(5.31) \quad \mathrm{Ext}_{\mathrm{MF}_{\mathbb{Q}_{p^{\mathrm{unr}}}}^{\mathrm{ad}}(\varphi, N)}^1(\mathbb{Q}_{p^{\mathrm{unr}}}, H_{\mathrm{dR}}^1(X_{\Gamma}, \mathcal{V}_n)[-m-1]).$$

**5.5. Heegner biextension.** In [IS03], the authors explicitly characterized these latter extension classes arising from Heegner cycles.

Let  $P$  be a Heegner point on  $X$  defined over  $K$  quadratic imaginary number field of discriminant  $D_K$ . Let  $\Psi : K \rightarrow B$  be an optimal embedding corresponding to  $P$  as in Lemma 3.3. Let  $z_{\bar{P}} \in \mathrm{CH}^m(\mathcal{A}_{\bar{P}}^m, \epsilon_n)$  be the Heegner class defined in Definition 3.6. Let  $P_{\Psi}(X, Y) \in \mathcal{P}_2$  the homogeneous polynomial in two variable defined by

$$(5.32) \quad P_{\Psi}(X, Y) = P_{\Psi(\sqrt{D_K})}(X, Y) = \mathrm{tr} \left( \Psi_{\sqrt{D_K}} \begin{pmatrix} X \\ Y \end{pmatrix} \begin{pmatrix} X & -Y \end{pmatrix} \right)$$

as in the isomorphism described in Lemma 3.2

**Lemma 5.1.** *The composition of the  $p$ -adic Abel-Jacobi map*

$$(5.33) \quad \mathrm{cl} : \mathrm{CH}^m(\mathcal{A}_{\bar{P}}^m, \epsilon_n) \rightarrow \mathbb{L}_{n, P}(m)$$

with the canonical isomorphisms of  $(\varphi, N)$ -modules

$$(5.34) \quad D_{\mathrm{st}, \mathbb{Q}_{p^{\mathrm{unr}}}}(\mathbb{L}_n(m)_{\bar{P}}) = (\mathcal{V}_n)_{\Psi}[-m]$$

sends  $z_{\bar{P}}^{(n)}$  to the element  $(-, P_{\Psi}^m) \in V_n = \mathcal{P}_n^{\vee}$ .

*Proof.* See [IS03, Lemma 8.4] □

We can now define the *Heegner biextension* we will study. Let  $K_1, K_2$  two quadratic imaginary fields inert at  $p$  and  $\Psi_i \in \mathrm{emb}(\mathcal{O}_{K_i}, R)$  for  $i = 1, 2$  two optimal embeddings to which correspond  $P_{\Psi_i}$  Heegner points. These points are defined over the Hilbert class fields  $H_i$ . Consider the Heegner classes  $z_{\bar{P}_i} \in \mathrm{CH}^m(\mathcal{A}_{\bar{P}_i}^m, \epsilon_n)$ . We consider the divisors

$$(5.35) \quad D_{\Psi_i} = (P_{\Psi_i}) + (P_{\bar{\Psi}_i}) \quad \text{for } i = 1, 2$$

and the cycle class maps

$$(5.36) \quad \begin{aligned} \mathrm{cl}_{\Psi_i} : \mathbb{Q}_p &\rightarrow (\mathbb{L}_n)_{P_{\Psi_i}}(m) \oplus (\mathbb{L}_n)_{P_{\bar{\Psi}_i}}(m) \\ 1 &\mapsto \left( \mathrm{cl}(z_{P_{\Psi_i}}^{(n)}), (-1)^{m+1} \mathrm{cl}(z_{P_{\bar{\Psi}_i}}^{(n)}) \right) \quad \text{for } i = 1, 2. \end{aligned}$$

It is important to remark that the point  $P_{\bar{\Psi}_i}$  corresponds to  $w_P(\bar{P}_{\Psi_i})$  where  $w_P$  is the Atkin-Lehner involution and  $\bar{P}_{\Psi_i}$  is the action of complex conjugation. See [BC91].

**Lemma 5.2.** *Let  $z_i, \bar{z}_i$  the points on  $\mathcal{H}_p$  corresponding to the optimal embeddings  $\Psi_i, \bar{\Psi}_i$ . Under the reduction map modulo  $p$ , the two points specialize to the same vertex and the cycle class map defines a morphism of filtered  $\varphi$ -modules*

$$(5.37) \quad \mathrm{cl}_{\Psi_i} : \mathbb{Q}_p^{\mathrm{unr}} \rightarrow (V_n)_{z_i}[-m] \oplus (V_n)_{\bar{z}_i}[-m]$$

such that, denoting  $Q_i = \mathrm{cl}_{\Psi_i}(1)$  we obtain

$$(5.38) \quad (Q_i)_{z_i} + (Q_i)_{\bar{z}_i} = 0.$$

*Proof.* See [IS03, Theorem 9.2]. □

The *Heegner biextension* is then obtained from (5.49) taking the pullback and pushout along the cycle class maps  $\mathrm{cl}_{\Psi_i}$ .

**5.6.  $p$ -adic heights as Poincaré pairing on de Rham cohomology.** In this section we will study how the  $p$ -adic pairing described in Subsection 5.3 can be explicitly computed in terms of the Poincaré pairing of special classes in de Rham cohomology. As an important assumption, as explained at the beginning of the previous Section, we will work over the maximal unramified field extension  $\mathbb{Q}_{p^{\text{unr}}}$ .

We start by fixing  $D_1, D_2 \hookrightarrow X_\Gamma$  two disjoint effective reduced divisors on  $X_\Gamma$ . Assume that every closed point of  $D_1, D_2$  is defined over  $\mathbb{Q}_{p^2}$ . A simple extension class of  $H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1]$  by  $\mathbb{Q}_{p^{\text{unr}}}$  can be constructed as the pullback along a map of filtered  $\varphi$ -modules  $\text{cl}_i : \mathbb{Q}_{p^{\text{unr}}} \hookrightarrow \bigoplus_{z \in D_i} (\mathcal{V}_n)_z^{\Gamma_z}[-m]$  of the Gysin exact sequence

$$(5.39) \quad \begin{array}{ccccccc} 0 & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) & \longrightarrow & L_i & \longrightarrow & \mathbb{Q}_{p^{\text{unr}}} \longrightarrow 0 \\ & & \parallel & & \downarrow \iota_{E_i} & & \downarrow \text{cl}_i \\ 0 & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) & \xrightarrow{\iota} & H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)[-m-1] & \xrightarrow{\text{Res}} & \bigoplus_{z \in D_i} (\mathcal{V}_n)_z^{\Gamma_z}[1] \longrightarrow 0. \end{array}$$

The notation  $\text{cl}$  is reminiscent of the fact that we will later construct such extensions from the Heegner classes previously described.

**Lemma 5.3.** *The exact sequence of filtered  $(\varphi, N)$ -modules*

$$(5.40) \quad 0 \longrightarrow H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \xrightarrow{\iota} H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)[-m-1] \xrightarrow{\text{Res}} \bigoplus_{z \in D_i} (\mathcal{V}_n)_z^{\Gamma_z}[-m] \longrightarrow 0.$$

*admits a unique Frobenius-monodromy equivariant splitting*

$$(5.41) \quad s_i : \bigoplus_{z \in D_i} (\mathcal{V}_n)_z^{\Gamma_z}[1] \rightarrow H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)$$

where  $s_i(Q) = \omega_Q \in H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)$  is the unique element satisfying

- (i)  $\omega_Q \in H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)_{m+1}$
- (ii)  $\text{Res}(\omega_Q) = Q$ ,
- (iii)  $N(\omega_Q) = 0$ .

*Proof.* First of all observe that  $\bigoplus_{z \in D_i} (\mathcal{V}_n)_z^{\Gamma_z}[1]$  is isotrivial of slope  $m+1$  since Frobenius acts by  $\det^{\otimes m}$  on  $\mathcal{V}_n\{m\}$  and has trivial monodromy. In particular, Frobenius acts on every element  $Q$  as multiplication by  $p^{m+1}$ . See Subsection 3.6. By the description in Proposition 4.4 and the constructions of the Frobenius stable maps  $I, P$  in Section 4, we have a commutative diagram with exact rows and columns:

$$(5.42) \quad \begin{array}{ccccccc} & & & 0 & & 0 & \\ & & & \downarrow & & \downarrow & \\ 0 & \longrightarrow & H^1(\Gamma, V_n) & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) & \xrightarrow{I} & C_{\text{har}}(V_n)^\Gamma \longrightarrow 0 \\ & & \parallel & \swarrow P & \downarrow \iota & & \downarrow \text{Res} \\ 0 & \longrightarrow & H^1(\Gamma, V_n) & \longrightarrow & H_{\text{dR}}^1(X_\Gamma - D, \mathcal{V}_n) & \xrightarrow{I_U} & C_U(V_n)^\Gamma \longrightarrow 0 \\ & & \swarrow P_U & & \downarrow \text{Res} & & \downarrow \\ & & & & \bigoplus_{z \in D} (\mathcal{V}_n)_z[1] & \longrightarrow & \bigoplus_{z \in D} \text{Ind}^\Gamma(V_n)^\Gamma \\ & & & & \downarrow & & \downarrow \\ & & & & 0 & & H^1(\Gamma, C_{\text{har}}(V_n)). \end{array}$$

From the diagram we then see that the intersection of the trivial monodromy  $N = 0$  and the slope  $m+1$  cuts out a non trivial subspace. The trivial monodromy will indeed land in  $T_U(C_U(V_n)^\Gamma)$

where  $T_U$  is the section of the middle row induced by  $P_U$ . Suppose  $\omega, \omega'$  are two elements satisfying the previous properties then their difference lie in

$$\omega - \omega' \in \iota(H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)_{m+1}) \cap \ker N = \{0\}$$

concluding the unicity by the fact that  $N$  defines a bijection on the global cohomology decomposition  $N : H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)_{m+1} \rightarrow H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)_m$ .  $\square$

**Lemma 5.4.** *Under the Bloch-Kato exponential map (2.10) we have an isomorphism*

$$(5.43) \quad \text{Ext}_{\text{MF}_{\mathbb{Q}_p^{\text{unr}}}^{\text{ad}}(\varphi, N)}^1(\mathbb{Q}_{p^{\text{unr}}}, H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1]) \cong (\text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n))^\vee \cong M_k(X_\Gamma)^\vee$$

and the extension class  $[L_i]$  of (5.39) is identified with the element  $(-, \eta_i) \in \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)^\vee$  where  $\eta_i \in H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)$  is the unique element  $\bmod \text{Fil}^{m+1}$  satisfying

$$(5.44) \quad \iota(\eta_i) \equiv s_i(\text{cl}_i(1)) \bmod \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)$$

where  $s_i(\text{cl}_i(1)) \in H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)$  is the element defined by the unique Frobenius-monodromy equivariant splitting of Lemma 5.3.

*Proof.* First of all, by the Bloch-Kato exponential the LHS of (5.43) is identified with  $H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)/\text{Fil}^{m+1}$  and by Proposition 4.5 it is isomorphic to  $\text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)^\vee$ . Furthermore we have the identification between filtered piece and modular forms thanks to [IS03, Proposition 6.1].

Consider the pullback diagram (5.39). The construction in Lemma 5.3 induces a Frobenius-monodromy splitting of the top row of the diagram

$$(5.45) \quad \begin{aligned} s : \mathbb{Q}_{p^{\text{unr}}} &\rightarrow L_i \\ 1 &\mapsto s_i(\text{cl}_i(1)). \end{aligned}$$

From the explicit construction of the Bloch-Kato discussed after Lemma 2.1, we have that  $\iota$  induces an isomorphism

$$(5.46) \quad H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)/\text{Fil}^{m+1} \cong L_i/\text{Fil}^{m+1} L_i$$

and  $\eta_i$  is defined to be the corresponding element in  $H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)$  such that

$$(5.47) \quad \iota(\eta_i) \equiv s_i(\text{cl}_i(1)) \bmod \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)$$

concluding the description of the extension class  $[L_i]$  under the isomorphism (5.43).  $\square$

Similarly, we can describe the extension class of the middle column of (5.29). Fix two classes  $\text{cl}_i : \mathbb{Q}_{p^{\text{unr}}} \rightarrow \bigoplus_{z \in D_1} (\mathcal{V}_n)_{\mathbb{Z}^z}^\Gamma[-m]$ , let  $Q_1, Q_2$  the respective images of the element 1. Consider the dual shifted extension of  $L_2$  that can be obtained as the push-out along the  $\text{cl}$  map

$$(5.48) \quad \begin{array}{ccccccc} 0 & \longrightarrow & \bigoplus_{z \in D_2} (\mathcal{V}_n)_{\mathbb{Z}^z}^\Gamma & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)[-m-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & \mathbb{Q}_{p^{\text{unr}}}[-1] & \longrightarrow & L_2^\vee[-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \longrightarrow 0 \end{array}$$

In the previous top exact sequence we have used the fact that the relative de Rham cohomology is dual to the open cohomology under the Poincaré pairing described in Section 4.5. Construct the

biextension obtained from the two classes  $\text{cl}_i$  pulling back and pushing out the diagram (5.29)

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & \mathbb{Q}_{p^{\text{unr}}}[-1] & \longrightarrow & L_2^\vee[-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \longrightarrow 0 \\
 & & \parallel & & \downarrow & & \downarrow \\
 (5.49) \quad 0 & \longrightarrow & \mathbb{Q}_{p^{\text{unr}}}[-1] & \longrightarrow & L & \longrightarrow & L_1 \longrightarrow 0 \\
 & & & & \downarrow & & \downarrow \\
 & & & & \mathbb{Q}_{p^{\text{unr}}} & \xlongequal{\quad} & \mathbb{Q}_{p^{\text{unr}}} \\
 & & & & \downarrow & & \downarrow \\
 & & & & 0 & & 0
 \end{array}$$

Lemma 2.2 explicitly identifies the extension class  $[L]$  as an element in

$$(5.50) \quad \text{Ext}_{\text{MF}_{\mathbb{Q}_{p^{\text{unr}}}^{\text{ad}}}(\varphi, N)}^1(\mathbb{Q}_{p^{\text{unr}}}, E_2^\vee[-1]) \cong \mathbb{Q}_p \oplus L_2^\vee[-1] / \text{Fil}^0(L_2^\vee[-1])$$

where  $L_2[-1]$  satisfies all of the assumptions of Lemma 2.2. Observe that, by the Poincaré pairing and the interplay with filtration (Proposition 4.5), we can identify the group of extensions with

$$(5.51) \quad L_2^\vee[-1] / \text{Fil}^0(L_2^\vee[-1]) \cong (\text{Fil}^0 L_2)^\vee.$$

The extension class corresponding to  $E$  in (5.49) can be determined as follows.

**Lemma 5.5.** *Let  $\iota_{L_2}$  be the injection of  $L_2$  in  $H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)$  (5.39) and  $\iota$  be the injection of  $H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)$  in  $H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)$ . The element  $[L]$  in the extension group of  $L_2^\vee[-1]$  by  $\mathbb{Q}_{p^{\text{unr}}}$  is given by*

$$(5.52) \quad (N_L, (\eta_L, \iota_{L_2}(-))) \in \mathbb{Q}_p \oplus (\text{Fil}^0 L_2)^\vee$$

where  $N_L$  is a certain element in  $\mathbb{Q}_{p^{\text{unr}}}$ , and  $\eta_L \in H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)$  satisfies the following properties

- (i)  $\iota(\eta_L) \equiv \omega_{Q_1} \pmod{\text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)}$  for  $\omega_{Q_1} \in H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)_{m+1}$ ,
- (ii)  $\text{Res}(\omega_{Q_1}) = Q_1$ ,
- (iii)  $N(j(\omega_{Q_1})) = 0$  in  $H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n)$ .

The cohomology class  $\omega_{Q_1}$  is uniquely characterized by these properties.

*Proof.* This time we need to apply Lemma 2.2, the general version of Lemma 2.1. First of all, we consider the pull-back of the last two columns of the biextension diagram (5.29) along the map

$\text{cl}_i : \mathbb{Q}_{p^{\text{unr}}} \hookrightarrow \bigoplus_{z \in D_i} (\mathcal{V}_n)_z^{\Gamma_z}[-m]$  obtaining the following diagram.

$$(5.53) \quad \begin{array}{ccccc} & 0 & & 0 & \\ & \downarrow & & \downarrow & \\ H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)[-m-1] & \xrightarrow{\quad} & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] & & \\ & \searrow & \swarrow & & \\ & H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)[-m-1] & \xrightarrow{\quad} & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] & \\ & \downarrow & & \downarrow & \\ H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)[-m-1] & \xrightarrow{\quad} & H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n)[-m-1] & & \\ & \swarrow & \nwarrow & & \\ & L' & & L_1 & \\ & \downarrow & & \downarrow & \\ \text{Res} & & \text{Res} & & \\ \downarrow & & \downarrow & & \\ \bigoplus_{z \in D_1} (\mathcal{V}_n)_z^{\Gamma_z}[-m] & \xrightarrow{\quad} & \bigoplus_{z \in D_1} (\mathcal{V}_n)_z^{\Gamma_z}[-m] & & \\ & \swarrow & \nwarrow & & \\ & \mathbb{Q}_{p^{\text{unr}}} & & \mathbb{Q}_{p^{\text{unr}}} & \\ & \downarrow & & \downarrow & \\ & 0 & & 0 & \end{array}$$

As defined in Lemma 5.3, we have a Frobenius-monodromy equivariant splitting of the extension corresponding to  $L_1$  given by (5.41). We cannot hope to define such a splitting for the extension corresponding to  $L'$  since the  $(\varphi, N)$ -module  $H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)[-m-1]$  contains a non-trivial submodule with trivial monodromy  $N = 0$ . In fact,  $H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)[-m-1]$  is obtained as an extension of  $H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1]$  by  $\bigoplus_{z \in D_2} (\mathcal{V}_n)_z^{\Gamma_z}[-m-1]$ . As in Lemma 2.2 and in the discussion that follows, we construct a  $\varphi$ -equivariant splitting of the surjective map  $\text{Res}$

$$(5.54) \quad s : \bigoplus_{z \in D_1} (\mathcal{V}_n)_z[1] \rightarrow H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)$$

that we can describe as follows. Observe that  $\bigoplus_{z \in D_1} (\mathcal{V}_n)_z^{\Gamma_z}[1]$  is isotrivial of slope  $m+1$  since Frobenius acts by  $\det^{\otimes m}$  on  $V_n\{m\}$  and trivial monodromy. See Subsection 3.6. For every  $Q \in \bigoplus_{z \in D_1} (\mathcal{V}_n)_z[1]$ , define  $s(Q) \in H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)$  to be the class  $\omega_Q$  satisfying

- $\text{Res}(\omega_Q) = Q$ ,
- $\varphi(\omega_Q) = p^{m+1}\omega_Q$ .

Such an element is unique because if  $\omega, \omega'$  are two elements satisfying such properties, then their difference lie in

$$\omega - \omega' \in \iota(H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)_{m+1})^{\varphi=p^{m+1}}.$$

Observe that as  $\varphi$ -modules we have a decomposition

$$(5.55) \quad H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n) = H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) \oplus \bigoplus_{z \in D_2} (\mathcal{V}_n)_z^{\Gamma_z}$$

and in particular

$$(5.56) \quad H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)_{m+1} = H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)_{m+1}$$

concluding the unicity by then noting that the

$$(5.57) \quad (H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)_{m+1})^{\varphi=p^{m+1}, N=0} = \{0\}.$$

By the commutativity of the diagram (5.53) and the compatibility of the Frobenius equivariant splitting, we have that the map

$$(5.58) \quad H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)[-m-1] \rightarrow H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n)[-m-1]$$

sends  $s(\text{cl}_1(1))$  to the cohomology class  $\omega$  satisfying

- (i)  $\omega \in H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n)_{m+1}$
- (ii)  $\text{Res}(\omega) = Q_1$ ,
- (iii)  $N(\omega) = 0$

that is the image of 1 under the Frobenius-monodromy equivariant map (5.45). In particular, observe that we can describe the element  $\omega_Q$  in a different equivalent way. From the fact that  $H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)_{m+1} \cong C_{U_1}(\mathcal{V}_n)^\Gamma \cong H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n)_{m+1}$  by Proposition 4.4 we have the following diagram

$$(5.59) \quad \begin{array}{ccccc} C_{U_1}(\mathcal{V}_n)^\Gamma & \xlongequal{\quad} & C_{U_1}(\mathcal{V}_n)^\Gamma & & \\ I_{U_1} \uparrow \scriptstyle \text{---} & & I_{U_1} \uparrow \scriptstyle \text{---} & & \\ H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n) & \xrightarrow{j} & H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n) & \longrightarrow & 0 \\ \uparrow \scriptstyle \text{---} & & \uparrow \scriptstyle \text{---} & & \\ H^1(\Gamma, \mathcal{K}_{D_2}^\bullet(\mathcal{V}_n)) & \longrightarrow & H^1(\Gamma, \mathcal{V}_n) & \longrightarrow & 0 \end{array}$$

where the maps are Frobenius equivariant. By the commutativity of the previous diagram we can define  $s(\text{cl}_1(1))$  to be the image of the element  $\omega_{Q_1} \in H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n)_{m+1}$  of Lemma 5.3 under the Frobenius splitting induced by the top commutative square in (5.59). It's important to recall that the  $\omega_{Q_1}$  has trivial monodromy in  $H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n)$  but  $s(\text{cl}_1(1))$  can acquire it in  $H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)$  justifying once again the distinction between the Bloch-Kato exponential maps.

Consider now the push-out of (5.53) along the map  $\text{cl}_2^\vee : \bigoplus_{z \in D_2} (\mathcal{V}_n)_z^{\Gamma_z}[-m-1] \rightarrow \mathbb{Q}_{p^{\text{unr}}}[-1]$

$$(5.60) \quad \begin{array}{ccccccc} 0 \rightarrow \bigoplus_{z \in D_2} (\mathcal{V}_n)_z^{\Gamma_z}[-m-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)[-m-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] & \longrightarrow & 0 \\ \downarrow & \searrow & \downarrow & \searrow & \downarrow & \searrow & \\ 0 & \longrightarrow & \mathbb{Q}_{p^{\text{unr}}}[-1] & \longrightarrow & L_2^\vee[-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \rightarrow 0 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \\ 0 \rightarrow \bigoplus_{z \in D_2} (\mathcal{V}_n)_z^{\Gamma_z}[-m-1] & \longrightarrow & L' & \longrightarrow & L_1 & \longrightarrow & 0 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \\ 0 & \longrightarrow & \mathbb{Q}_{p^{\text{unr}}}[-1] & \longrightarrow & E & \longrightarrow & L_1 \longrightarrow 0 \end{array}$$

Observe that we have a Frobenius stable decomposition of  $L$  that is given by

$$(5.61) \quad L = \mathbb{Q}_{p^{\text{unr}}}[-1] \oplus H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \oplus \mathbb{Q}_{p^{\text{unr}}}$$

and the element  $s(\text{cl}_1(1))$  previously considered lands in the  $\mathbb{Q}_{p^{\text{unr}}}$ -part. We now observe that under the inclusion

$$(5.62) \quad \iota : H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)[-m-1] \rightarrow L'$$

we have an isomorphism of  $\mathbb{Q}_{p^{\text{unr}}}$ -vector spaces given by

$$(5.63) \quad H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n) / \text{Fil}^{m+1} \cong L' / \text{Fil}^0(L')$$

that means that there exists a class  $\eta_L \in H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)$  satisfying the following property

$$(5.64) \quad \iota(\eta_L) \equiv s(\text{cl}_1(1)) \pmod{\text{Fil}^0(L')}.$$

We can then define  $\tilde{\eta}_L$  to be the image of  $\eta_L$  under the surjection map

$$(5.65) \quad H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n) / \text{Fil}^{m+1} \rightarrow L_2^\vee[-1] / \text{Fil}^0(L_2^\vee[-1]).$$

The RHS of the last equation is isomorphic to  $(\text{Fil}^{m+1} L_2)^\vee$  under Poincaré duality. Recall that  $L_2$  injects into

$$(5.66) \quad \iota_{L_2} : L_2 \hookrightarrow H_{\text{dR}}^1(X_\Gamma - D_{2,\Gamma}, \mathcal{V}_n)[-m-1]$$

as filtered  $(\varphi, N)$ -modules. Recall that we have a Poincaré pairing (4.11)

$$(5.67) \quad H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)[-m-1] \otimes H_{\text{dR}}^1(X_\Gamma - D_{2,\Gamma}, \mathcal{V}_n)[-m] \rightarrow \mathbb{Q}_{p^{\text{unr}}}$$

This implies that element  $\tilde{\eta}_L$  can be identified with the functional

$$(5.68) \quad \tilde{\eta}_L = (\tilde{\eta}_L, -) = (\eta_L, \iota_{L_2}(-)) \in (\text{Fil}^0 L_2)^\vee.$$

Under the Bloch-Kato exponential map we can then identify the class  $[E]$  with

$$(5.69) \quad [L] = (N(s(\text{cl}_1(1))), (\eta_L, \iota_{L_2}(-))) \in \mathbb{Q}_p \oplus (\text{Fil}^0 L_2)^\vee$$

concluding the proof.  $\square$

Recall from Proposition 4.1 that we have a decomposition of the de Rham cohomology by

$$(5.70) \quad H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) = \text{Fil}^{m+1} \oplus H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)_m.$$

This will be our choice of decomposition of the de Rham realization needed to define the  $p$ -adic height pairing.

**Definition 5.2.** For every  $Q \in \bigoplus_{z \in D_i} (\mathcal{V}_n)_{z^{\Gamma_z}}^{\Gamma_z}$  define the Coleman-Gross differential associated to  $Q$  to be the unique class

$$(5.71) \quad \Psi_Q \in \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)$$

such that  $\text{Res}(\Psi_Q) = Q$  and  $N(\Psi_Q) = 0$ .

**Theorem 5.1.** The local  $p$ -adic height pairing of the biextension  $[E]$  obtained from the class maps  $\text{cl}_i$  is given by

$$(5.72) \quad h(e_1, e_2)_{E,v} = -\text{Tr}_{\mathbb{Q}_{p^2}/\mathbb{Q}_p}(\eta_L, \Psi_{Q_2})$$

where  $\eta_L \in H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)$  is the class described in Lemma 5.5.

*Proof.* Recall from Subsection 5.3 that the local  $p$ -adic height of the class of a biextension  $[E] \in H_{\text{st}}^1(\mathbb{Q}_{p^2}, E_2)$  is defined by

$$(5.73) \quad h(e_1, e_2)_{E,v} = l_v(w([E_v])).$$

where  $w$  is the splitting of the bottom exact sequence

$$(5.74) \quad \begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Q}_p \oplus \mathbb{Q}_{p^2} & \longrightarrow & \mathbb{Q}_p \oplus \text{DR}(E_2)/\text{Fil}^0 & \longrightarrow & \text{DR}(V)/\text{Fil}^0 \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & H_{\text{st}}^1(\mathbb{Q}_{p^2}, \mathbb{Q}_p(1)) & \longrightarrow & H_{\text{st}}^1(\mathbb{Q}_{p^2}, E_2) & \longrightarrow & H_{\text{st}}^1(\mathbb{Q}_{p^2}, V) \longrightarrow 0 \end{array}$$

$\swarrow \quad \quad \quad \nwarrow$   
 $w \quad \quad \quad w'$

where the  $p$ -adic Galois representations are given by

$$(5.75) \quad V = H_{\text{ét}}^1(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n)^G(m+1)$$

and  $E_2$  is the push-out of the dual of the extension

$$(5.76) \quad 0 \rightarrow H_{\text{ét}}^1(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow H_{\text{ét}}^1((X_M - D_2) \otimes \bar{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(m) \rightarrow 0$$

under the class map  $\text{cl}_2$ . The map  $w$  is induced by an explicit splitting of the upper exact sequence. Recall that  $\mathbb{Q}_{p^2}$  is mapped isomorphically to itself through the first horizontal map (5.17). Furthermore, we can identify under Poincaré duality the quotients in the upper exact sequence as

$$(5.77) \quad 0 \longrightarrow \mathbb{Q}_{p^2} \longrightarrow (\text{Fil}^0 \text{DR}(E_2))^\vee \longrightarrow (\text{Fil}^{m+1} \text{DR}(V))^\vee \longrightarrow 0.$$

In order to study and explicitly describe the unique splitting of this exact sequence, we now pass to the maximal unramified extension  $\mathbb{Q}_{p^{\text{unr}}}$ . In this way, we can apply the explicit description in terms of the analytic quotient  $X_\Gamma$ . In particular we obtain a sequence

$$(5.78) \quad 0 \longrightarrow \mathbb{Q}_{p^{\text{unr}}} \longrightarrow (\text{Fil}^0 L_2)^\vee \longrightarrow (\text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n))^\vee \longrightarrow 0.$$

inducing a dual exact sequence contained in the usual Gysin extension

$$(5.79) \quad \begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Fil}^{m+1} H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n) & \xrightarrow{\quad} & \mathrm{Fil}^0 E_2 & \longrightarrow & \mathbb{Q}_{p^{\mathrm{unr}}} \longrightarrow 0 \\ & & \parallel & \nwarrow w^{\vee} & \downarrow \iota_{E_2} & & \downarrow \mathrm{cl}_2 \\ 0 & \longrightarrow & \mathrm{Fil}^{m+1} H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n) & \xrightarrow{\iota} & \mathrm{Fil}^{m+1} H_{\mathrm{dR}}^1(X_\Gamma - D_2, \mathcal{V}_n) & \xrightarrow{\mathrm{Res}} & \bigoplus_{z \in D_2} \mathrm{Fil}^m(\mathcal{V}_n)_{\Gamma_z} \longrightarrow 0. \end{array}$$

By Lemma 5.5 we have that the class of  $[L]$ , under the Bloch-Kato isomorphism, is sent to

$$(5.80) \quad (N_L, (\eta_L, \iota_{L_2}(-))) \in \mathbb{Q}_p \oplus (\mathrm{Fil}^0 L_2)^\vee$$

with  $\eta_L \in H_{\mathrm{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)$  satisfying prescribed properties of the Lemma. In Subsection 5.3 we have described a section of the top exact sequence that Nekovář constructs in [Nek93]. Explicitly this is given by

$$(5.81) \quad w^{\vee} : \mathrm{Fil}^0 L_2 \longrightarrow L_2 \xrightarrow{\alpha} H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n) \longrightarrow \mathrm{Fil}^{m+1} H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n)$$

where the first map is the canonical inclusion,  $\alpha$  is the unique  $(\varphi, N)$ -equivariant splitting of the canonical inclusion and the last morphism is the projection according to the decomposition (5.70). This map exists in more generality as the composition of the equivalent maps

$$(5.82) \quad \begin{array}{ccccccc} \mathrm{Fil}^0 L_2 & \longrightarrow & L_2 & \xrightarrow{\alpha} & H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n) & \longrightarrow & \mathrm{Fil}^{m+1} H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n) \\ \downarrow & & \downarrow & & \parallel & & \parallel \\ \mathrm{Fil}^{m+1} H_{\mathrm{dR}}^1(X_\Gamma - D_2, \mathcal{V}_n) & \longrightarrow & H_{\mathrm{dR}}^1(X_\Gamma - D_2, \mathcal{V}_n) & \longrightarrow & H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n) & \longrightarrow & \mathrm{Fil}^{m+1} H_{\mathrm{dR}}^1(X_\Gamma, \mathcal{V}_n). \end{array}$$

The Frobenius-monodromy equivariant splitting is given by Lemma 5.3. Let  $\omega_{Q_2} = s_2(Q_2)$  be the image of the element  $Q_2$  under the  $(\phi, N)$ -equivariant splitting. The splitting of the dual filtered exact sequence (5.79) can be then given as

$$(5.83) \quad \mathbb{Q}_{p^{\mathrm{unr}}} \rightarrow \mathrm{Fil}^0 E_2$$

$$(5.84) \quad 1 \mapsto \Psi_{Q_2} = \omega_{Q_2} - (\iota \circ w^{\vee})(\omega_{Q_2})$$

where  $\Psi_{Q_2}$  is then Coleman-Gross differential associated to  $Q_2$ . We can then see that the splitting of the exact sequence (5.74) is given by

$$(5.85) \quad \begin{aligned} \mathbb{Q}_p \oplus DR(E_2)/\mathrm{Fil}^0 &\rightarrow \mathbb{Q}_p \oplus \mathbb{Q}_{p^2} \\ (n, (\omega, \iota_{L_2}(-))) &\mapsto (n, (\omega, \Psi_{Q_2})). \end{aligned}$$

Under assumption (5.1)  $l_v = \mathrm{Tr}_{\mathbb{Q}_{p^2}/\mathbb{Q}_p} \circ \log$  and the Kummer isomorphism (5.21), we conclude that the  $p$ -adic height is given by the projection of the second component of (5.85) composed with the trace map. This concludes the proof.  $\square$

We conclude this part giving an explicit description of the Poincaré pairing, and then of the height pairing, in some favorable conditions.

**Theorem 5.2.** *Consider the class maps  $\mathrm{cl}_i : \mathbb{Q}_{p^{\mathrm{unr}}} \rightarrow \bigoplus_{z \in D_i} (\mathcal{V}_n)_{\Gamma_z}^{\Gamma_z} [1]$  with  $\mathrm{cl}_i(1) = Q_i$  and suppose that  $D_1$  reduces to a unique vertex and  $\sum_{z \in D_1} Q_{1,z}/\#\Gamma_z = 0$ . The local  $p$ -adic height of the biextension  $[E]$  can be computed by*

$$(5.86) \quad h(e_1, e_2)_{E,v} = \mathrm{Tr}_{\mathbb{Q}_{p^2}/\mathbb{Q}_p} \left( \int_{D_1} (\Psi_{Q_2}, Q_1) dz \right) = \mathrm{Tr}_{\mathbb{Q}_{p^2}/\mathbb{Q}_p} \left( \sum_{z \in D_1} \frac{(F_{\omega_{Q_2}}(z), Q_{1,z})_{\Gamma}}{\#\Gamma_z} \right).$$

*Proof.* By Lemma 4.1 We have

$$(5.87) \quad (\eta_L, \Psi_{Q_2}) = (P_{D_2}(\eta_L), I_{U_2}(\Psi_{Q_2}))_{\Gamma} - (I(\eta_L), P(\Psi_{Q_2}))_{\Gamma}.$$

Since we have a decomposition  $H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n) = \text{Fil}^{m+1} \oplus H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)_m$  and modifying  $\eta_L$  up to  $\text{Fil}^{m+1}$  will not change the cup product. We can assume

$$(5.88) \quad \eta_L \in H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)_m = \ker(I)$$

simplifying the previous expression to

$$(5.89) \quad (\eta_L, \Psi_{Q_2}) = (P(\eta_L), I_{U_2}(\Psi_{Q_2}))_\Gamma.$$

Consider now  $\Psi_{Q_1} \in \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)$  the element that, under condition (i) of Lemma 5.5, is given by

$$(5.90) \quad \Psi_{Q_1} = \omega_{Q_1} - \iota(\eta_L).$$

This element satisfies the following properties:

- $\text{Res}(\Psi_{Q_1}) = \text{Res}(\omega_{Q_1}) = Q_1$ ,
- $P(\Psi_{Q_1}) = -P(\eta_L)$ ,
- $I_{U_1}(\Psi_{Q_1}) = I_{U_1}(\omega_{Q_1})$ ,

and it is the Coleman-Gross differential associated to the differential  $Q_1$  of Definition 5.2 under the identification

$$(5.91) \quad \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n) = \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n).$$

Since the Poincaré pairing is orthogonal with respect to the filtration  $\text{Fil}^{m+1}$  we obtain

$$(5.92) \quad 0 = (\Psi_{Q_1}, \Psi_{Q_2}) = (P(\Psi_{Q_1}), I_{U_2}(\Psi_{Q_2}))_\Gamma - (I_{U_1}(\Psi_{Q_1}), P(\Psi_{Q_2}))_\Gamma =$$

$$(5.93) \quad = -(P(\eta_L), I_{U_2}(\Psi_{Q_2}))_\Gamma - (I_{U_1}(\omega_{Q_1}), P(\Psi_{Q_2}))_\Gamma$$

where we have used the identification

$$(5.94) \quad \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_2, \mathcal{V}_n) = \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_2, D_1, \mathcal{V}_n).$$

This simplifies the initial pairing (5.87) to the computation

$$(5.95) \quad (\eta_L, \Psi_{Q_2}) = -(I_{U_1}(\omega_{Q_1}), P_{D_1}(\Psi_{Q_2}))_\Gamma.$$

Explicitly we have

$$(5.96) \quad P_{D_1}(\Psi_{Q_2}) = ([F\omega_{Q_2}], (F\omega_{Q_2}(z))_{z \in D_1}),$$

$$(5.97) \quad I_{U_1}(\omega_{Q_1})(\gamma_z)_{z \in D_1} = ((\text{Res}_e(\omega_{Q_1}))_{e \in \mathfrak{E}}, (\text{Res}_{\gamma_z(z)}(\omega_{Q_1}))_{z \in D_1}).$$

It's important to note that from the construction of Lemma 5.5 we have that the element  $\omega_{Q_1}$  maps under  $j$  to the unique element in  $H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n)_{m+1} = \ker(P) \cong C_{U_1}(V_n)^\Gamma$  with trivial monodromy

$$(5.98) \quad \begin{array}{ccc} C_{U_1}(V_n)^\Gamma & \xlongequal{\quad} & C_{U_1}(V_n)^\Gamma \\ I_{U_1} \uparrow \text{dotted} & & I_{U_1} \uparrow \text{dotted} \\ H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n) & \xrightarrow{j} & H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n) \longrightarrow 0 \\ \uparrow \text{dotted } P_{D_2} & & \uparrow \text{dotted } P \\ H^1(\Gamma, \mathcal{K}_{D_2}^\bullet(V_n)) & \longrightarrow & H^1(\Gamma, V_n) \longrightarrow 0 \end{array}$$

where the dotted arrows are Frobenius equivariant.

Fix  $\Gamma' \subseteq \Gamma$  a free subgroup with generators  $\gamma_1, \dots, \gamma_g$ . Let  $\mathcal{F}$  be a good fundamental domain for  $\mathcal{T}_p/\mathcal{F}$  and  $b_1, \dots, b_g, c_1, \dots, c_g$  its free edges in the sense of de Shalit [Sha89]. The pairing is then given by the explicit formula (4.13)

$$(5.99) \quad (\eta_L, \Psi_{Q_2}) = -\frac{1}{[\Gamma : \Gamma']} \sum_{i=1}^g ([F\omega_{Q_2}](\gamma_i), (\text{Res}_{c_i}(\omega_{Q_1})))_\Gamma - \sum_{z \in D_1} \frac{(F\omega_{Q_2}(z), Q_{1,z})_\Gamma}{\#\Gamma_z}.$$

Observe now that under the condition that  $D_1$  reduce to only one vertex and

$$(5.100) \quad \sum_{z \in D_1} Q_{1,z} / \#\Gamma_z = \sum_{z \in D_1} \text{Res}_z(\omega_{Q_1}) / \#\Gamma_z = 0,$$

the element  $(\text{Res}_e(\omega_{Q_1}))_{e \in \mathcal{E}} \in C^1(V_n)$  lies in  $C^1_{\text{har}}(V_n)^\Gamma$  by (4.8). Since  $N(j(\omega_{Q_1})) = 0$  then the harmonic part has to be trivial, concluding  $(\text{Res}_e(\omega_{Q_1}))_{e \in \mathcal{E}} = 0$ . The first term of the pairing (5.99) is then trivial giving the simplified expression

$$(5.101) \quad (\eta_L, \Psi_{Q_2}) = - \sum_{z \in D_1} \frac{(F\omega_{Q_2}(z), Q_{1,z})_\Gamma}{\#\Gamma_z}.$$

As a sanity check, conclude observing that the formula is invariant under rescaling of the Coleman integral of  $\omega_{Q_2}$  since the residues of  $Q_1$  add up to 0.  $\square$

**Remark 5.1.** *The explicit computation in the previous proof shows that the height pairing is symmetric. In particular, we have used the fact that the pairing between the Coleman differentials  $\Psi_{Q_1}, \Psi_{Q_2}$  is 0 in  $H^1_{\text{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n) \times H^1_{\text{dR}}(X_\Gamma - D_2, D_1, \mathcal{V}_n)$  as in (5.92) leading to the equality (5.93)*

$$(5.102) \quad (P(\Psi_{Q_1}), I_{U_2}(\Psi_{Q_2}))_\Gamma = (I_{U_1}(\Psi_{Q_1}), P(\Psi_{Q_2}))_\Gamma.$$

**5.7. Local heights away from  $p$ .** Let  $V = H^1_{\text{et}}(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n)^G(m+1)$  be the  $p$ -adic Galois representation. We recall from [Bry89], [Sch94] and [Nek95] the known results on the local heights away from  $p$ .

**Lemma 5.6.** *Let  $F$  be a finite extension of  $\mathbb{Q}_l$  for  $l \neq p$  then*

$$(5.103) \quad H^i(F, V) = 0 \quad \text{for } i = 0, 1, 2.$$

*Proof.* Let  $\mathbb{T}$  be the subalgebra of  $\text{End}(M_k(X, \mathbb{Q}))$  generated by the Hecke operators  $T_l$  for  $l \nmid N$  and  $U_l$  for  $l \mid N$ . The  $p$ -adic representation  $V$  is a free  $\mathbb{T}_{\mathbb{Q}_p}$ -module of rank 2 that decomposes in eigenspaces  $V_p(f)$  for  $f \in M_k(X, \mathbb{Q})$  eigenforms. Under the Jacquet-Langlands correspondence, Iovita and Spiess [IS03, Lemma 5.8] proved that these eigenspaces are isomorphic to  $p$ -adic representations  $V_p(f_\infty)$  coming from the associate elliptic modular form  $f_\infty$ . The proof of the Lemma follows then from [Nek95, Lemma 2.12] in the elliptic modular case.  $\square$

Let now  $\mathcal{X}$  be model for  $X_M \otimes K_v^{\text{unr}}$  i.e. a proper flat scheme over  $\mathcal{O}_{K_v}^{\text{unr}}$  such that

$$(5.104) \quad \mathcal{X} \times \text{Spec}(K_v^{\text{unr}}) \cong X_M \otimes K_v^{\text{unr}}.$$

Consider once again two reduced effective divisors  $D_1, D_2$  defined over  $K_1, K_2$  with disjoint support. Consider  $\underline{D}_1, \underline{D}_2$  the Zariski closure of  $D_1, D_2$  in the model. Assume that both divisors are contained in the regular locus  $\mathcal{X}_M^{\text{reg}} \subset \mathcal{X}_M$ . Consider

$$(5.105) \quad \text{cl}_i : \mathbb{Q}_p \rightarrow \bigoplus_{x \in D_i} (\mathbb{L}_n)_x^{G_x}(m)$$

and let  $Q_i = \text{cl}_i(1)$ . The maps  $\text{cl}_i$  induce two extensions  $e_1, e_2$  and the natural biextension  $E$  splits as  $p$ -adic local Galois representation at  $v$  as  $E_v = V_v \oplus U_v$ . Assume now that there exists cohomology classes on the model  $Q_i \in H^2_{\text{et}, \underline{D}_i}(\mathcal{X}, \mathbb{L}_n(m+1))^G$  restricting to

$$(5.106) \quad Q_i \in H^2_{\text{et}, D_i}(X \otimes K_v^{\text{unr}}, \mathbb{L}_n(m+1))^G \cong H^0_{\text{et}}(D_i, \iota_{D_i*} \mathbb{L}_n(m))^G$$

under the pullback to the generic fiber. We can then express the height pairing in terms of these classes.

**Lemma 5.7.** *Assume that the class  $Q_i$  has image zero in  $H^2_{\text{et}}(\mathcal{X}, \mathbb{L}_n(m+1))^G$  under the connecting morphism of the long exact sequence of relative cohomology. Then*

$$(5.107) \quad h(e_1, e_2)_{E,v} = -l_v(\pi_v) \cdot \text{Tr}(Q_1 \cup Q_2)$$

where the cup product is an element of  $H^4_{\text{et}, c}(\mathcal{X}, \mathbb{Q}_p(2))^G$  and  $\text{Tr}$  is the trace map

$$(5.108) \quad \text{Tr} : H^4_{\text{et}, c}(\mathcal{X}, \mathbb{Q}_p(2))^G \rightarrow \mathbb{Q}_p.$$

*Proof.* See [Bry89, Proposition 4.10] and [Sch94]. As pointed out by [Nek95, II.2.16], there is a minus sign correction that does not appear in the work of [Bry89].  $\square$

We would like to explicitly characterize the cup product of the extended cohomology class in terms of intersection of divisors. To do this, assume that  $Q_i$  extend to sections

$$(5.109) \quad \underline{Q}_i \in H_{\text{ét}}^0(D_i, \iota_{D_i,*} \mathbb{L}_n(m))^G.$$

If  $y, z$  are two points of  $\underline{D}_1, \underline{D}_2$ , following [Nek95], we define the *local intersection index* at a point  $x$  in  $\mathcal{X}$  to be

$$(5.110) \quad (\underline{Q}_1, \underline{Q}_2)_x := (\underline{y}, \underline{z})_x (\underline{Q}_{1,x}, \underline{Q}_{2,x})$$

where  $(\underline{y}, \underline{z})_x$  denotes the usual intersection number on  $\mathcal{X}^{\text{reg}}$ .

**Theorem 5.3.** *Assume that the cohomology classes*

$$(5.111) \quad \mathcal{Q}_i = \underline{Q}_i \cup \text{cl}(\underline{D}_i) \in H_{\text{ét}, \underline{D}_i}^2(\mathcal{X}, \mathbb{L}_n(m+1))^G$$

*satisfy the assumption of Lemma 5.7. The local height pairing is given by*

$$(5.112) \quad h(e_1, e_2)_{E,v} = -l_v(\pi_v) \cdot \sum_{x \in \underline{D}_1 \cap \underline{D}_2} (\underline{Q}_1, \underline{Q}_2)_x$$

*Proof.* See [Nek95, II.2.16] and [Sch94], [Bry89, Theorem 4.3]. □

**Remark 5.2.** *The picture simplifies when the global cohomology group  $H_{\text{ét}}^2(\mathcal{X}, \mathbb{L}_n(m+1)) = 0$ . In this case, the assumption of Lemma 5.7 is satisfied implying that the local height computation simplifies to the expression (5.112). Under smooth proper base change, this is always the case when  $v$  is a place of good reduction. This is indeed the same obstacle that the explicit local height with trivial coefficients has when dealing with bad reduction. In that case, a correction by a vertical divisor needs to be introduced.*

To study the global cohomology we will adapt some of the arguments from [KM85, 14.5.5]. Consider  $l \mid \text{disc}(B)$  a prime different from  $p$  and let  $v$  a place above  $l$ . Thanks to Čerednik-Drinfeld Theorem 3.2 we can explicitly characterize the bad reduction of the special fiber  $\bar{C} = \mathcal{X} \times \text{Spec}(\mathbb{F}_l)$

$$(5.113) \quad \begin{array}{ccc} \bar{C} & \xrightarrow{j} & \mathcal{X} \\ \downarrow f & & \downarrow g \\ \text{Spec}(\bar{\mathbb{F}}_l) & \hookrightarrow & \text{Spec}(\mathcal{O}_v^{\text{unr}}). \end{array}$$

In particular, the irreducible components  $C_i$  of  $\bar{C}$  are isomorphic to  $\mathbb{P}_{\bar{\mathbb{F}}_l}^1$ . By the proper base change theorem and the property of strictly henselian bases, we have

$$(5.114) \quad H_{\text{ét}}^2(\mathcal{X}, \mathbb{L}_n(m+1)) \cong (R^2 g_* \mathbb{L}_n(m+1))_v \cong H_{\text{ét}}^2(\bar{C}, j^* \mathbb{L}_n(m+1)).$$

Consider now the normalization of the special fiber

$$(5.115) \quad \nu : \coprod_j C_j \rightarrow \bar{C},$$

this gives us a long exact sequence in cohomology that identifies the second étale cohomology

$$(5.116) \quad H_{\text{ét}}^2(\bar{C}, \mathbb{L}_n(m+1)) \cong \bigoplus_j H_{\text{ét}}^2(C_j, \mathbb{L}_n(m+1)).$$

Under Poincaré duality, the last term can be identified with

$$(5.117) \quad \bigoplus_j H_{\text{ét}}^0(C_j, \mathbb{L}_n(m)) \cong \bigoplus_j \mathbb{L}_n(m)_{z_j}$$

where the  $z_j$  are points on distinct components of  $C_j$ . The following Proposition will help us simplify the computation of the local height pairing in the case of Heegner cycles.

**Proposition 5.1.** *Let  $\underline{Q}_i \in H_{\text{ét}}^0(D_i, \iota_{D_i,*} \mathbb{L}_n(m))^G$  the extended cohomology class supported on the divisor  $\underline{D}_i$ . Supposed that the specialization on ever irreducible components  $C_j$  satisfies the property*

$$(5.118) \quad \sum_{\bar{x} \in C_j} \underline{Q}_{i,\bar{x}} / \# \Gamma_{z_j} = 0.$$

*Then the cohomology class  $\mathcal{Q}_i$  defined by (5.111) satisfies the assumption of Lemma 5.7.*

*Proof.* Under the previous characterization the connecting map in relative cohomology is given by (5.119)

$$H_{\text{ét}}^1(\mathcal{X} - \underline{D}_i, \mathbb{L}_n(m+1))^G \rightarrow H_{\text{ét}, \underline{D}_i}^2(\mathcal{X}, \mathbb{L}_n(m+1))^G \rightarrow H_{\text{ét}}^2(\mathcal{X}, \mathbb{L}_n(m+1))^G \cong \bigoplus_j \mathbb{L}_n(m)_{z_j}^{\Gamma_{z_j}}$$

where the right most map sends

$$(5.120) \quad \mathcal{Q}_i \mapsto \left( \sum_{\bar{x} \in C_j} \frac{\mathcal{Q}_{i,x}}{\#\Gamma_{z_j}} \right)_j.$$

From this we deduce that the cohomology class is map to zero exactly when the weighted sum of the stalks is 0.  $\square$

**5.8. Hecke action on extensions.** Let us briefly recall the Hecke action on the motive  $h(\mathcal{A}^m)$ . For an integer  $l$  prime to  $M$  and  $N$ , consider  $X_M(l)$  the moduli space of abelian surfaces over  $\mathbb{Q}$  parametrizing abelian surfaces  $(A, i, C, v, \lambda)$  where the first triple is the data parametrized by  $X_M$  by Definition 3.3 and  $\lambda : A \rightarrow A'$  is an isogeny of degree  $l$ . There are two finite maps

$$(5.121) \quad \begin{array}{ccc} & X_M(l) & \\ \pi_1 \swarrow & & \searrow \pi_2 \\ X_M & & X_M \end{array}$$

given by the moduli action

$$(5.122) \quad \pi_1(A, i, C, v, \lambda) = (A, i, C, v), \quad \pi_2(A, i, C, v, \lambda) = (A', \lambda_*(i), \lambda_*(C), \lambda_*(v)).$$

At the level of the universal abelian surface we have a commutative diagram of the form

$$(5.123) \quad \begin{array}{ccccccc} \mathcal{A}^m & \xleftarrow{\phi_1} & \mathcal{A}^m \times_{X_M, \pi_1} X_M(l) & \xrightarrow{\psi} & \mathcal{A}^m \times_{X_M, \pi_2} X_M(l) & \xrightarrow{\phi_2} & \mathcal{A}^m \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ X_M & \xleftarrow{\pi_1} & X_M(l) & \xlongequal{\quad} & X_M(l) & \xrightarrow{\pi_2} & X_M \end{array}$$

where the map  $\psi$  is induced by the universal isogeny and the left and right diagram are cartesian. We can then define the  $m$ -th Hecke correspondence to be the graph  $\Gamma_{\phi_1, \phi_2 \psi}$  of

$$(5.124) \quad (\phi_1, \phi_2 \circ \psi) : \mathcal{A}^m \times_{X_M, \pi_1} X_M(l) \rightarrow \mathcal{A}^m.$$

Fix now  $D$  a reduced divisor on  $X_M$  defined over a number field  $K$  and a choice of a map  $\text{cl} : \mathbb{Q}_p \rightarrow \bigoplus_{x \in D} \mathbb{L}_n(m)^{G_x}$ . Consider now an extension of the  $p$ -adic Galois representation  $V = H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G$  of  $G_K = \text{Gal}(\bar{K}/K)$

$$(5.125) \quad 0 \rightarrow V \rightarrow H_{\text{ét}}^1((X_M - |D|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(m) \rightarrow 0$$

An Hecke operator  $T_l$  then sends this sequence to the exact sequence

$$(5.126) \quad 0 \rightarrow V \rightarrow H_{\text{ét}}^1((X_M - |D'|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow \bigoplus_{x \in D'} (\mathbb{L}_n)_x^{G_x}(m) \rightarrow 0$$

with  $D' = \pi_1 \circ \pi_2^{-1}(D)$ .

**Lemma 5.8.** Consider  $x$  a closed point of  $X_M$ , let  $Q \in (\mathbb{L}_n)_x^{G_x}(m)$ . Let  $(A, i, C, v)$  be the data associated to  $x$ . Then the action of the Hecke operator  $T_l$  on  $Q$  is given by

$$(5.127) \quad T_l Q = \sum_{\substack{\mu: A' \rightarrow A \\ \deg(\mu)=l}} \mu^*(Q) \in \bigoplus_{y \in \pi_1 \pi_2^{-1}\{x\}} (\mathbb{L}_n)_y^{G_y}(m).$$

*Proof.* This follows verbatim from the modular curve computation treated by Nekovář [Nek95, II.2.10].  $\square$

The previous lemma extends by linearity to a divisor  $D$  and a class map  $\text{cl} : \mathbb{Q}_p \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(m)$ . Fixing this data, consider now  $T$  a linear combination of Hecke operators acting by 0 on the  $p$ -adic Galois representation  $V$  of  $G_K$ , this gives us a diagram

(5.128)

$$\begin{array}{ccccccc} 0 & \longrightarrow & V & \longrightarrow & H_{\text{ét}}^1((X_M - |D|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G & \longrightarrow & \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(m) \longrightarrow 0 \\ & & \downarrow T=0 & & \downarrow & \nearrow & \downarrow \\ 0 & \longrightarrow & V & \longrightarrow & H_{\text{ét}}^1((X_M - |D'|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G & \longrightarrow & \bigoplus_{x \in D'} (\mathbb{L}_n)_x^{G_x}(m) \longrightarrow 0 \end{array}$$

and since the action of  $T$  on  $V$  is 0, this means that the central vertical maps factors through

$$(5.129) \quad T : \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(m) \rightarrow H_{\text{ét}}^1((X_M - |D'|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G.$$

Let  $e$  the extension given by pulling back the top exact row of the previous diagram along the class map  $\text{cl}$

$$(5.130) \quad e : \quad 0 \longrightarrow V \longrightarrow E_1 \longrightarrow \mathbb{Q}_p \longrightarrow 0$$

then under the action  $T$  we can construct a split extension  ${}^T e$  given by the pullback of the bottom row of the diagram (5.128) by the composition of the class map  $\text{cl}$  with the map  $T$  given by (5.129)

(5.131)

$$\begin{array}{ccccccc} 0 & \longrightarrow & V & \longrightarrow & E_1 & \longrightarrow & \mathbb{Q}_p \longrightarrow 0 \\ & & \downarrow T=0 & & \downarrow & \nearrow & \downarrow \\ 0 & \longrightarrow & V & \longrightarrow & {}^T E_1 & \longrightarrow & \mathbb{Q}_p \longrightarrow 0 \\ & & \downarrow T=0 & & \downarrow & \nearrow & \downarrow \\ 0 & \longrightarrow & V & \longrightarrow & H_{\text{ét}}^1((X_M - |D|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G & \longrightarrow & \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(m) \longrightarrow 0 \\ & & \downarrow T=0 & & \downarrow & \nearrow & \downarrow \\ 0 & \longrightarrow & V & \longrightarrow & H_{\text{ét}}^1((X_M - |D'|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G & \longrightarrow & \bigoplus_{x \in D'} (\mathbb{L}_n)_x^{G_x}(m) \longrightarrow 0 \end{array}$$

where then we have a canonical decomposition of  $p$ -adic Galois representations  ${}^T E_1 = V \oplus \mathbb{Q}_p$ . This picture promotes to the biextension setting. Fix now two reduced divisors  $D_1, D_2$  with two corresponding class maps  $\text{cl}_i$ . Instead of using the divisor  $D_1$ , we consider the biextension given by the divisors  $D'_1$  and  $D_2$

(5.132)

$$\begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \bigoplus_{x \in D_2} (\mathbb{L}_n)_x^{G_x}(m+1) & \longrightarrow & H_{\text{ét}, |D_2|}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G & \longrightarrow & H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \bigoplus_{x \in D_2} (\mathbb{L}_n)_x^{G_x}(m+1) & \longrightarrow & H_{\text{ét}, |D_2|}^1((X_M - |D'_1|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G & \longrightarrow & H_{\text{ét}}^1((X_M - |D'_1|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \longrightarrow 0 \\ & & & & \downarrow & & \downarrow \\ & & & & \bigoplus_{x \in D'_1} (\mathbb{L}_n)_x^{G_x}(m) & \xlongequal{\quad} & \bigoplus_{x \in D'_1} (\mathbb{L}_n)_x^{G_x}(m) \\ & & & & \downarrow & & \downarrow \\ & & & & 0 & & 0 \end{array}$$

We can now take the push-out along the dual of  $\text{cl}_2$  and the pullback along the composition  $T \circ \text{cl}_1$  as described in the previous extension (5.131) obtaining

$$(5.133) \quad \begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathbf{Q}_p(1) & \longrightarrow & E_2^\vee(1) & \longrightarrow & V \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbf{Q}_p(1) & \longrightarrow & {}^T E & \longrightarrow & {}^T E_1 \longrightarrow 0 \\ & & & & \downarrow & & \downarrow \scriptstyle \tau \\ & & & & \mathbf{Q}_p & \xlongequal{\quad} & \mathbf{Q}_p \\ & & & & \downarrow & & \downarrow \\ & & & & 0 & & 0 \end{array} .$$

Pulling back the central horizontal map under the dotted section gives us a simple extension

$$(5.134) \quad u : 0 \rightarrow \mathbf{Q}_p(1) \rightarrow U \rightarrow \mathbf{Q}_p \rightarrow 0.$$

We conclude observing that under the exact sequence of global Selmer group

$$(5.135) \quad 0 \rightarrow H_{\text{st}}^1(K, \mathbf{Q}_p(1)) \rightarrow H_{\text{st}}^1(K, E_2^\vee(1)) \rightarrow H_{\text{st}}^1(K, V) \rightarrow 0$$

the class  $[{}^T E]$  is sent to the class  $[{}^T E_1] = [V \oplus \mathbf{Q}_p] = 0$  that means that the class  $[{}^T E]$  comes from  $[U] \in H_{\text{st}}^1(K, \mathbf{Q}_p(1))$  as defined in equation (5.134).

**Proposition 5.2.** *Suppose that  $D_1$  reduces to a unique vertex and  $\sum_{z \in D_1} Q_{1,z} / \# \Gamma_z = 0$ . The local  $p$ -adic height of the biextension  $[{}^T E]$  can be computed by*

$$(5.136) \quad h({}^T e_1, e_2)_{T E, v} = \text{Tr}_{K_v / \mathbf{Q}_p} \left( \int_{\pi_1 \pi_2^{-1} D_1} (\Psi_{Q_2}, T Q_1) dz \right) = \text{Tr}_{K_v / \mathbf{Q}_p} \left( \sum_{z \in \pi_1 \pi_2^{-1} D_1} \frac{(F_{\omega_{Q_2}}(z), (T Q_1)_z)_\Gamma}{\# \Gamma_z} \right).$$

*Proof.* The result follows from Theorem 5.2 observing that by linearity we can decompose the height along  $D'_1 = \pi_1 \pi_2^{-1} D_1 = D_1^{(1)} + \dots + D_1^{(r)}$  where  $D_1^{(i)}$  reduce to a unique vertex and for each  $i$  we have  $\sum_{z \in D_1^{(i)}} (T Q_1)_z / \# \Gamma_z = 0$ .  $\square$

## 6. HIGHER $p$ -ADIC GREEN FUNCTIONS

**6.1. Poincaré series.** Following [BF25], we will define Poincaré series in this section, whose periods compute heights of Heegner cycles. Let  $\tilde{z}_1, \tilde{z}_2$  be any two points belonging to  $X_\Gamma(\mathbf{Q}_{p^2})$  and define the divisor  $D_1 = (\tilde{z}_1) + (\tilde{z}_2)$ . Assume further that both  $\tilde{z}_1$  and  $\tilde{z}_2$  reduce modulo  $p$  to smooth points lying on the same irreducible component of the special fiber of  $\mathfrak{X}_\Gamma$ . It follows that there exists lifts  $z_1, z_2 \in \mathcal{H}_p \cap \mathbf{Q}_{p^2}$  of  $\tilde{z}_1$  and  $\tilde{z}_2$ , respectively, such that

$$\text{red}_p(z_1) = \text{red}_p(z_2).$$

We fix such lifts.

Fix a distinct pair of points  $w_1, w_2 \in \mathcal{H}_p$ , and  $i, j \in \mathbb{N}$  such that  $i + j = n$ , define the Poincaré series

$$(6.1) \quad \Psi_\Gamma^{i,j}(z; w_1, w_2) := \sum_{\gamma \in \Gamma} \frac{1}{j(\gamma, z)^k} \cdot \frac{w_1 - w_2}{(\gamma z - w_1)^{i+1} (\gamma z - w_2)^{j+1}}.$$

**Proposition 6.1.** *The infinite sum  $\Psi_\Gamma^{i,j}(z; z_1, z_2)$  converges to a rigid meromorphic modular form of weight  $k$  for  $\Gamma$ .*

*Proof.* To prove convergence of this power series, we introduce some preliminary definitions. For  $z \in \mathcal{H}_p$ , define

$$\mathrm{Im}(z) = \inf\{|z - t|_p \mid t \in \mathbb{Q}_p\}.$$

Given two points  $x, y \in \mathcal{H}_p$ , let  $d_{\mathcal{T}_p}(x, y)$  denote the distance on  $\mathcal{T}_p$  between the reduction vertices  $\mathrm{red}_p(x)$  and  $\mathrm{red}_p(y)$ .

**Lemma 6.1.** *If  $x, y \in \mathcal{H}_p$  reduce to different vertices of  $\mathcal{T}_p$ , then*

$$d_{\mathcal{T}_p}(x, y) = v_p \left( \frac{|x - y|_p^2}{\mathrm{Im}(x)\mathrm{Im}(y)} \right)$$

*Proof.* This is [Wei18, Lemma 2.4] □

**Lemma 6.2.** *If  $x \in \mathcal{H}_p$ , then for every  $\gamma \in \mathrm{GL}_2(\mathbb{Q}_p)$ , we have*

$$\mathrm{Im}(\gamma \cdot x) = \det(\gamma) \frac{\mathrm{Im}(x)}{|j(\gamma, x)|^2}$$

*Proof.* This is [GR96, (1.1.5) Lemma]. □

Using the lemmas above we see that if  $\mathrm{red}_p(\gamma z) \neq \mathrm{red}_p(w_1)$ , then

$$\begin{aligned} v_p(j(\gamma, z)^k (\gamma z - w_1)^{i+1} (\gamma z - w_2)^{j+1}) = \\ \frac{-1}{2} ((i+1)d_{\mathcal{T}_p}(\gamma z, w_1) + (j+1)d_{\mathcal{T}_p}(\gamma z, w_2) + v_p(\mathrm{Im}(w_1)^{i+1} \cdot \mathrm{Im}(w_2)^{j+1})) \end{aligned}$$

for all but finitely many  $\gamma \in \Gamma$ . Furthermore, since  $\Gamma$  acts discretely on  $\mathcal{H}_p$ , it follows that for each  $N \in \mathbb{N}$ , there exists only finitely many  $\gamma \in \Gamma$  such that

$$v_p(j(\gamma, z)^k (\gamma z - w_1)^{i+1} (\gamma z - w_2)^{j+1}) \geq -N.$$

Therefore, the sum converges. The modularity of the resulting function is automatic from its definition as a Poincaré series. □

The differential  $\Psi_\Gamma^{i,j}(z)(X - zY)^n dz$  descends to a global section of

$$H^0(X_\Gamma, \Omega_{X_\Gamma}^1(\log(D_1)) \otimes \mathcal{V}_n) = F_{U_1}^{m+1} \subset H_{\mathrm{dR}}^1(U_1, \mathcal{V}_n).$$

The calculation of the residues of the differential  $\Psi_\Gamma^{i,j}(z)(X - zY)^n dz$  follows precisely as in [BF25, Corollary 3.11].

**Lemma 6.3.** *For all  $i, j \geq 0$ , such that  $i + j = n$ , let*

$$\kappa_{\Gamma, w}^{i,j} := \sum_{\gamma \in \Gamma_{w_1, w_2}} \frac{1}{j(\gamma, w_1)^i j(\gamma, w_2)^j}.$$

*Then*

$$\mathrm{Res}_{z=w_1} \left( \Psi_\Gamma^{i,j}(z; w_1, w_2) (X - zY)^k dz \right) = \frac{(-1)^i \kappa_{\Gamma, w}^{i,j}}{(w_1 - w_2)^n} \binom{n}{i} (X - w_2 Y)^i (X - w_1 Y)^j.$$

**Lemma 6.4.** *The differential*

$$\Psi_\Gamma^{i,j}(z)(X - zY)^n dz$$

*is a Coleman–Gross differential (Definition 5.2).*

*Proof.* It suffices to show that

$$N \left( (\Psi_\Gamma^{i,j}(z)(X - zY)^n dz) \right) = 0.$$

Observe that  $\Psi_\Gamma^{i,j}$  can be written as a truncated limit

$$\Psi_\Gamma^{i,j}(z; w_1, w_2) = \lim_{d \rightarrow \infty} \left( \sum_{\gamma \in \Gamma_{\leq d}} \frac{1}{j(\gamma, z)^k} \frac{w_1 - w_2}{(\gamma z - w_1)^{i+1} (\gamma z - w_2)^{j+1}} \right).$$

Each of the sums inside the limit is a finite sum. Note that

$$\frac{1}{j(\gamma, z)^k} \frac{1}{(\gamma z - w_1)^{i+1} (\gamma z - w_2)^{j+1}} = \frac{1}{j(\gamma^{-1}, w_1)^{i+1} j(\gamma^{-1}, w_2)^{j+1}} \frac{1}{(z - \gamma^{-1} w_1)^{i+1} (z - \gamma^{-1} w_2)^{j+1}}.$$

Since  $\gamma^{-1} w_1$  and  $\gamma^{-1} w_2$  both reduce to the same vertex of  $\mathcal{T}_p$  and have opposite residues, it follows that for every edge  $e \in \mathcal{T}_p^1$ ,

$$\text{res}_e \left( \frac{1}{(z - \gamma^{-1} w_1)^{i+1} (z - \gamma^{-1} w_2)^{j+1}} (X - zY)^n dz \right) = 0.$$

Therefore,

$$\text{res}_e(\Psi_{\Gamma}^{i,j}(z)(X - zY)^n dz) = 0,$$

and so the result follows.  $\square$

**6.2. An explicit formula.** Let  $D_2$  be the divisor  $(w_1) + (w_2)$  and  $D_1$  be the divisor  $(\sigma_1) + (\sigma_2)$ . Assume for simplicity that  $w_1, w_2, \sigma_1, \sigma$  have trivial stabilizers. For any two elements  $x, y \in \mathbb{C}_p$ , define

$$c_{x,y}^{i,j} = \frac{(X - xY)^i (X - yY)^j}{(x - y)^n} \in V_n.$$

Let  $Q_2 \in \mathcal{V}_{w_1} \oplus \mathcal{V}_{w_2} = V_n \oplus V_n$  be the residue class defined by

$$Q_2^{i,j} = (c_{w_1, w_2}^{i,j}, -c_{w_1, w_2}^{i,j}).$$

Similarly, let  $Q_2 \in (\mathcal{V}_n)_{\sigma_1} \oplus (\mathcal{V}_n)_{\sigma_2} = V_n \oplus V_n$  be the residue class defined by

$$Q_1^{r,s} = (c_{\sigma_1, \sigma_2}^{r,s}, -c_{\sigma_1, \sigma_2}^{r,s}).$$

These choices of divisors lead to a bi-extension  $E_{r,s}^{i,j}$  as described in ... . By ... the associated local  $p$ -adic height is given by

$$h_{[E_{r,s}^{i,j}]} = (F_{\Psi_{Q_2}}(\sigma_1) - F_{\Psi_{Q_2}}(\sigma_2), c_{\sigma_1, \sigma_2}^{r,s}).$$

It follows by Lemma 6.3 and 6.4 that

$$\Psi_{Q_2}(z) = \frac{(-1)^i \binom{n}{i}^{-1}}{2^{???}} \Psi_{\Gamma}^{j,i}(z; w_1, w_2) (X - zY)^n dz$$

To obtain an explicit description of the primitive  $F_{\Psi_{Q_2}}$ , we describe how to relate primitives of the weight  $k$  meromorphic modular form  $\Psi_{\Gamma}^{i,j}(z)$  to primitives of the vector valued differential  $\Psi_{\Gamma}^{i,j}(z)(X - zY)^n dz$ .

**Lemma 6.5.** *For any locally meromorphic function,  $F(z)$ , on  $\mathcal{H}_p$ . Let  $F^{(l)}(z)$  denote the  $l$ -th derivative of  $F$ , and let*

$$\beta(F)(z) := n! \sum_{l=0}^n \frac{F^{(l)}(z)}{l!} (X - zY)^l Y^{n-l}.$$

*It holds that*

$$\frac{d}{dz} \beta(F)(z) = F^{(n+1)}(z) (X - zY)^n.$$

*Proof.* The left hand side is a telescoping sum the reduces to the right hand side.  $\square$

**Lemma 6.6.** *For  $i, j \geq 0$  and  $n = i + j$ , let*

$$F_{i,j}(z; w_1, w_2) = \frac{(z - w_1)^j (z - w_2)^i}{(w_1 - w_2)^n} \log_{\mathcal{L}} \left( \frac{z - w_1}{z - w_2} \right),$$

*then*

$$F_{i,j}^{(n+1)}(z; w_1, w_2) = (-1)^i i! j! \frac{w_1 - w_2}{(z - w_1)^{i+1} (z - w_2)^{j+1}}$$

*Proof.* This statement holds by induction on  $i + j = n$ .  $\square$

**Remark 6.1.** It follows that a Coleman primitive of

$$\frac{w_1 - w_2}{(z - w_1)^{i+1}(z - w_2)^{j+1}} (X - zY)^n dz$$

is given by  $\frac{(-1)^i}{i!j!} \beta(F_{i,j}(z; w_1, w_2))$ .

**Theorem 6.1.** The local height pairing for the bi-extension  $E_{r,s}^{i,j}$  is given explicitly by

$$\begin{aligned} h_{[E]_{r,s}}^{i,j} &= \frac{1}{2^{???} n!} \sum_{\gamma \in \Gamma} c_{\sigma_1, \sigma_2}^{r,s} (\beta(F_{i,j}(\gamma z; w_1, w_2))|_{z=\sigma_1} - \beta(F_{i,j}(\gamma z; w_1, w_2))|_{z=\sigma_2}) \\ &= \sum_{\gamma \in \Gamma} \left( \sum_{l=0}^n \frac{1}{l!} \left( \frac{d^l}{dz^l} \left[ \frac{(\gamma z - w_1)^i (\gamma z - w_2)^j}{(w_1 - w_2)^n} \log_{\mathcal{L}} \left( \frac{\gamma z - w_1}{\gamma z - w_2} \right) \right]_{z=\sigma_1} \right. \right. \\ &\quad \left. \left. - \frac{d^l}{dz^l} \left[ \frac{(\gamma z - w_1)^i (\gamma z - w_2)^j}{(w_1 - w_2)^n} \log_{\mathcal{L}} \left( \frac{\gamma z - w_1}{\gamma z - w_2} \right) \right]_{z=\sigma_2} \right) (X - \sigma_1 Y)^l, \frac{(X - \sigma_1 Y)^r (X - \sigma_2 Y)^s}{(\sigma_1 - \sigma_2)^n} \right). \end{aligned}$$

*Proof.* For every  $V_n$ -valued meromorphic differential,  $\omega = f(z)(X - zY)^n dz$ , on  $\mathcal{H}_p$ , let  $F(\omega)$  denote a choice of Coleman primitive of  $\omega$ . For every  $P \in \mathcal{P}_n$ , the value given by

$$(F(\sigma_1) - F(\sigma_2), P^\vee),$$

is equal to the value given by the Morita pairing

$$\int_{\mathbb{P}^1(\mathbb{Q}_p)} P(t, 1) \log_{\mathcal{L}} \left( \frac{t - \sigma_1}{t - \sigma_2} \right) d\mu_f(t),$$

where  $\mu_f$  is the measure on  $\mathbb{P}^1(\mathbb{Q}_p)$  associated to  $f$  [Cite Teitelbaum](#). The Morita pairing is a topological duality. It follows that the local height

$$\left( F_{\Psi_{Q_2}}(\sigma_1) - F_{\Psi_{Q_2}}(\sigma_2), \frac{(X - \sigma_1 Y)^r (X - \sigma_2 Y)^s}{(\sigma_1 - \sigma_2)^n} \right)$$

can be computed term by term inside the convergent sum (6.1) by

$$\begin{aligned} &\left( \sum_{\gamma \in \Gamma} \left[ F \left( \frac{1}{j(\gamma, z)^k} \cdot \frac{w_1 - w_2}{(\gamma z - w_1)^{i+1}(\gamma z - w_2)^{j+1}} (X - zY)^n dz \right) \right]_{z=\sigma_1} \right. \\ &\quad \left. - F \left( \frac{1}{j(\gamma, z)^k} \cdot \frac{w_1 - w_2}{(\gamma z - w_1)^{i+1}(\gamma z - w_2)^{j+1}} (X - zY)^n dz \right) \right]_{z=\sigma_2} \right), \frac{(X - \sigma_1 Y)^r (X - \sigma_2 Y)^s}{(\sigma_1 - \sigma_2)^n}. \end{aligned}$$

The result follows by Remark 6.1.  $\square$

We will now relate the expression give in Theorem 6.1 to the archimedean expression given for the higher Green's functions in [BF25]. Following loc. cit. it will be helpful to introduce formal variables  $U, V, A, B$ . Define

$$\psi^k(z; w_1, w_2) = \sum_{\substack{i+j=n \\ i,j \geq 0}} \frac{w_1 - w_2}{(z - w_1)^{i+1}(z - w_2)^{j+1}} (U - w_1 V)^i (U - w_2 V)^j,$$

and

$$c_1^k = \sum_{\substack{r+s=n \\ r,s \geq 0}} (-1)^r \binom{n}{r} c_{\sigma_1, \sigma_2}^{s,r} (A - \sigma_1 B)^r (A - \sigma_2 B)^s.$$

As before, let  $F_k(z; w_1, w_2)$  be a vector valued Coleman primitive of  $\psi^k(z; w_1, w_2)(X - zY)^n dz$ . By Remark 6.1 it follows that, up to a constant polynomial,

$$F_k(z, w_1, w_2) = \sum_{\substack{i+j=n \\ i,j \geq 0}} \frac{(-1)^i}{i!j!} \beta(F_{i,j})(z; w_1, w_2) (U - w_1 V)^i (U - w_2 V)^j.$$

In [BF25, Proposition 4.3], Brown and Fonseca define a choice of primitive of  $\psi^k(z; w_1, w_2)(X - zY)^k dz$  as follows: define

$$p_k(z, w) = - \sum_{i=0}^n \frac{1}{i!} \left( \frac{(X - zY)(U - wV)}{z - w} \right)^i (VX - UY)^{n-i},$$

and

$$l_k(z; w_1, w_2) = (VX - UY)^n \log_{\mathcal{L}} \left( \frac{z - w_1}{z - w_2} \right),$$

then

$$p_k(z, w_1) - p_k(z, w_2) + l_k(z; w_1, w_2)$$

is a primitive of  $\psi^k(z; w_1, w_2)(X - zY)^n dz$ . Therefore,

$$\begin{aligned} \sum_{\substack{i+j=n \\ i,j \geq 0}} \frac{(-1)^i}{i!j!} (\beta(F_k)(\sigma_1; w_1, w_2) - \beta(F_k)(\sigma_2; w_1, w_2)) (U - w_1V)^i (U - w_2V)^j = \\ p_k(\sigma_1, w_1) - p_k(\sigma_1, w_2) - p_k(\sigma_2, w_1) + p_k(\sigma_2, w_2) + l_k(\sigma_1; w_1, w_2) - l_k(\sigma_2; w_1, w_2) \end{aligned}$$

It follows that

$$\begin{aligned} (F_k(\sigma_1; w_1, w_2) - F_k(\sigma_2; w_1, w_2), c_1^k) &= \sum_{\substack{i+j=n \\ i,j \geq 0}} \frac{(-1)^i}{i!j!} \sum_{\substack{r+s=n \\ r,s \geq 0}} \left[ (\beta(F_{i,j})(\sigma_1; w_1, w_2) - \beta(F_{i,j})(\sigma_2; w_1, w_2), c_{\sigma_1, \sigma_2}^{r,s}) \right. \\ &\quad \left. \times (U - w_1V)^i (U - w_2V)^j (A - \sigma_1B)^r (A - \sigma_2B)^s \right]. \end{aligned}$$

On the other hand, by the binomial theorem

$$c_1^k = (XB - YA)^n.$$

It follows that

$$\begin{aligned} (F_k(\sigma_1; w_1, w_2) - F_k(\sigma_2; w_1, w_2), c_1^k) &= \sum_{\substack{i+j=n \\ i,j \geq 0}} \frac{(-1)^i}{i!j!} \left[ (\beta(F_{i,j})(\sigma_1; w_1, w_2) - \beta(F_{i,j})(\sigma_2; w_1, w_2), (XB - YA)^n) \right. \\ &\quad \left. \times (U - w_1V)^i (U - w_2V)^j \right] \\ &= \sum_{\substack{i+j=n \\ i,j \geq 0}} \left[ (\beta(F_{i,j})(\sigma_1; w_1, w_2) - \beta(F_{i,j})(\sigma_2; w_1, w_2))^{X=A, Y=B} (U - w_1V)^i (U - w_2V)^j \right] \\ &= \left[ p_k(\sigma_1, w_1) - p_k(\sigma_1, w_2) - p_k(\sigma_2, w_1) + p_k(\sigma_2, w_2) + l_k(\sigma_1; w_1, w_2) - l_k(\sigma_2; w_1, w_2) \right]^{X=A, Y=B} \end{aligned}$$

This last expression formally agrees with  $g_k(z, w)$  of [BF25, Definition 4.1], after substituting  $X = A$  and  $Y = B, z = \sigma_1, \bar{z} = \sigma_2, w = w_1, \bar{w} = w_2$ , and  $\log$  is replaced with  $\log_{\mathcal{L}}$ . **I think the following works only when we have conjugates.**

**Definition 6.1.** Let  $Q_{s, \mathcal{L}}$  be the  $p$ -adic analogue of the Legendre function of the second kind, defined by

$$Q_{s, \mathcal{L}}(x) := \begin{cases} \frac{1}{2} \log_{\mathcal{L}} \left( \frac{1+x}{1-x} \right) & s = 0 \\ x Q_{0, \mathcal{L}}(x) - 1 & s = 1 \\ \frac{2s-1}{s} x Q_{s-1, \mathcal{L}}(x) - \frac{s-1}{s} Q_{s-2}(x) & s \geq 2. \end{cases}$$

**Corollary 6.1.** Let

$$f_{n/2}(\sigma_1, \sigma_2, w_1, w_2) = -2Q_{(k-2)/2, \mathcal{L}} \left( 1 - 2 \frac{(\sigma - w)(\bar{\sigma} - \bar{w})}{(\sigma - \bar{\sigma})(w - \bar{w})} \right).$$

Then

$$h_{[E]_{n/2,n/2}}^{n/2} = * \frac{f_{n/2}(\sigma_1, \sigma_2, w_1, w_2)}{(w_1 - w_2)^{n/2}(\sigma_1 - \sigma_2)^{n/2}}.$$

*Proof.* This is the statement of [BF25, Proposition 4.13], which implies the statement in our  $p$ -adic setting as explained above.  $\square$

## 7. A $p$ -ADIC GROSS–KOHLEN–ZAGIER RESULT

In this section we present a  $p$ -adic analogue of the Gross–Kohnen–Zagier conjecture [GKZ87] of the log-algebraicity of heigher Green functions evaluated at Heegner points. Let  $\mathcal{G}_k$  denote the higher  $p$ -adic Green function constructed in Section 6, with the branch  $\log_{\mathcal{L}}$  and the normalization conventions fixed there.

**Theorem 7.1** ( $p$ -adic Gross–Kohnen–Zagier). *Fix the Shimura curve  $X$  and the positive even integer  $k \geq 4$ . Consider a non-zero finite linear combination of Hecke operators*

$$T = \sum_{r \geq 1} c_r T_r, \quad c_r \in \mathbb{Q},$$

*which kills  $V$ . Let  $z_0, w_0$  be CM points associated to imaginary quadratic fields  $K_1, K_2$  with coprime discriminants satisfying the Heegner hypothesis and the local hypotheses of Sections 5 and 6. Consider the Heegner biextension described in Section 5.5. If  $H_i$  denotes the Hilbert class field of  $K_i$ , then there is an element  $\alpha \in H_1 H_2$  for which the sum over the Galois orbits of Heegner points  $z_0, w_0$  of the evaluation of the higher Green's function is equal to*

$$(7.1) \quad \mathrm{Tr}_{H_v/\mathbb{Q}_p} \left( \sum_{\sigma, \tau} \mathcal{G}_k(\sigma \cdot z_0, T(\tau \cdot w_0)) \right) = C_{k, K_1, K_2} \log_p(N_{H/\mathbb{Q}}(\alpha))$$

*with  $\sigma \in \mathrm{Gal}(H_1/K_1)$ ,  $\tau \in \mathrm{Gal}(H_2/K_2)$  and an explicit constant  $C_{k, K_1, K_2}$ .*

*Proof.* Let  $T$  be a combination of Hecke operators annihilating  $V$ , let  $H = H_1 H_2$ , and let  $E$  be the biextension considered in Section 5. By Section 5.8, the transformed biextension  $[^T E]$  reduces to an element

$$(7.2) \quad [U] \in \mathrm{Ext}_{\mathrm{Rep}_{G_H}}^1(\mathbb{Q}_p, \mathbb{Q}_p(1)) = H^1(H, \mathbb{Q}_p(1)) \cong H^\times \hat{\otimes} \mathbb{Q}_p$$

where the last isomorphism is the Kummer isomorphism (5.21). Let  $\alpha \in H^\times \hat{\otimes} \mathbb{Q}_p$  denote the class corresponding to  $[U]$ . With the local sign convention fixed in Section 5, the height identity is

$$(7.3) \quad \sum_{v|p} h(^T e_1, e_2)_{E,v} = - \sum_{v|p} l_v(\alpha) = \sum_{v|p} l_v(\alpha) = - \sum_{v \nmid p} \mathrm{ord}_v(^T E) \log_p(N_{H/\mathbb{Q}}(v)),$$

where the second equality follows from the global relation among the local characters. Now observe that by the compatibility with the Galois action on the places we have

$$(7.4) \quad h(e_1^\sigma, e_2^\sigma)_{\sigma(E), \sigma(v)} = h(e_1, e_2)_{E,v} \quad \text{for all } \sigma \in \mathrm{Gal}(H/\mathbb{Q}).$$

Since the discriminants are coprime, we obtain that  $H^1 \cap H^2 = \mathbb{Q}$  and the Galois group is given by the product of the two dihedral groups associated with  $H_1, H_2$ . Recall that  $p$  is inert in  $K_1, K_2$  so it is totally split on the Hilbert class fields  $H_1, H_2$  so there are  $[H : \mathbb{Q}]/2$  places above  $p$ . Fixing a place  $v$ , the formula then becomes

$$(7.5) \quad \sum_{v|p} h(^T e_1, e_2)_{E,v} = \frac{1}{2} \sum_{\sigma \in \mathrm{Gal}(H/\mathbb{Q})} h(^T e_1^\sigma, e_2^\sigma)_{E,v}$$

In particular, the right sum amounts of taking a sum over a Galois orbit of the Heegner cycles in  $\mathrm{emb}(\mathcal{O}_{K_1}, R)$ ,  $\mathrm{emb}(\mathcal{O}_{K_2}, R)$  under the compatibility of the Galois action described in Section 3.10. Now observe that by Theorem 5.3 the local contribution away from  $p$  is given by

$$(7.6) \quad \mathrm{ord}_v(^T E) = \sum_{x \in \underline{D}_{1,v} \cap \underline{D}_{2,v}} (\underline{Q}_1, \underline{Q}_2)_x.$$

We can then rewrite the equation (7.3)

$$(7.7) \quad \mathrm{Tr}_{H_v/\mathbb{Q}_p} \left( \sum_{\sigma, \tau} \mathcal{G}_k(\sigma \cdot z_0, T(\tau \cdot w_0)) \right) = \sum_{\sigma \in \mathrm{Gal}(H/\mathbb{Q})} h(e_1^\sigma, {}^T e_2^\sigma)_{E,v} =$$

$$(7.8) \quad = -2 \sum_{v \nmid p} \log_p(N_{H/\mathbb{Q}}(v)) \sum_{x \in \underline{D}_{1,v} \cap \underline{D}_{2,v}} (\underline{y}, \underline{z})_x(\underline{Q}_{1,x}, \underline{Q}_{2,x})$$

concluding the proof. □

**Remark 7.1.** *The explicit factorization of the rational number could be studied in terms of local deformation theory for the intersection pairing of the two Heegner divisors as done by [Zha97].*

**Remark 7.2.** *The motivic conjecture of Brown and Fonseca would give a second route. It predicts that  $\alpha$  lies in*

$$\mathrm{Ext}_{\mathrm{DM}}^1(\mathbb{Q}, \mathbb{Q}(1)) = H^1(H, \mathbb{Q}(1)) \cong H^\times \otimes \mathbb{Q}.$$

## REFERENCES

- [BC91] Jean-François Boutot and Henri Carayol. Uniformisation  $p$ -adique des courbes de Shimura : les théorèmes de Čerednik et de Drinfeld. *Astérisque*, pages 45–158, 1991.
- [BD96] M. Bertolini and H. Darmon. Heegner points on Mumford–Tate curves. *Inventiones mathematicae*, 126(3):413–456, 1996.
- [BD98] Massimo Bertolini and Henri Darmon. Heegner points,  $p$ -adic  $L$ -functions, and the Čerednik–Drinfeld uniformization. *Inventiones mathematicae*, 131(3):453–491, 1998.
- [BDIS02] Massimo Bertolini, Henri Darmon, Adrian Iovita, and Michael Spiess. Teitelbaum’s exceptional zero conjecture in the anticyclotomic setting. *Amer. J. Math.*, 124(2):411–449, 2002.
- [Bes04] Amnon Besser. The  $p$ -adic height pairings of Coleman–Gross and of Nekovář. In *Number Theory*, volume 36 of *CRM Proceedings and Lecture Notes*, pages 13–25. American Mathematical Society, Providence, RI, 2004.
- [Bes22] Amnon Besser.  $p$ -adic heights and Vologodsky integration. *Journal of Number Theory*, 239:273–297, 2022.
- [BEY21] Jan Hendrik Bruinier, Stephan Ehlen, and Tonghai Yang. CM values of higher automorphic Green functions for orthogonal groups. *Inventiones mathematicae*, 225(3):693–785, 2021.
- [BF25] Francis Brown and Tiago J. Fonseca. Single-valued periods of meromorphic modular forms and a motivic interpretation of the Gross–Zagier conjecture, 2025.
- [Bry89] Jean-Luc Brylinski. Heights for local systems on curves. *Duke Mathematical Journal*, 59(1):1 – 26, 1989.
- [BZ23] Amnon Besser and Sarah Livia Zerbes. Vologodsky integration on curves with semi-stable reduction. *Israel Journal of Mathematics*, 253(2):761–770, 2023.
- [CF00] Pierre Colmez and Jean-Marc Fontaine. Construction des représentations  $p$ -adiques semi-stables. *Inventiones mathematicae*, 140(1):1–43, 2000.
- [CG89] Robert F. Coleman and Benedict H. Gross.  $p$ -adic heights on curves. In John Coates, editor, *Algebraic Number Theory*, volume 17 of *Advanced Studies in Pure Mathematics*, pages 73–81. Academic Press, Boston, 1989.
- [CI10] Robert Coleman and Adrian Iovita. Hidden structures on semistable curves. In *Représentations  $p$ -adiques de groupes  $p$ -adiques III : méthodes globales et géométriques*, number 331 in *Astérisque*, pages 179–254. Société mathématique de France, 2010.
- [Dis17] Daniel Disegni. The  $p$ -adic Gross–Zagier formula on Shimura curves. *Compositio Mathematica*, 153(10):1987–2074, 2017.
- [Dis23] Daniel Disegni. The  $p$ -adic Gross–Zagier formula on Shimura curves, II: nonsplit primes. *Journal of the Institute of Mathematics of Jussieu*, 22(5):2199–2240, 2023.
- [DM91] Ch. Deninger and Jacob Murre. Motivic decomposition of abelian schemes and the Fourier transform. *Journal für die reine und angewandte Mathematik*, 422:201–219, 1991.
- [Fal89] Gerd Faltings. Crystalline cohomology and  $p$ -adic Galois-representations. In Jun-Ichi Igusa, editor, *Algebraic Analysis, Geometry, and Number Theory (Baltimore, MD, 1988)*, pages 25–80. Johns Hopkins University Press, Baltimore, MD, 1989.
- [Fal92] Gerd Faltings. Crystalline cohomology of semistable curves, and  $p$ -adic Galois-representations. *J. Algebraic Geom.*, 1(1):61–81, 1992.
- [Fal02] Gerd Faltings. Almost étale extensions. In Berthelot Pierre, Fontaine Jean-Marc, Illusie Luc, Kato Kazuya, and Rapoport Michael, editors, *Cohomologies  $p$ -adiques et applications arithmétiques (II)*, number 279 in *Astérisque*, pages 185–270. Société mathématique de France, 2002.
- [FM87] Jean-Marc Fontaine and William Messing.  $p$ -adic periods and  $p$ -adic étale cohomology. In Kenneth A. Ribet, editor, *Current Trends in Arithmetical Algebraic Geometry (Arcata, Calif., 1985)*, volume 67 of *Contemporary Mathematics*, pages 179–207. American Mathematical Society, Providence, RI, 1987.
- [Fon79] Jean-Marc Fontaine. Modules galoisiens, modules filtrés et anneaux de Barsotti–Tate. In *Journées de géométrie algébrique de Rennes - (Juillet 1978) (III) - Groupes formels, représentations galoisiennes et cohomologie des variétés de caractéristique positive*, number 65 in *Astérisque*, pages 3–80. Société mathématique de France, 1979.

- [Fon94] Jean-Marc Fontaine. Exposé II : Le corps des périodes  $p$ -adiques. In Jean-Marc Fontaine, editor, *Périodes  $p$ -adiques - Séminaire de Bures, 1988*, number 223 in Astérisque, pages 59–101. Société mathématique de France, 1994. talk:2.
- [GKZ87] B. Gross, W. Kohnen, and Don Zagier. Heegner points and derivatives of  $L$ -series, II. *Mathematische Annalen*, 278:497–562, 1987.
- [GR96] E.U. Gekeler and M. Reversat. Jacobians of Drinfeld modular curves. *Journal für die reine und angewandte Mathematik*, 476:27–94, 1996.
- [GZ86] Benedict H. Gross and Don B. Zagier. Heegner points and derivatives of  $L$ -series. *Inventiones mathematicae*, 84(2):225–320, 1986.
- [IS03] Adrian Iovita and Michael Spiess. Derivatives of  $p$ -adic  $L$ -functions, Heegner cycles and monodromy modules attached to modular forms. *Inventiones mathematicae*, 154(2):333–384, 2003.
- [KM85] Nicholas M. Katz and Barry Mazur. *Arithmetic Moduli of Elliptic Curves. (AM-108)*. Princeton University Press, 1985.
- [Kün94] Klaus Künnemann. On the Chow motive of an abelian scheme. *Motives (Seattle, WA, 1991)*, 55(Part 1):189–205, 1994.
- [Li23] Yingkun Li. Algebraicity of higher Green functions at a CM point. *Inventiones mathematicae*, 234(1):375–418, 2023.
- [Mas11] Marc Masdeu. CM cycles on Shimura curves, and  $p$ -adic  $L$ -function. *Compositio Mathematica*, 148, 10 2011.
- [Mel08] Anton Mellit. Higher Green’s functions for modular forms. *arXiv: Number Theory*, 2008.
- [Nek93] J Nekovář. On  $p$ -adic height pairing. *Séminaire de Théorie des Nombres, Paris 1990/91. Progress in Mathematics*, 108:127–202, 1993.
- [Nek95] Jan Nekovář. On the  $p$ -adic height of Heegner cycles. *Mathematische Annalen*, 302(4):609–686, 1995.
- [PR87] Bernadette Perrin-Riou. Points de Heegner et dérivées de fonctions  $L$   $p$ -adiques. *Inventiones Mathematicae*, 89(3):455–510, 1987.
- [Sch88] Norbert Schappacher. *Periods of Hecke Characters*. Lecture Notes in Mathematics. Springer, Berlin, Germany, 1988 edition, February 1988.
- [Sch90] A. J. Scholl. Motives for modular forms. *Inventiones mathematicae*, 100(1):419–430, 1990.
- [Sch94] A. J. Scholl. Height pairings and special values of  $L$ -functions. In *Motives (Seattle, WA, 1991)*, volume 55, Part 1 of *Proc. Sympos. Pure Math.*, pages 571–598. Amer. Math. Soc., Providence, RI, 1994.
- [Sha89] Ehud de Shalit. Eichler cohomology and periods of modular forms on  $p$ -adic Schottky groups. *Journal für die reine und angewandte Mathematik*, 400:3–31, 1989.
- [Shn16] Ariel Shnidman.  $p$ -adic heights of generalized Heegner cycles. *Annales de l’Institut Fourier*, 66(3):1117–1174, 2016.
- [The26] The Stacks Project Authors. The stacks project, section 50.20: Poincaré duality. <https://stacks.math.columbia.edu/tag/0FW3>, 2026.
- [Tsu99] Takeshi Tsuji.  $p$ -adic étale cohomology and crystalline cohomology in the semi-stable reduction case. *Inventiones mathematicae*, 137(2):233–411, 1999.
- [Via11] Maryna Viazovska. CM values of Higher Green’s functions, 2011.
- [Wei18] Fu-Tsun Wei. Green’s functions on Mumford curves. *Math. Ann.*, 370(3-4):1571–1605, 2018.
- [Zha97] Shouwu Zhang. Heights of Heegner cycles and derivatives of  $L$ -series. *Inventiones mathematicae*, 130(1):99–152, 1997.